

Welcome to the Nutrition Care Manual - Diet Manual

Academy of Nutrition and Dietetics

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The Nutrition Care Manual® (NCM) Diet Manual is a publication of the Academy of Nutrition and Dietetics.

For questions about materials to be included or participating in future updates as a contributor and/or reviewer, please contact the editor at

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The client education handouts in the Academy's Nutrition Care Manual® are not intended to substitute for nutrition counseling with a registered dietitian. The information is meant to serve as a general guideline and may not meet the unique nutritional needs of individual patients. All medical professionals should consult with a registered dietitian before providing handouts to clients or patients.

With the exception of adding patient-specific recommendations in the Notes section clearly marked in the client education handouts, no part of the handouts may be modified without prior written consent of the publisher.

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Diet Manual

The NCM Diet Manual provides practical, usable knowledge that can be easily implemented in a multitude of nutrition care settings. It is intended to assist food and nutrition services departments with planning nutritionally adequate meals for their patients.

As a communication tool, the NCM Diet Manual assists registered dietitian nutritionists (RDNs) and nutrition and dietetics technicians, registered (NDTR) working with other health care professionals to provide menu choices that are consistent and accurate with regard to optimal nutritional care. Simultaneously, these menus respect patients' individual, ethnic, religious, and cultural needs.

The NCM Diet Manual focuses on how to design and implement nutritionally balanced menus while customizing to individual medical nutrition therapy needs and preferences. It incorporates the clinical care updates found within the NCM, while supporting the "nutrition care" part of the nutrition therapy. The step-by-step process for implementation of the Diet Manual can be saved electronically or printed as hard copy, which represents multiple options for meeting the regulation to have the diet manual available to the health care team. Much of the NCM Diet Manual content can be adapted for staff, patient, and family education. See the Implementation section.

Diet Liberalization and patient choice are emphasized throughout the NCM Diet Manual. The importance of individualizing diet for each patient is essential. Expanding healthy food choices to meet patient requests as well as continuing to meet their varying nutrition needs poses food management and foodservice challenges. Concise, organized menu guidelines are included to facilitate consistent, accurate planning and serving of diets. This NCM Diet Manual also provides direction when stricter diets are essential or are desired by the patient.

Note: The term "patient" will be used throughout the NCM Diet Manual except in sections where regulations of the Centers for Medicare & Medicaid Services (CMS) are discussed. Per the language of CMS regulations, the term "resident" is used in those sections.

Diet Liberalization

In the NCM Diet Manual, a House diet is the most liberal diet and is not considered a therapeutic or mechanically altered diet. The majority of patients will benefit from a House Diet designed to meet the Dietary Reference Intakes (DRI) for their age group. Texture-modified diets are indicated for patients with difficulty chewing or swallowing. Therapeutic diets controlling carbohydrates or restricting specific nutrients such as protein, sodium, or potassium may be indicated for patients with impaired metabolic capacity for specific nutrients or to prevent the progression of certain diseases. Unless a therapeutic diet is required to manage a medical condition, it may be preferable to start with a House Diet and monitor a patient's individual tolerance.

Diet liberalization is essential in the registered dietitian nutritionist's assessment and evaluation process to determine the least restricted therapeutic and texture-modified diet order for a patient. The benefits of liberalizing diets include:

- Increased patient food and beverage choices
- Simplified meal-ordering procedures
- Improved dining experience
- Improved flavor and acceptance of foods served
- Added options to address inadequate intake
- Simplified foodservice
- Decreased potential meal service errors
- Streamlined guidance on diet definitions (who can have what foods/fluids)
- Increased staff time for calm interaction during meal selection and service
- Additional supports for a successful survey process

Conversely, heavily restricted diets may result in the following:

- Frustration and dissatisfaction with meal service
- Reduced intake related to variety of offerings
- Fatigue
- Weight loss and muscle breakdown
- Underfeeding that increases risk for or exacerbating existing malnutrition
- Poor wound healing
- Compromised immune function
- Delayed recovery
- Reduce quality of life

The RDN can provide education on the importance of liberalizing diets to facility management and the health care team. Some menu software programs allow users to run diet order frequency reports that provide information on individual diets and combination diets (ie, more than one diet restriction) ordered over a given period. These reports can be used in quality improvement initiatives to evaluate if therapeutic diets are overprescribed and to improve patient meal satisfaction. The reports help providers analyze the extent of combination diets ordered and whether they are appropriate for a given patient population. Use the Letter to the Health Care Team or Diet Manual Approval Committee to

summarize the annual changes made to the NCM and effectively communicate the importance of diet liberalization at your facility. Review the Regulations section to choose which supportive documents to include.

One of the most commonly cited references that support liberalizing diets is the Academy of Nutrition and Dietetics position paper on individualized nutrition approaches for older adults:

It is the position of the Academy of Nutrition and Dietetics “...that the quality of life and nutritional status of older adults in long-term care, post-acute care, and other settings can be enhanced by individualized nutrition approaches. The Academy advocates that as part of the interprofessional team, registered dietitian nutritionists assess, evaluate, and recommend appropriate nutrition interventions according to each individual’s medical condition, desires, and rights to make health care choices. Nutrition and dietetic technicians, registered assist registered dietitian nutritionists in the implementation of individualized nutrition care, including the use of least restrictive diets. Health care practitioners must assess risks vs benefits of therapeutic diets, especially for frail older adults. Food is an essential component of quality of life; an unpalatable or unacceptable diet can lead to poor food and fluid intake, resulting in malnutrition and related negative health effects. Including older individuals in decisions about food can increase the desire to eat and improve quality of life” (Dorner, 2018).

References

Dorner B, Friedrich EK. Position of the Academy of Nutrition and Dietetics: Individualized nutrition approaches for older adults: Long-term care, post-acute care, and other settings. J Acad Nutr Diet. 2018;118(4):724-735. Abstract.

Adapting a new diet manual presents the opportunity to connect patient needs and evidence-based nutrition. A diet manual provides effective, consistent communication with diet order terminology and definitions and emphasizes individualization.

Diet Manual Approval

It is generally best practice to review and approve an institution's diet manual annually. The NCM Diet Manual provides resources to simplify this process for the registered dietitian nutritionist (RDN) and foodservice staff. The diet manual approval process requires the approval committee to review updated nutrition resources and information in order to implement evidence-based nutrition care within health care settings.

Quality Assurance and Performance Improvement (QAPI)

Successful implementation of the NCM Diet Manual can be guided through the QAPI process, which includes documenting patient and staff input. To review the QAPI process and access sample forms, refer to the CMS QAPI Resources, Academy MQii page, or Malnutrition Care Score.

Menu Planning

The NCM Diet Manual provides menu planning guidance that reflects evidence-based nutrition recommendations that can assist foodservice departments in accommodating patient preferences. RDNs and foodservice managers customize the diet manual to establish consensus on what food items are appropriate for each diet. Implementing and customizing the diet manual ensures that menu choices are accurate according to the diet orders, consistent with patient requests, and congruous with regulations.

Menu planning requires determining which foods are acceptable for each diet type. The NCM Diet Manual provides information and/or links to the appropriate client education handouts that provide general meal-planning tips and Foods Lists based on the major food groups. These resources combined with your NCM Diet Manual Crosswalk create the recommendations for what foods to include on or omit from a patient's menu options.

Communicating menu modifications to foodservice staff is essential for safe, efficient and accurate food production and meal service. Some facilities may use diet guide sheets (also referred to as diet breakout sheets or menu spread) to communicate menu modifications to foodservice staff. These sheets provide specific guidance on food, fluid, and texture modifications for facility diets. These sheets are generally less complex when a facility follows a more liberalized diet approach.

Regulations

Regulations dictate that the diet manual be accessible to the health care staff at your facility. Nutrition Care Manual products (NCM, PNCM) are consistent with

the Centers for Medicare & Medicaid Services' Interpretive Guidelines and the Joint Commission's Hospital Accreditation Standards. Nutrition Care Manual products (NCM, PNCM, SNCM) are also consistent with the Academy's Evidence Analysis Library, Academy position papers, and the Nutrition Care Process.

Follow these seven steps annually to implement and customize your NCM Diet Manual.

1. Collaborate on Diet Manual Philosophy and Goals

Review and agree on diet manual philosophy and menu planning goals along with members of the health care team. Include registered dietitian nutritionist (RDN), Food and Nutritional Services team, nursing, speech therapists, administration, and other team members. The following resources from the NCM Diet Manual will assist you with achieving a successful diet manual and menu implementation:

- Focus on diet liberalization
- Emphasize patient choice and patient-directed care
- Incorporate considerations that respect religious, cultural, ethnic, and usual eating practices
- Revisit therapeutic diets and texture-modified diets essential for medical nutrition therapy
- Review NCM Update Summary to discuss the most recent updates to NCM

2. Identify All Diet Orders Used at Your Facility

- Review NCM Diet Order Terminology & Definitions to determine which diet order terms are available for use at your facility.
- Obtain a list of diets approved by your medical committee for use at your facility.
 - Expert tip: A list of diets used by your foodservice department can be generated from your diet office software.
 - Compare your facility's current diet order terminology and definitions with the NCM Diet Manual names (performed by clinical nutrition staff).
 - Identify NCM diets to be used at your facility in consultation with medical staff leadership and medical director, referring to the evidence to support use of diets.
 - As needed, clinical nutrition staff will create its own unique folder of diets.
- Record the names for all facility diets and NCM diets in the NCM Diet Manual Crosswalk Worksheet. **Expert tip: Use the exact names of NCM**

Diets/Client Education materials at your facility to simplify this process.

- Review Obsolete Diets to ensure you avoid using these diet terms in your facility's accepted diet terminology.
- Diet names used in all areas should match: medical record documentation (electronic or paper), printed menus or tray tickets, diet manual, documents used by the kitchen staff (i.e. menu spreadsheets), diet guide sheets, and policies and procedures.

3. Make Changes to Facility Diets as Needed

- Identify current diet names used at your facility that may require a change and/or diet names that require clarification of foods served.
- Discuss modifications to existing diet order names with health care team and agree on final names for all diets. Document and date this discussion.
 - Consider involving speech therapy with diet order terminology and definitions of texture-modified diets.
- Change existing diet terminology to match new, updated diet names.

4. Customize the NCM Diet Manual to Your Facility

The facility administrator completes this step which will connect the NCM diets with the names of the diets offered at your facility. Customizing the NCM Diet Manual crosswalk is an important step with meeting regulations concerning diet manuals.

- Use the NCM Diet Manual Crosswalk Guide and the information gathered in NCM Diet Manual Crosswalk Worksheet completed in step two to customize the NCM Diet Manual to your facility.
- Create and update Facility Pages for unique diets used at your facility (for those that have not been published among NCM Diets). See FAQ on Creating Facility Pages.

5. Obtain Diet Manual Approval

- Select one of the NCM Diet Manual Approval Form templates (1, 2, or 3) for your facility (depending on the participants of the approval committee). There are three templates to choose from, but only one template needs to be completed.
- Provide Diet Manual updates at committee meetings for awareness. Access the News/Updates for the most recent NCM Update Summary page

- which includes information about the rolling updates made to the NCM.
- Obtain required signatures from all relevant parties.

6. IP Authentication

Work with your facility's IT department to set up IP authentication, which will allow users at your facility to access the NCM without a username and password. A custom URL will be automatically generated, and can be included on your facility's home page or wherever is convenient. You may choose to use the NCM as the access icon. Anyone accessing the NCM within the facility's IP range will be able to launch directly into the NCM. The NCM will be accessible in all patient care areas, which is a regulatory requirement. The FAQ titled "How do I access the NCM products without logging in each time?" provides detailed instructions to complete this process. Contact NCMSupport@eatright.org if further assistance is needed.

7. Backup Diet Manual

Create an abridged backup diet manual binder composed of relevant and frequently used diet manual materials. This can be stored in a common area within the food and nutrition department where it can be accessed by staff. Determine if multiple copies of the abridged backup diet manual binder are needed in other locations within your facility. Creating this resource will ensure that you have access to important NCM content in the event of a power outage or inability to access the internet. Additionally, this will meet annual survey expectations. The abridged backup diet manual should include the following items:

Electronic

- The back-up PDF of the NCM which includes the NCM and the NCM Diet Manual.
 - An electronic backup is recommended because printing all these NCM resources would require several thousand pages to print.
 - To download the backup PDF of the NCM please go to "How to download the back-up PDF of NCM or PNCM " listed under Frequently Asked Questions. You must login as the administrator to do so.

Printed

- Frequently used and relevant NCM Client Education handouts
- Commonly used therapeutic diets and texture-modified diets
- NCM Diet Order Terminology & Definitions
- Customized facility pages including your Facility's Diet Manual Crosswalk and any added diets unique to your facility
- NCM Table of Contents
- Current NCM Update Summary
- A signed copy of one of the NCM Diet Manual Approval Forms (1, 2, or 3)

Regulations

Regulations and standards of practice dictate the importance of routine nutrition assessment and evaluation of the least restrictive therapeutic and texture-modified diet for each patient. The culture change movement and focus on patient-directed care and decision making continue to influence regulatory expectations, which affect nutrition and foodservice department policy, procedures, and operations. Refer to federal, state, and local regulations for your area of practice.

Meeting Regulations and Preparing for Survey

NCM Diet Manual users must adhere to facility policies and procedures and local, state, and national regulations. The NCM Diet Manual is designed to meet the unique regulatory needs in multiple settings and across state lines through its customizable features.

To assist in facility customization and to prepare for the survey, follow the step-by-step process found in the Implementation section of the diet manual. Repeat the Diet Manual review and implementation process each year. This exercise will ensure continued use of the NCM Diet Manual at your facility and proactively keep the manual current with regard to regulatory requirements. Update the customized NCM Diet Manual annually to meet your facility's specific needs and to reflect actual diet terminology and content used at your facility.

Creating an abridged back-up diet manual of the NCM using printed copies of the NCM meets regulations that require that it be available to the health care team. However, retaining electronic copies on multiple desktops, CD, or USB drives is the economical option. The health care team should be aware of the NCM Diet Manual resources for standard and emergency processes and how to locate them.

During a facility survey, a surveyor may be interested in viewing the facility-specific backup manual (printed resource). This backup manual includes the following resources stored in an accessible, labeled binder:

Electronic

- The back-up PDF of the NCM which includes the NCM, the NCM Diet Manual, and Client Education.
 - An electronic back-up is recommended because printing all these NCM resources would require several thousand pages to print.
 - To download the back-up PDF of the NCM please go to “How to download the back-up PDF of NCM or PNCM “ listed under Frequently Asked Questions. You must login as the administrator to do so.

Printed

- Frequently used and relevant NCM Client Education handouts
- Commonly used therapeutic diets and texture-modified diets
- NCM Diet Order Terminology & Definitions
- Customized facility pages including your Facility's Diet Manual Crosswalk and any added diets unique to your facility
- NCM Table of Contents
- Current NCM Update Summary
- A signed copy of one of the NCM Diet Manual Approval Forms (1, 2, or 3)
- Letters to the health care team or diet manual approval committee

Locating Regulations for Health Care Facilities

Health care facility regulations are updated frequently and vary significantly depending on the setting. You can find regulations applicable for your setting below:

Acute Care

- Joint Commission Standards, On-Site Survey Process, Requirements, Manuals and Related Resources
- See CMS Guidance to Laws and Regulations for Appendix A of the State Operations Manual which includes CMS §482.28 Condition of Participation: Food and Dietetic Services

Post-Acute Care

- Joint Commission Standards, On-Site Survey Process, Requirements, Manuals and Related Resources
- CMS Final Rule 2016
- CMS Regulations and Guidance Manuals
- See CMS Guidance to Laws and Regulations for Appendix PP State Operations Manual, LTC Survey Pathways (dining, kitchen, resident council, dialysis, nutrition, hydration, tube feeding, and dementia care), and New Long-term Care Survey Process
- CMS QAPI Tools
- Assisted Living
 - Regulated on a state by state basis
- Corrections

Check your individual state regulations as well.

Where applicable, regulations are discussed in detail in subsections of the NCM Diet Manual.

Acute Care

Other factors that may affect menu content include:

- CMS Regulations and Interpretive Guidelines
- The Joint Commission

Skilled Nursing Facilities

The Centers for Medicare & Medicaid Services (CMS) regulations mandate that skilled nursing facilities must provide each resident with a "nourishing, palatable, well-balanced diet that meets the daily nutritional and special dietary needs of each resident" (CMS, 2006). The regulations (CMS, 2006) currently state that the menu:

- Must meet the Recommended Dietary Allowances (RDA) of the Food and Nutrition Board of the National Research Council, National Academy of Sciences
- Is not intended to meet the nutrition needs of all residents and, therefore, must be adjusted to consider individual differences
- Uses a meal planning guide
- Is planned in advance
- Is followed
- Offers substitutions of similar nutritional value

(Note: Because these are CMS regulations regarding skilled nursing facilities, the term "resident" is used here.)

Menus are planned in coordination with the registered dietitian nutritionist, who adjusts the menu for residents' individual needs. There are many meal planning guides available online, including the following:

- 2020-2025 Dietary Guidelines for Americans
- MyPlate
- MyPlate for Older Adults
- USDA Food Patterns
- Dietary Approaches to Stop Hypertension (DASH) Eating Plan

The House Diet Menu Planning Guide outlines the nutritional components of all healthy eating plans above, which can be used as a meal planning guide consistent with regulation requirements.

References: Regulations

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Emergency Preparedness

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Emergency preparedness involves strategic planning, preparation, and documentation of procedures to effectively respond to events that may disrupt normal operations, including natural disasters. Comprehensive emergency response plans should be developed based on the type of facility, location, and likelihood of potential disasters such as fire, hurricane, and cyber attack. Conducting regular drills, training staff, and ensuring resource availability to manage each type of crisis may help to minimize the impact on food and clinical nutrition services and provide for employee and patient safety. Without proper preparation, organizations may face prolonged disruptions and financial losses. By developing a robust emergency preparedness program, organizations can navigate crises more efficiently, recover faster, and maintain stability in the face of adversity.

Regulations regarding emergencies vary by site and by state, but health care agencies including the Centers for Medicaid and Medicare Services (CMS), the Joint Commission, and the Centers for Disease Control and Prevention offer continuously updated guidelines on how to plan for emergencies in foodservice operations. CMS regulations require that emergency plans at inpatient facilities be prepared, implemented, reviewed, and then updated every 2 years (CFR, 2025). The requirement for long-term facilities is an annual review. Check local regulations, including state, region, and municipality, for additional guidance (CMS, 2021).

The primary focus for emergency preparedness in a dietetics setting is to ensure the safe and adequate provision of food and fluid along with critical medical nutrition therapy.

General Emergency Planning Guidance

Emergency planning guidance can help you determine protocols for concerns related to communication, staffing, supply chain and food production, and patient meal service.

Communication

Create a phone tree to delineate the hierarchy of communication for emergency alerts. The phone tree should include names, titles, and work and cell/home phone numbers.

- Designate a person in charge (PIC) who can commit to the responsibility for continued departmental and clinical operations during an emergency. Make sure this appointment is clearly communicated to all staff in the organization.
- Designate who is the upline leader to serve as an interim PIC if the assigned PIC is unable to perform these duties.

Create a contact list that includes the names, titles, emails, and phone numbers for human resources, food vendors, and suppliers:

- Foodservice and clinical nutrition (inpatient and outpatient) employees
- All vendors, including suppliers providing water, refrigeration, ice, cleanup support, technology assistance including menu software, nutrition products, and central admixture pharmacy services (CAPS) (if parenteral nutrition is outsourced)
- Local and regulatory offices, including health department and emergency management offices
- City building department (if structural safety needs to be determined)
- Utility companies to verify the status of gas, electric, and telephone
- Insurance company (to begin claim process)

Foodservice Messaging

Develop an operations communication plan for messaging related to any public service meal disruptions such as the closure of retail establishments or specialty dining services.

Communicate announcements regarding any patient meal service disruptions or delays and if the meal service is switching to the use of a preselected, emergency, cold food or limited hot menu.

Clinical Nutrition Messaging

Develop a clinical nutrition communications plan for messaging any changes to the following:

- Patient screening and prioritizing
- Outpatient services
- Formulary options or delivery times for oral nutritional supplements (ONS), enteral nutrition, and parenteral nutrition.

Staffing

Create a minimum staffing plan for foodservice operations and inpatient and outpatient clinical nutrition. Consider plans that establish modified staffing such as four 10-hour shifts or three 12-hour shifts. Post a printout of the schedule in key service areas, even if technology is still available, to ensure maximum visibility.

Cross-train staff to ensure that tasks can be prioritized according to need when reassignments are necessary.

During an emergency situation

Identify any gaps in the ability to provide normal service levels. Communicate these gaps to upline leadership and affected areas as previously noted.

Provide resources to assist staff in coping mentally and physically as their work and personal lives are disrupted.

Supply Chain and Food Production

Develop a food production strategy for potential disruptions in the supply chain that could affect the following areas:

- Food delivery schedules and storeroom staffing
- Food and beverages, pack sizes, and ingredients that affect dietary compliance and allergens
- ONS, enteral, and parenteral products that may require changes to nutrition formulary

Assess available emergency food and water in terms of anticipated emergency duration and identify purchasing needs to close gap.

Ensure the amount of emergency food and water aligns with the number of days needed per regulatory requirements.

Develop a contingency plan for equipment alternatives if there are disruptions to utilities (gas, electricity, and water).

Patient Meal Service

For patient meal service: Ensure the following information is available in your emergency reference documentation.

- The minimum type of diets that will be required based on patient population (eg, regular, clear liquid, full liquid, facility-appropriate IDDSI levels, gluten-free, and lactose-free).
- A printed copy of the most frequently ordered diets for use by the call center or diet office if there is a disruption in access to the computerized system.
- Emergency hot and cold menus.
- A diet crosswalk for emergency menus based on the standard menus.
- A document that delineates allergens present in emergency menu options.
- Food, water, and supplies that have been purchased for the

emergency menus. Check expiration dates on products and rotate as needed.

- Tray tickets for the emergency menus.

Disruptions in Phone Service, Electronic Health Record (EHR), and Foodservice Software

Develop a protocol for operations if there are disruptions to phone, menu software, and/or EHR software.

Phone Service Disruption

A backup plan should be created for disruptions to phone service if it is the primary means for communicating diet orders, menu requests, and nutrition consults. If staffing allows, this backup plan may include rounding to each unit with a paper or spoken menu before mealtime. Staff should use an updated list that includes patient room number, name, birthdate, diet order, and food allergies. If staffing levels are insufficient, default to a preselect menu with preexisting tray tickets for emergency purposes.

Utilize online ordering for food, supplies, and communications with vendors.

Menu Software Disruption

Use EHR to create a patient list that includes room number, name, birthdate, diet order, and any food allergy. If staffing allows during a meal software disruption, consider rounding to each unit with a paper or spoken menu before mealtime. Staff should use an updated list that includes patient room number, name, birthdate, diet order, and food allergies. If staffing levels are insufficient, default to a preselect menu with preexisting tray tickets for emergency purposes.

Create a manual labeling system for ONS and enteral and parenteral nutrition. Store these in an emergency preparedness binder so they are readily available should menu software disruption occur.

EHR Disruption

If the EHR is disrupted, a downtime computer may be used to retrieve documentation with baseline data such as:

- Diet orders and allergies listed by patient name, room number, and birthdate.
- ONS orders listed by patient name, room number, birthdate, and dosage
- Enteral nutrition orders listed by patient name, room number, birthdate, formula type, rate, and volume.
- Parenteral nutrition orders listed by patient name, room number, birthdate, formula type, rate, and volume.

Ensure the foodservice and clinical nutrition staff have access to the downtime computer in advance. If a downtime computer does not exist, all four reports should be run at the end of each day in case downtime procedures are needed. Access the manual labeling system for ONS, enteral and parenteral nutrition if these are printed out of your EHR.

Develop and implement a system to obtain the following:

- Updated diet orders, allergies, transfers, admissions and discharges before each meal
- Enteral nutrition and parenteral nutrition updates daily or more frequently as needed
- ONS updates daily

EHR downtime has a significant impact on RDN workflow. Creating a standardized manual template for documenting clinical nutrition assessments, enteral and parenteral nutrition orders, and outpatient education is important. Standardized manual templates for specialty units such as pediatrics, NICU, burn or transplant may also be desirable. Documents can be provided to the clinical nutrition PIC to retain until scanning into the EHR can occur. Follow HIPAA guidelines for data collection and future scanning into the EHR.

Simultaneous Disruption in Menu Software and EHR

Access either the downtime computer for baseline diet and enteral and parenteral nutrition orders or the reports that are printed at the end of each day. Either resource can be used as your initial means to manually track patients, admissions, discharges, transfers, diet orders, diet changes, and food allergies.

Default to a preselect menu unless staffing allows for rounding with paper menus.

Planning an Emergency Menu

Planning disaster/emergency menus involves creating meal plans that are practical and nutritious when access to regular food sources may be disrupted. The type of emergency and availability of energy for heating equipment dictates whether a cold or hot menu will be provided.

General guidance for planning emergency menus:

- **Assess needs:** Consider dietary restrictions, preferences, and nutrition needs of patients and staff. Plan for a variety of meals to avoid monotony and ensure balanced nutrition.
- **Choose nonperishable foods:** Maintain an emergency stock of items that have a long shelf life (eg, canned goods, dried fruits, nuts, and grains), which is ideal for emergencies where refrigeration or frequent food purchases may not be possible.
- **Incorporate easy-to-prepare items:** Select foods that are easy to prepare with minimal resources, such as canned soups, ready-to-eat meals, and instant grains like rice or oatmeal.
- **Plan for cooking methods:** If cooking is necessary, ensure you have the means to do so safely. Consider items that can be prepared with minimal equipment or use alternative cooking methods if power is unavailable.
- **Store wisely:** Keep your emergency food supply easily accessible and rotate items regularly to ensure freshness and usability.
- **Include comfort foods:** Adding some common comfort foods may help to maintain morale during stressful times.

The menus in this section are intended as a tool to provide examples of emergency meal planning. Emergency menus should be individualized by facility.

Diet Manual > Emergency Preparedness > Planning an Emergency Menu > Emergency Cold Menus

Emergency Cold Menus

Notes:

- For a sodium-restricted diet: Omit ham and add one slice of cheese, replace mustard packet with lite mayonnaise packet, and replace salt packet with a salt substitute.
- Consider consolidating your IDDSI menus to Level 6 Soft and Bite-Sized and Level 4 Pureed. Medical staff approval for consolidating IDDSI menus for emergency purposes is necessary. Your facility's crosswalk should clarify which IDDSI levels are served to patients whose IDDSI level is not available during emergencies. Ensure that the texture-modified foods and liquids meet the patient's prescribed IDDSI level.

	Regular and Consistent Carbohydrate	IDDSI Level 6 Soft and Bite Sized	IDDSI Level 4 Pureed
Day 1: Breakfast	1 ea Hard-boiled egg 1 bowl Ready-to-eat whole grain cereal 1 ea Muffin 8 oz 2% Milk 4 oz Orange juice 1 ea Margarine 1 ea Salt 1 ea Pepper	1 ea Hard-boiled egg (chopped) 1/2 c Diced canned pears 1 c Yogurt 8 oz 2% milk 4 oz Orange juice 1 ea Salt 1 ea Pepper	1 ea Hard-boiled egg (pureed with 1/8 cup milk) 1/2 c Pureed pears 1 c Yogurt, smooth 8 oz 2% milk 4 oz Orange juice 1 ea Salt 1 ea Pepper
Day 1: Lunch	1 ea Chicken salad on wheat bread sandwich 1 ea Lettuce and tomato 1 ea Lite mayonnaise 1 ea Baked potato chips	1 ea Ground chicken salad on pureed bread 1 ea Lite mayonnaise 1/2 c Diced canned peaches 1/2 c Pudding	1 ea Pureed chicken salad on pureed bread 1/2 c Pureed peaches 1/2 c Pudding 8 oz 2% milk

	1/2 c Grapes 1/2 c Carrots and celery 1 ea Ranch dressing 8 oz 2% milk	8 oz 2% milk	
Day 1: Dinner	1 ea Turkey on wheat bread sandwich 1 ea Lettuce and tomato 1 ea Lite mayonnaise 1/2 c Fresh fruit cup 1 c Tossed salad 1 ea Italian dressing 1 ea Chocolate chip cookie 8 oz 2% milk	1 ea Ground turkey salad on pureed bread 1 ea Lite mayonnaise 1/2 c Low-sodium vegetable juice 1/2 c Applesauce 1/2 c Gelatin 8 oz 2% milk	1 ea Pureed turkey salad on pureed bread 1/2 c Low-sodium vegetable juice 1/2 c Applesauce 1/2 c Gelatin 8 oz 2% milk
Day 2: Breakfast	1 bowl Ready-to-eat whole grain cereal 1 ea Bagel 2 ea Peanut butter 8 oz 2% Milk 4 oz Orange juice 1 ea Margarine 2 ea Jelly	1/2 c Cottage cheese 1/2 c Diced canned pears 1 c Yogurt 8 oz 2% milk 4 oz Orange juice 1 ea Salt 1 ea Pepper	1/2 c Pureed cottage cheese 1/2 c Pureed peaches 1 c Yogurt 8 oz 2% milk 4 oz Orange juice 1 ea Salt 1 ea Pepper

Day 2: Lunch	1 ea Tuna salad on wheat bread sandwich 1 ea Lettuce and tomato 1 ea Lite mayonnaise 1 ea Baked potato chips 1 ea Apple 1/2 c Carrots and celery 1 ea Ranch dressing 8 oz 2% milk	1 ea Tuna salad on pureed bread 1 ea Lite mayonnaise 1/2 c Applesauce 1/2 c Gelatin 8 oz 2% milk	1 ea Pureed tuna salad on pureed bread 1/2 c Applesauce 1/2 c Gelatin 8 oz 2% milk
Day 2: Dinner	1 ea Roast beef on white bread 1 ea Lettuce and tomato 1 ea Lite mayonnaise 1/2 c Canned peaches 1 c Tossed salad 1 ea Italian dressing 1/2 c Pudding 8 oz 2% milk	1 ea Egg salad on pureed bread 1 ea Lite mayonnaise 1/2 c Low-sodium vegetable juice 1/2 c Diced canned peaches 1/2 c Pudding 8 oz 2% milk	1 ea Pureed egg salad on pureed bread 1/2 c Low-sodium vegetable juice 1/2 c Pureed peaches 1/2 c Pudding 8 oz 2% milk
Day 3: Breakfast	1 ea Hard-boiled egg 1 bowl Ready-to-eat whole grain cereal 1 ea Muffin 8 oz 2% milk 4 oz Orange juice	1 ea Hard-boiled egg (chopped) 1/2 c Applesauce 1 c Yogurt 8 oz 2% milk 4 oz Orange juice 1 ea Salt	1 ea Hard-boiled egg (pureed with 1/8 cup milk) 1/2 c Applesauce 1 c Yogurt 8 oz 2% milk 4 oz Orange juice

	1 ea Margarine 1 ea Salt 1 ea Pepper	1 ea Pepper	1 ea Salt 1 ea Pepper
Day 3: Lunch	1 ea Turkey on wheat bread sandwich 1 ea Lettuce and tomato 1 ea Lite mayonnaise 1 ea Baked potato chips 1 ea Orange 1/2 c Carrots and celery 1 ea Ranch dressing 8 oz 2% milk	1 ea Ground turkey salad on pureed bread 1 ea Lite mayonnaise 1/2 c Diced canned peaches 1/2 c Pudding 8 oz 2% milk	1 ea Pureed turkey salad on pureed bread 1/2 c Pureed peaches 1/2 c Pudding 8 oz 2% milk
Day 3: Dinner	1 ea Ham and cheese on wheat bread sandwich 1 ea Lettuce and tomato 1 ea Mustard 1/2 c Canned pears 1 c Tossed salad 1 ea Italian dressing 1 ea Chocolate chip cookie 8 oz 2% milk	1 ea Ground ham salad on pureed bread 1/2 c Low-sodium vegetable juice 1/2 c Diced canned pears 1/2 c Gelatin 8 oz 2% milk	1 ea Pureed ham salad on pureed bread 1/2 c Low-sodium vegetable juice 1/2 c Pureed pears 1/2 c Gelatin 8 oz 2% milk

Emergency Hot Menus

Notes:

- Provide condiments to complement meal offerings as available
- For sodium-restricted diets: Remove salt packet
- For consistent-carbohydrate diets: Limit foods with added sugars
- Consider streamlining your IDDSI menus to Level 6 Soft and Bite-Sized and Level 4 Pureed. Medical staff approval for consolidating IDDSI menus for emergency purposes is necessary. Your facility's crosswalk should clarify which IDDSI levels are served to patients whose IDDSI level is not available during emergencies. Ensure that the texture-modified foods and liquids meet the patient's prescribed IDDSI level.

	Regular and Consistent Carbohydrate	IDDSI Level 6 Soft and Bite Sized	IDDSI Level 4 Pureed
Day 1: Breakfast	1 bowl ready-to-eat cereal 1 slice Bread 1 Hard-boiled egg 8 oz 2% milk 4 oz Juice 8 oz Coffee/tea	8 oz Cooked cereal (no lumps) 1/2 c Starch (chopped) 1 ea Hard-boiled egg (chopped) 8 oz 2% milk 8 oz Coffee/tea 4 oz Juice	8 oz Pureed cooked cereal (no lumps) 1/2 c Starch (pureed) 1 ea Hard-boiled egg (pureed with 1/8 cup milk) 8 oz 2% milk 8 oz Coffee/tea 4 oz Juice
Day 1: Lunch	2-3 oz Chicken breast 1/2 c Rice 1/2 c Green beans ½ c Canned pears 1 ea Margarine 1 ea Salt 1 ea Pepper 8 oz 2% milk 8 oz Water	2-3 oz Chopped chicken breast, with 1 oz poultry gravy 1/2 c Starch (chop) 1/2 Chopped green beans 1/2 c Diced canned pears 1 ea Margarine 1 ea Salt	2-3 oz Pureed chicken breast with 1 oz poultry gravy 1/2 c Pureed starch 1/2 c Pureed green beans 1/2 c Pureed canned pears 8 oz 2% milk

		1 ea Pepper 8 oz 2% milk 8 oz Water	8 oz Water
Day 1: Dinner	2-3 oz Turkey 1/2 c Noodles 1/2 c Baby carrots ½ cup Canned peaches 1 ea Margarine 1 ea Salt 1 ea Pepper 8 oz 2% milk 8 oz Water	2-3 oz Chopped turkey with 1 oz poultry gravy 1/2 c Chopped buttered noodles ½ cup Chopped cooked carrots 1/2 c Chopped canned peaches 1 ea Margarine 1 ea Salt 1 ea Pepper 8 oz 2% milk 8 oz Water	2-3 oz Pureed turkey with 1 oz poultry gravy 1/2 c Pureed pasta 1/2 c Pureed cooked carrots 1/2 c Pureed peaches 8 oz 2% milk 8 oz Water
Day 2: Breakfast	3 oz Scrambled egg 8 oz Cooked cream of wheat 1 Pancake 1 oz Syrup 1 ea Margarine 1 ea Salt 1 ea Pepper 4 oz Orange juice 8 oz 2% milk 8 oz Coffee/tea	3 oz Scrambled egg 8 oz Cooked cream of wheat (no lumps) 1 Chopped pancake with syrup 1 ea Margarine 1 ea Salt 1 ea Pepper 4 oz Orange juice 8 oz 2% milk 8 oz Coffee/tea	3 oz Pureed scrambled egg 8 oz Pureed cooked cereal (no lumps) 3 oz Pureed pancake with syrup 4 oz Orange juice 8 oz 2% milk 8 oz Coffee/tea

Day 2: Lunch	1 ea Tuna salad on wheat bread sandwich 1 ea Lettuce and tomato 1 ea Lite mayonnaise 1 ea Baked potato chips 1 ea Apple 1/2 c Carrots and celery 1 ea Ranch dressing 8 oz 2% milk	1 ea Tuna salad on pureed bread 1 ea Lite mayonnaise 1/2 c Applesauce 1/2 c Gelatin 8 oz 2% milk	1 ea Pureed tuna salad on pureed bread 1/2 c Applesauce 1/2 c Gelatin 8 oz 2% milk
Day 2: Dinner	2-3 oz Chicken breast 1/2 c Mashed potatoes 1/2 c Green beans 1/2 c Canned pears 1 ea Margarine 1 ea Salt 1 ea Pepper 8 oz 2% milk 8 oz Water	2-3 oz Chopped Chicken with 1 oz poultry gravy 1/2 c Mashed potatoes 1/2 c Chopped green beans 1/2 c Diced canned pears 1 ea Margarine 1 ea Salt 1 ea Pepper 8 oz 2% milk 8 oz Water	2-3 oz Pureed chicken with 1 oz poultry gravy 1/2 c Mashed potatoes (no lumps) 1/2 c Pureed green beans 1/2 c Pureed pears 8 oz 2% milk 8 oz Water

Water Calculations for Emergencies

The amount of daily drinking water and all-purpose (non-potable) water needed at a hospital during an emergency for patients and staff is calculated based on the number of staff and the number of beds in the facility.

The following water calculations are based on state guidelines; Centers for Medicare & Medicaid Services federal emergency preparedness guidelines; and the emergency water supply guidance jointly published by the US Department of Health and Human Services, Centers for Disease Control and Prevention, and the Environmental Protection Agency.

In this example, a 4-day water supply is needed at a 100-bed hospital with a 3:1 ratio of staff to patients. Each person needs approximately 1 gallon of water per day.

Drinking water

1. To calculate the amount of drinking water needed for patients, multiply the number of beds x number of emergency days
 - o $100 \text{ beds} \times 4 \text{ days} = 400 \text{ gallons of water for patients}$
2. To calculate the amount of drinking water needed for staff, multiply the number of beds x staff ratio x number of emergency days
 - o $100 \text{ beds} \times 3 \text{ staff} \times 4 \text{ days} = 1200 \text{ gallons of water for staff}$
3. To determine the total drinking water needed, add the amount of drinking water needed for staff and patients
 - o $400 \text{ gallons} + 1200 \text{ gallons} = 1600 \text{ gallons of total drinking water}$

Bottled water

Use the total drinking water needed for patients and staff (1,600 gallons) to determine how many 20-oz units of bottled water are needed on-hand in case of an emergency where boiling water is not possible.

1. Convert gallons to ounces (1 gallon = 128 oz)
 - o $1,600 \text{ gallons} \times 128 \text{ oz} = 204,800 \text{ oz total drinking water}$
2. Divide total drinking water in ounces by 20 oz (per bottle)
 - o $204,800 \text{ total drinking water} / 20 \text{ oz} = 10,240 \text{ bottles of drinking water}$

All Purpose Water: Non-Potable Water

The calculation to determine non-potable water needs is based on the duration of the emergency and the number of patient beds and staff. Each person needs approximately 5 gallons of non-potable water per day.

1. Multiply the total number of people x number of emergency days x 5 gallons of non-potable water/per person/day.
 - a. $400 \text{ people (staff and patients)} \times 4 \text{ days} \times 5 \text{ gallons of non-potable}$

water = 8,000 gallons (or 1,024,000 oz) non-potable water.

References

Centers for Medicare & Medicaid Services, Center for Clinical Standards and Quality/Quality, Safety & Oversight Group. Updated Guidance for Emergency Preparedness-Appendix Z of the State Operations Manual (SOM). Updated March 2021. Accessed April 1, 2025. Available at: Website

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Planning a House Diet

The goal of planning the House diet is to meet multiple patient nutrition requirements and expectations as well as to provide robust, appealing choices. The House diet in the NCM Diet Manual is based on the General, Healthful Nutrition Therapy client education handout and General and Lifecycle section of the Nutrition Care Manual. Facilities use many names for this diet including, house diet, or general, healthy diet. When completing the NCM Diet Manual Crosswalk Worksheet, identify the terminology your facility uses for the House Diet which will connect your facility's terminology to the NCM diet definition. Diet orders include two components: The therapeutic component (house) and the texture component.

The goal for facilities should be to have a majority of residents receive the House Diet order. The House Diet order allows for freedom of meal choice, especially at the time of service. This diet order, when written effectively, can serve the needs of most residents, including those with chronic illnesses. Develop House menus to provide an eating pattern consistent in carbohydrates with recommended healthy levels of energy, protein, and fat. Including several alternate menu items and culturally sensitive choices will provide options. The need to order the strict, consistent carbohydrate diet order for most patients with diabetes is thus eliminated. If the House diet is also planned to meet heart-healthy eating principles, the need to order a strict “cardiac”-type diet is also eliminated.

The House Diet can lead to simple adaptations to meet the needs of residents who require altered textures. See the Texture Modification section to review foods allowed on different mechanically altered diets and consider texture modifications while planning the overall menu.

Designing a varied House menu that meets the nutrition needs of most patients provides the ease to liberalize diets and decrease the number of therapeutic diets required. Potential positive outcomes include:

- Improving patient satisfaction via additional menu choices
- Creating a more relaxed dining atmosphere
- Decreasing clerical costs associated with managing multiple therapeutic diets
- Enhancing positive patient interactions (without concern for therapeutic diet restrictions)
- Allowing focus on serving meals that accurately correspond to individual residents' texture diets
- Decreasing labor costs and food costs

- Allows for menu choices at point of service
- Adds control of anticipated inventory
- Adhering to federal and state regulations by:
 - Improving dining services and atmosphere
 - Increasing patient choice options
 - Improving intake, potentially reducing patient weight loss

Developing a House Diet

1. Review all Federal and State Regulations. Refer to the Regulations section for additional resources.

- Review pertinent CMS documents
- Review state regulations
 - State regulations often include more specific details on menu planning
 - Review guidelines specific to the patient population such as acute care, assisted living, hospice, or corrections

2. Determine Nutrition Content

- Determine kilocalorie range for facility diet. Many facilities set their range at 2000 kcal to 2400 kcal per day.
 - Research the average, range, and most common weight of your residents. These data can give you a general idea of your resident population's energy needs

Example:

If the average weight of your residents is 70 kg requiring 1750 kcal to 2100 kcal per day (25 kcal/kg to 30 kcal/kg), you may want to set a menu goal for providing 2000 kcal/day to 2400 kcal/day. Keep in mind that residents may not eat 100% of the planned menu or may be choosing alternatives, therefore increasing the importance of healthy eating plan.

- Determine protein range
 - Setting range for protein: 10%-20% of total energy from protein
- Determine carbohydrate range
 - Setting range for carbohydrates: carbohydrates should provide 45%-65% of total energy and be evenly distributed throughout the day
 - Keep the carbohydrate consistent and balanced throughout the day to meet many patients' nutrition needs and medical nutrition therapy.
- Determine fat range
 - Setting range for fat: 20%-35% of total energy from fat

Menu Planning Example

Determining Menu Components for Energy, Protein, Carbohydrates, and Fat:

Energy: You have determined the average and most common weight for patients at your facility is 70 kg; daily energy needs for people at this weight are estimated at 1750 kcal to 2100 kcal (25-30 kcal/kg). You recognize that many patients will not eat all the foods on the planned menu; may choose the alternative; or may have additional needs related to healing, weight gain, and other health concerns. A reasonable energy goal per day for menus would be 2100 kcal.

Protein: Using 2100 kcal as your base, you have determined that 10%-20% estimated energy from protein would range from 210 kcal to 420 kcal. This calculates as 53 g to 105 g protein per day.

Fat: Using 2100 kcal as your base, the upper end of 35% of energy from fat is 735 kcal from fat, or 82 g fat per day.

Carbohydrates: Using 2100 kcal as your base, 45% to 65% of energy from carbohydrate would amount to 945 kcal to 1365 kcal from carbohydrate, or 236 g to 341 g carbohydrate per day.

Summary:

- 2100 kcal per day
- 30 g protein per meal, 3 meals per day=90 g protein=360 kcal from protein
- 25 g fat per meal, 3 meals per day=75 g fat=675 kcal from fat
- 90 g carbohydrate, 3 meals per day=270 g carbohydrate=1080 kcal from carbohydrate

Total= 2115 kcal*:

- **17% protein**
- **32% fat**
- **51% carbohydrate**

*Approximately 300 kcal remaining for snacks and beverages

3. Set goal for sodium intake

- The 2020-2025 US Dietary Guidelines encourages people to decrease their sodium intake (USDA/USDHHS, 2020).
 - This may not always be possible for some patients.
 - Facilities may choose to plan a house diet that includes a sodium intake greater than 2300 mg/day. In this instance, low-sodium choices should be available for patients who request or need these dietary modifications.
- Malnutrition may be a patient's primary concern, and restricting sodium may not be indicated.
- The House Diet must be planned with multiple menu options to allow residents to choose lower-sodium items. Refer to Sodium section and Dietary Guidelines for Americans for additional resources.

4. Determine Food Preferences of Patients

The goal is always to write a menu that your patients will enjoy eating and enhance their dining experience. The best written, most nutritionally sound, cost-effective menus will be of no use if they do not include items your patients will eat. Planning, researching, and anticipating patient food requests will help ensure the House Diet is well accepted.

Common Preferences

Many patients may prefer a lower-carbohydrate eating pattern. Those patients may still have a House Diet order but the registered dietitian/nutritionist can individualize those patients' meal plans to modify the carbohydrate content. See Consistent Carbohydrate Diet and Sodium-Restricted Diet for examples of how to modify the house menu to address therapeutic diet needs.

Many patients may follow a vegetarian eating pattern, possibly requiring some individualization (Melina, 2016). Use the Nutrition Care Manual Vegetarian Nutrition section to individualize meal plan preferences and to assist with planning vegetarian options as part of the House menu.

Regional Food Preferences

It is important to familiarize yourself with the types of food, style of cooking, and flavoring most desired in the geographical area of your facility. Some examples of regional foods in the United States include:

- Biscuits/gravy or muffins/butter
- Grits or other hot cereal such as oatmeal, cream of wheat, and Maypo

- Style of eggs and condiments served (salsa, hot sauce)
- Style and frequency of serving potatoes
- Chowders, cream soups, cold soups and broth soups
- Use of different types of bread, rolls and wraps for sandwiches
- Favorite local entrees such as “French pork pie,” bratwurst, or “lobster roll”

Examples of Regional Food Preference Questions to Consider:

- How often does the population in a given region expect certain foods (sausage, bacon, ice cream)?
- Are potatoes served with gravy?
- What types of desserts are preferred: pie, custards, pudding, cookies, cakes or cupcakes; fruit cobblers?

Demographic preferences

Examples of Demographic Questions to Consider:

- In what age groups are patients at your facility?
- Do you serve more baby boomers (born 1946-1964) or traditionalists (born 1925-1945)?
- Are your patients accustomed to using the terms “dinner” or “supper”?
- Which meal do they prefer as the main (largest) meal?
- Are your patients largely “meat and potatoes” demographic?
- Are there many patients who would prefer a menu with more salads and wraps?

Religious Holidays and Special Occasions

Patients often look forward to special meals to celebrate a variety of important events. Consider the individual needs of the facility/center as well as the variety of patient expectations. Through working with other department managers and patients, identify weekly or monthly events that affect the menu. By building these patterns into your menu, there will be fewer changes needed for the routine House menu.

Suggested Special Meals and Events Menu Offerings:

- Ice cream floats every Thursday (do not serve ice cream as dessert on Thursdays; serve “light” dessert)
- Fish option every Friday at both lunch and supper (fulfills some religious needs)
- Pasta/Italian meal every Wednesday
- Pizza dinner (evening meal) every other Monday
- Monthly barbeque on Tuesdays
- Weekly “Sundae on Sunday” activity every Sunday at 2pm

Plan one alternative vegetarian meal to anticipate meeting patient religious, ethnic, and cultural needs/requests.

Review religious holidays and decide in advance what menu options to make available. If possible, create special celebrations for some holiday mealtimes.

Writing the House Diet

Steps for writing a menu

1. Determine which meal planning guide will be used at your facility. It may be necessary to create a menu checklist to ensure all components are met. Adjust as needed to meet state regulations and patient requests.

See Menu Planning Checklist for Nutritionally Balanced Meals for a template

2. Determine menu cycle length. A cycle menu between 2 to 5 weeks is most common.
 - An additional consideration is how many menu cycles throughout year will change to accommodate seasonal items

3. Separate basic entrée types throughout week and from week to week (eg, beef, pork, chicken, fish, pasta, vegetarian)

4. Complement entrees with patient's preferred sides, incorporating fruits and vegetables and whole grains. Establish a consistent system for ensuring variety—including foods, shapes, and textures—across days/weeks in meal plans.

5. Consider the carbohydrate content of desserts to keep carbohydrate amounts consistent throughout day. One suggestion is to serve a higher carbohydrate and energy-dense dessert at one meal (for example, at lunch) and a fruit-based dessert at the other meal where dessert is served.

6. Provide menu alternates to give patients the most choice possible.

- Offer alternatives that are lower in fat, lower in sodium, and have more subtle flavors
- Consider texture-friendly alternates (such as mashed potatoes)
- Create an “always offered” quick order menu so residents can always select a meal that meets their preferences
- Offer vegetarian alternatives at all meals.
- Examples of entrée options:

Chicken breast	Chicken salad	Salmon filet
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Turkey breast	Hamburger	Tuna salad
Chef salad	Baked cod	Grilled cheese
Pasta	Lentils/rice	Beans/rice

- Examples of options for sides:

Green bean casserole	Green beans
Fruit salad	Fresh sliced apples
Potato salad	Baked potato
Mixed vegetables	Steamed carrots

7. Determine recipes for each menu item, including texture-modified and therapeutic recipes.

- Consider ingredient availability, quantity, form, and necessity
- Provide precise ingredient measurements and preparation directions and include hazardous analysis and critical control points (HACCP) for holding and storage.
- Include serving sizes
- Review preparation methods to ensure that not all items need to be prepared in the oven, stove top, or steamer
- Ensure that cooking staff skill level matches preparation requirements
- Consider visual appeal of items on the plate or tray
- Assess inventory to ensure you have the appropriate dishware, including casserole dishes, and other necessary supplies for service

8. Analyze cost of menu execution, factoring in both food and labor costs.

9. Perform broad review of menu, including analysis of the factors below to ensure these requirements are addressed on the menu for each day of the week. Mark off each item on the checklist once it has been reviewed/confirmed.

Additional Checks and Balances:	Meal	Sun	Mon	Tue	Wed	Thur	Fri	Sat
Is there at least one serving of a fruit and a vegetable in the meal?	B							
	L							
	D							
Is there a variety of color in the meal?	B							
	L							
	D							
Is there a variety in texture and shapes in the meal?	B							
	L							
	D							
Is there a variety in flavor in the meal?	B							
	L							
	D							
Do flavors of the meal complement each other?	B							
	L							
	D							
Is there a variety in preparation process for the meal?	B							
	L							
	D							
Will foods maintain appropriate temperature during service?	B							
	L							
	D							
Can the day's menu be prepared within	B							
	L							

vvvvvvvv budget?	D						
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10. Review house diet draft and make adjustments for carbohydrates and sodium content so the house diet offerings can also be used for consistent carbohydrate and sodium-restricted diets.

- Can be done for entire facility menu or on an individual basis
- Consider items patients frequently request for daily menus:
 - Cereal
 - Yogurt
 - Soup
 - Ice cream
 - Fruit or fresh banana
 - Juice

Benchmarking and Usage Reports

Accurately monitoring portion sizes served and plate waste may also provide insights on acceptance of what you are serving and whether portion sizes served are as specified in the daily menu. There are several objective tools for gathering and assessing data on resident menu preferences:

• Usage data

The goal of usage data is to track which items are ordered the most. The registered dietitian nutritionist (RDN) can record at each meal how many residents choose the main entrée, how many choose each alternative, and so on.

Example:

Entree	Number of residents who ordered
Main: Beef Stroganoff	85
Alternative 1: Baked chicken	20
Alternative 2: Salmon	15
Alternative 3: Turkey sandwich	5

• Meal Satisfaction Surveys

The goal of meal satisfaction surveys is to receive written feedback from residents/patients on quality, variety, and temperature of foods as well as the general service.

• Focus Groups:

Conduct focus groups to expand your understanding of trends identified by the written survey data. Good focus groups have a defined purpose, ask meaningful questions, use well-developed scripts, and are overseen by an experienced facilitator.

• Resident Council Meetings:

These meetings, often conducted in extended residential settings, offer an opportunity for participating residents to provide broad-based information on preferred foods, snacks, and dining choices on behalf of all residents.

- The RDN can assist in the development of a resident food committee that is independent of the resident council.
- Residents can also vote on the time they would like the meals served, serving style choices, dining options, and what snacks they want to be offered.
- These meetings provide an opportunity to remind residents that they always have other choices if they do not like the food that is served and that they always have access to snacks.
- Refer to CMS QIS tools to monitor effectiveness of these meetings:
 - Resident Council Interview
 - Dining Observation Survey

● **Trial New Menu Items**

Pilot new menu ideas by having cooks prepare sample items for participating residents and staff to taste test at resident council meetings or by inviting vendors to offer residents a taste of new products. Include description of new menu items when residents are taste testing.

● **Determine Food Preparation Methods and Plate Presentation**

Research shows that adding flavor and visual appeal to foods makes the dining experience more enjoyable and reduces the need for added sodium and fats in the preparation process (Huisman, 2016; Michel, 2015). See Sodium-Free Flavoring Tips.

- Choose spices and fresh herbs that enhance the natural flavor of the food to reduce sodium and limit kilocalories from non–nutrient-dense foods to meet the recommended dietary guidelines.
- Flavor is a combination of how taste interacts with the other senses. Flavor is also affected by how food is held before service. For example, warming or chilling a food affects its overall taste, texture, and appearance.
- Consider garnishing and other plating presentation strategies while planning menus. Serving some items in a small dish on the plate or serving stews in a casserole dish can improve visual appeal.

References: Planning a House Diet

Huisman G, Bruijnes M, Heylen DJK. A moving feast: Effects of color, shape, and animation on taste associations and taste perceptions. Proceedings of the 13th International Conference on Advances in Computer Entertainment Technology. Osaka, Japan. November 9-12, 2016. Available at: Website

Melina V, Craig W, Levin S. Position of the Academy of Nutrition and Dietetics: Vegetarian diets. J Am Nutr Diet. 2016;116(12):1970-1980. Abstract.

Michel C, Velasco C, Fraemohs P, Spence C. Studying the impact of plating on ratings of the food served in a naturalistic dining context. Appetite. 2015;90:45-50. Abstract.

US Department of Agriculture and US Department of Health and Human Services. Dietary Guidelines for Americans, 2020-2025. 9th ed. Updated December 2020. Accessed on October 25, 2022. Available at:Website

House Diet

Definition: The House diet is the most liberal diet and is not considered a therapeutic or texture modified diet. It provides freedom of food choices for those that have no therapeutic restrictions. Occasionally a regular diet terminology is used to describe a House diet, however a regular diet refers to the texture part of the diet order. See IDDSI Levels for texture modified diet options.

The House diet is based on general, healthful nutrition from a variety of foods. It is planned using the Dietary Guidelines for Americans and serves as the basis for all other diets (USDA, 2020). The House diet can be individualized to accommodate patient food preferences and needs. See Planning the House Diet for more information.

Indication: The House diet is indicated for individuals without nutrition-related problems or those with nutrition-related problems who need a liberalized diet to allow more menu choices to improve intake.

Nutritional Adequacy: The House diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Use the blue "Related Client Ed" link to the right of the title of this page to access the General, Healthful Nutrition Therapy to view the Sample 1-Day Menu with nutrient analysis and the Foods List.

References

US Department of Agriculture, US Department of Health and Human Services. Dietary Guidelines Advisory Committee. Dietary Guidelines for Americans, 2020-2025. Updated 2020. Accessed March 24, 2022. Available at: Website

Fortified Foods

Definition: Fortified foods have protein, carbohydrates, and/or fats added to increase the total nutritional value of the food. Often, the serving size and volume of food is the same as standard menu items, but the fortified food provides more calories and/or protein.

Indication: Fortified foods are indicated for individuals who have inadequate intake, who are at risk for malnutrition, or who have increased energy and/or protein needs. Energy-dense and protein-dense foods can increase patients' nutrient intake and weight (Sossen, 2021; Mills, 2018). Fortified foods are especially effective when an individual does not tolerate or accept oral nutritional supplements (ONS) and for patients with impaired cognitive function (Mills, 2018).

Individualize the patient's meal plan to meet their nutrition needs by incorporating snacks, ONS, and fortified foods (Dorner, 2018). Add different flavors to increase the variety and to minimize palate fatigue.

Tips for Managing a Successful Fortified Foods Program

- Review facility policies and procedures regarding fortified foods. Some policies offer a specific fortified food meal plan and some require individualization by ordering "fortified foods" as needed.
- Individualize the fortified foods meal plan based on patient preferences and subsequent acceptance of items provided.
- Liberalize the diet as much as possible to allow for more menu choices and to increase the fortified food options.
- Maintain a list of regular menu items (eg pudding, ice cream, yogurt, or custard) and include nutritional content. Although these items may not be fortified with nutrients, they can be included in a fortified foods diet because they still provide calories, and the patient may accept these items more readily.
- Keep an updated list of purchased fortified foods and include their nutritional content.
- Develop fortified food recipes for well accepted items (eg hot cereal, mashed potatoes, pudding, custard, ice cream, and soups) with noted nutritional value.
- Offer the fortified foods either with or between meals. ONS are another way to add concentrated nutrition or to increase a snack's protein/calorie content.
- Evaluate consumption and acceptance of fortified foods by observing meal and snack time service.
- Enhance meals to add flavor and variety to increase intake at mealtime:
 - Add butter, cream, sour cream, nut butters, or avocado
 - Serve extra gravy, sauces, mayonnaise, or ketchup
 - Provide individual small cups of condiments and sauces for easy-access and dipping
 - Switch to full-fat dairy products if applicable and accepted
 - Increase sweetness and flavor if needed by adding sugar to hot

- beverages, syrup or honey to hot cereal, or toppings to desserts.
- Identify food items and methods to add protein to a meal plan. For example:
 - Add Greek-style yogurt or offer extra eggs for breakfast
 - Serve a side of cottage cheese/fruit, extra serving of sandwich filling, or a large portion of the protein entrée
 - Replace whole or low-fat milk with skim milk to provide more protein without extra calories. Increase skim milk servings throughout the day and include with snacks.
 - Offer peanut butter, yogurt, cheese or higher-protein items as a snack
 - Obtain order to add a protein supplement to favorite beverages or food items

Commercial Supplements

ONS are frequently used to increase energy and protein intake for those at nutritional risk. Some patients may prefer commercially available ONS because of their convenience. Commercial ONS may also be used as ingredients in homemade shakes. Various types of commercial ONS are available, including liquids, powders, disease specific formulations, and nutrient-dense products.

Supplement Choices for Patients Receiving Thickened Liquids

Patients receiving thickened liquids may require modifications to homemade supplement recipes. Identify homemade liquid supplement choices available at your facility and use the International Dysphagia Diet Standardization Initiative (IDDSI) testing methods to ensure safety for patients on thickened liquids. Refer to the IDDSI Levels section for guidance.

A commercial supplement's label shows the product's IDDSI level and often will state it is intended to be served chilled. The serving temperature affects the liquid thickness. Follow the manufacturer's directions and use the IDDSI testing methods to assure safety for those on thickened liquids. Contact the supplement manufacturer with any questions regarding product thickness. Consult with the speech language pathologist when introducing new commercial products.

Use the blue "Related Client Ed" link to the right of the title of this page to access the High-Calorie, High-Protein Nutrition Therapy handout. The handout provides additional tips for adding calories and protein to meals and snacks, and a sample 1-Day Menu with nutrient analysis. If your facility has a high-calorie, high-protein diet, the menu can be customized to calculate the extra calories and protein to serve.

References

Dorner B, Friedrich EK. Position of the Academy of Nutrition and Dietetics: Individualized nutrition approaches for older adults: Long-term care, post-acute care, and other settings. J Acad Nutri Diet. 2018;118(4):724-735. Full text.

Mills SR, Wilcox CR, Ibrahim K, Roberts HC. Can fortified foods and snacks increase the energy and protein intake of hospitalised older patients? A

systematic review. J Hum Nutr Diet. 2018;31(3):379-389. Full text.

Sossen L, Bonham M, Porter J. Can fortified, nutrient-dense and enriched foods and drink-based nutrition interventions increase energy and protein intake in residential aged care residents? A systematic review with meta-analyses. Int J Nurs Stud. 2021;124:104088. Abstract.

Vegetarian Diets

Definition: Vegetarian diets encompass a variety of plant-derived foods and exclude some or all foods derived from animals. More specifically, vegetarian diets exclude all flesh foods (such as meat, poultry, wild game, seafood, and their products). Within the broad category of vegetarian diets, there are several common subtypes (Raj, 2025):

- Vegan diets, which exclude eggs and dairy products
- Lacto-vegetarian diets, which include dairy products but not egg products
- Lacto-ovo-vegetarian diets, which include eggs and dairy products
- Ovo-vegetarian diets, which include egg products but not dairy products

Indication: Vegetarian diets are indicated for any individual upon request.

Nutritional Adequacy: Comprehensively planned vegetarian diets meet the Dietary Reference Intakes (DRI) for most macronutrients and micronutrients. Nutrients of concern may include calcium, choline, protein, iodine, vitamin D, and vitamin B12 (Raj, 2025).

Use the blue "Related Client Ed" link to the right of the title of this page to access the General, Healthful Vegetarian Nutrition Therapy to view the Sample 1-Day menus with nutrient analysis and the Foods List.

References

Raj S, Guest N, Landry M, et al. Vegetarian Dietary Patterns for Adults: A Position Paper of the Academy of Nutrition and Dietetics. J Acad Nutr Diet. 2025;125 (6):831-846. Full text.

Kosher Diet

Definition: Jewish people who observe the rules of kosher follow specific dietary guidelines that fit into their religious beliefs. Kosher foods meet the regulations that are laid out by Jewish dietary law. These laws define appropriate techniques for animal slaughter, food preparation, and meal planning.

- **Meat and Poultry:** In the rules of kosher related to meat, the animal must have cloven hooves and chew its cud. Any animal (meat or poultry) must be slaughtered in a humane way based on Jewish dietary law. Any animal that died of natural causes, was killed by other animals, or has disease or flaws (like a missing limb) are not allowed for consumption. Even if an animal is allowed to be eaten, not all parts are allowed. Certain parts of animals that are close to the sciatic nerve and may contain blood (such as cuts for filet mignon) are not allowable. Animals must be slaughtered by a shokhet, a specially trained and "pious" man who is certified in the rules of Jewish dietary law. If an animal's meat is not allowed, neither are its organs, eggs, or milk. Rodents, reptiles, amphibians, and insects are all forbidden (Jewish Virtual Library, 2022, Orthodox Union, 2022).
- **Fish:** Acceptable fish must have fins and scales. Fish cannot be served on the same plate or during the same course as meat. It can, however, be served at the same meal.
- **Fruits and Vegetables:** All must be inspected for insects prior to consumption (insects are forbidden)
 - **Grapes:** Any food derived from grapes (wine, grape jelly, grape juice, candies) must be supervised from start to finish. Only these grape products can be certified as kosher.
- **Cheese:** Most cheese manufacturers use rennet, an enzyme used to harden cheese that is typically obtained from non-kosher animals. Cheeses must use kosher forms of rennet in order to be certified as kosher.
- **Separating Dairy and Meat:** Foods are divided into three categories, which in Yiddish are: fleishik (meat), milchik (dairy), and pareve (neutral). Pareve foods can be eaten with fleishik foods or milchik foods. However, milchik foods can never be combined with fleishik foods. Furthermore, a person must wait a certain amount of time between eating milchik and fleishik foods. The time between eating fleishik to eating milchik varies between 3 to 6 hours (due to differing beliefs and practices). The time to wait between eating milchik and then eating fleishik is typically shorter; sometimes one only needs to rinse out their mouth. Again, the exact wait time depends on beliefs and opinions.
- **Kitchen Equipment:** Utensils and cookware must also be separated as

“for fleishik” and “for milchik.” The pots and pans used to prepare the food, plates and flatware used to eat the food, and dishwashers, sinks, and sponges used to clean the food also all need to be designated for either dairy or meats. Kosher households will have two sets of dishes, flatware, utensils, pots, pans, and even dishwashers and sinks. Oftentimes, different color plates are used to indicate if it belongs to the milk or meat category (typically red for meat and blue for milk). If equipment designated for dairy comes in contact with meat, it is not considered kosher any longer. The equipment would then need to be replaced or be made kosher again by following specific practices set by kosher dietary law (Chabad, 2022, Jewish Virtual Library, 2022)

Indication: The kosher diet is indicated for any individual upon request.

Nutritional Adequacy: The kosher diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Foods to Include and Foods to Omit

Food Group	Foods to Include (Kosher)	Foods to Omit (Non-Kosher)
Grains	Grains and grain products, such as bread, rolls, prepared breakfast cereals, crackers, rice, oats, quinoa and pasta	Grain products made with non-kosher ingredients or prohibited foods.
Protein Foods (fleishik)	Kosher red meat and poultry Kosher fish (with scales and a fin) Eggs and egg substitutes Kosher beans, lentils, and peas Kosher nuts and nut butter, seeds and seed butter Kosher tofu, tempeh, and soy nuts Kosher plant protein-based meat alternatives	Non-kosher meats or parts, or prohibited cuts of meat such as filet mignon. All pork or foods containing pork ingredients. Seafood or fish without scales and a fin

Dairy (milchik)	Milk, yogurt, and cottage cheese Kosher cheeses made from kosher rennet Fortified soymilk and soy yogurt	Dairy products when meat is served Cheese made from non-kosher rennet
Vegetables	All fresh, frozen, and canned insect-free vegetables; vegetable juices	Any vegetable with insects
Fruits	All fresh, frozen, and canned insect-free fruits; fruit juices Grape products that have been certified as kosher (wine, grape jelly, grape juice, candies)	Grape products that have not been certified as kosher
Oils	Vegetable oils, including olive, peanut, and canola oils Vegetable oil-based margarines and spreads Butter (substitute with vegetable oil-based margarine in meals where meat is served)	Lard
Beverages	Coffee, tea, and water	Non-kosher certified grape juice
Other	Kosher prepared foods, including soups, casseroles, salads, baked goods, and snacks made from recommended ingredients	Non-kosher certified foods or foods made from prohibited ingredients.

View the Kosher Sample 1-Day Menu with nutrient analysis. View the Jewish Cultural Food Practices in the Resources section for additional information.

References

Chabad.org. What is a kosher kitchen? Accessed May 23, 2022. Available at: Website

Jewish Virtual Library. Jewish dietary laws (Kashrut): Overview of laws and regulations. Accessed March, 28, 2022. Available at: Website

Orthodox Union. Kosher food: The kosher primer. Accessed May 23, 2022. Available at: Website

Halal Diet

Definition: Muslims follow specific dietary guidelines of Islam and may vary with the degree to which these guidelines are observed. Halal foods meet the regulations that are laid out by Islamic dietary law. These laws define permitted (halal) and prohibited (haram) foods as well as food preparation practices. If it cannot be determined if a food is halal or haram, that food, called mashbooh, is forbidden. Slaughtering must be completed by a Muslim familiar with Islamic slaughtering practices, in accordance with Islamic law. Notable foods common in North America that are not permitted include (IFANCA, [2022](#)).

- Animal foods: pigs, boars, frogs, poisonous aquatic animals, blood
- Plant foods: hazardous plants except those where toxin can be destroyed during processing
- Beverages: alcohol
- Food Additives: derived from prohibited animal foods, plant foods, or alcohol

Indication: The halal diet is indicated for any individual upon request.

Nutritional Adequacy: The halal diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Foods to Include and Foods to Omit

Food Group	Foods to Include (Halal)	Foods to Omit (Haram)
Grains	Grains and grain products, such as bread, rolls, prepared breakfast cereals, crackers, rice, oats, quinoa and pasta	Grains made with mashbooh ingredients such as: Bread with lecithin or monoglycerides/diglycerides Bagels with cysteine hydrochloride, enzymes, folic acid, or niacin Cake with artificial/natural flavors or monoglycerides/diglycerides Cereal with artificial/natural flavors, vitamins A, B12, C, or D

		Cookies with folic acid or thiamine Granola bars with added flavorings Donuts or pastries with monoglycerides/diglycerides, added flavoring
Protein Foods	Halal-certified (zhabiha) red meat and poultry Halal fish and seafood Eggs and egg substitutes Beans, lentils, and peas Nuts and nut butter, seeds and seed butter Tofu, tempeh, and soynuts Plant protein-based meat alternatives	Pork, bacon, lard, hydrolyzed porcine collagen, gelatin, animal shortening, hydrolyzed animal protein (from pig), rennet Non-halal certified* (zhabiha) red meat, poultry, fish, and seafood Tallow
Dairy	Halal milk, yogurt Halal cheeses made without rennet enzymes Fortified soymilk and soy yogurt	Cheeses or cheese products made with rennet derived from pig sources Dairy foods made from mashbooh ingredients such as: Coffee creamer with natural/artificial flavors or monoglycerides/diglycerides Cheese with enzymes

		Ice cream with whey, artificial flavor, or monoglycerides/diglycerides Pudding with gelatin or artificial/natural flavors Yogurt with flavors, gelatin, or wheym
Vegetables	All fresh, frozen, and canned vegetables; fruit juices	Vegetables prepared with haram ingredients
Fruits	All fresh, frozen, dried and canned fruits; fruit juices	Fruits prepared with haram ingredients Fruits or fruit desserts prepared with mashbooh ingredients: gelatin of pig origin
Oils	Vegetable oils, including olive, peanut, and canola oils Vegetable oil-based margarines and spreads	Fats and oils with mashbooh ingredients: fatty acids or fatty acid esters, hydrolyzed bovine collagen, gelatin of pig origin or artificial/natural flavors
Beverages	Coffee, tea, water	Alcohol including beer, wine, liqueur Foods made with alcohol as an ingredient such as wine sauces, flavor extracts (vanilla, etc), or ethyl alcohol
	Gelatin from fish origin	Artificial coloring/flavoring Enzymes Extracts (such as vanilla) Fatty acids or fatty acid

Other	esters Gelatin (of pork origin), rennet (of pork origin) Glycerin, Glycerides (monoglycerides and diglycerides) Gum base Hydrolyzed bovine collagen Phospholipids Polysorbates Sodium lauryl sulfate Stearates (glycerol, magnesium, potassium, and sodium; stearic acid, sorbitan monostearate, propylene glycol monostearate) Tallow
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*Level of strictness to halal-certification adherence varies among Muslims

View the Halal Sample 1-Day Menu with nutrient analysis. View the Halal Cultural Food Practices in the Resources section for additional information.

References

ifanca.org. A quick halal guide for nutrition and foodservice professionals. Accessed May 23, 2022. Available at: Website

Finger Foods

Definition: Finger foods can be easily picked up with the hands without falling apart. A finger food meal plan may specify eating strategies that promote independence with eating. For example, serving soup in a mug eliminates the need for a spoon. A finger food meal plan may include the use of adaptive equipment to facilitate or develop self-eating skills .

Indication: A finger foods meal plan may be indicated for individuals who have limited use of hands and arms due to conditions such as arthritis, stroke, acute injury, or dementia. Using fingers to pick up foods enables self-feeding, independence, and diet adequacy. Utensils may become harder to use depending on the individual's physical condition and cognition function. Most foods can be eaten without utensils but doing so requires creativity and planning by foodservice personnel.

Nutritional Adequacy: A finger foods meal plan meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Tips for Adjusting a Meal Plan to Promote Self-Eating Skills:

- Work with occupational therapy staff to identify adaptive feeding devices including cups, utensils, specialized plates, and bowls that would be appropriate in your facility.
- Use covers, straws, cups, or other adaptive devices to decrease spillage.
- Provide cereals with milk, pureed foods and soups in a mug
- Serve condiments, gravies, and sauces (for example, hummus, guacamole, salad dressings) in a small cup on the plate for dipping to eliminate the need to open packages. Offer extra dipping sauces based on patient preference to encourage greater intake.
- Serve each part of the meal in individual bowls or mugs, making it easier for patients to hold the food closer to their mouth and require less distance for the utensil to travel
 - Example: Provide individual mugs containing macaroni and cheese, cooked vegetable, soup, and dessert at the same meal
- Use paper products (as they are much lighter in weight) to serve meals if it safely improves eating skills
- When possible, cut foods into quarters or bite-sized pieces.

Examples of Finger Foods

Menu items that are easy to pick up and may accommodate patient preferences include the following:

- Hard-boiled egg, quiche, or egg sandwich
- Plain cold cereal, milk on the side
- Peanut butter or cheese crackers
- Cheese sticks
- Dinner rolls
- Raw vegetables and fruit, cut in bite-sized pieces
- Cooked vegetables (eg broccoli, green beans, or carrots) cut up into

bit-sized pieces

- Canned fruit with the liquid drained
- Potato wedges, tots/puffs
- Ziti and small pastas with sauce to dip
- Chicken nuggets, fish sticks, burgers, veggie burgers, pizza, quesadillas, tofu chunks, or sandwiches

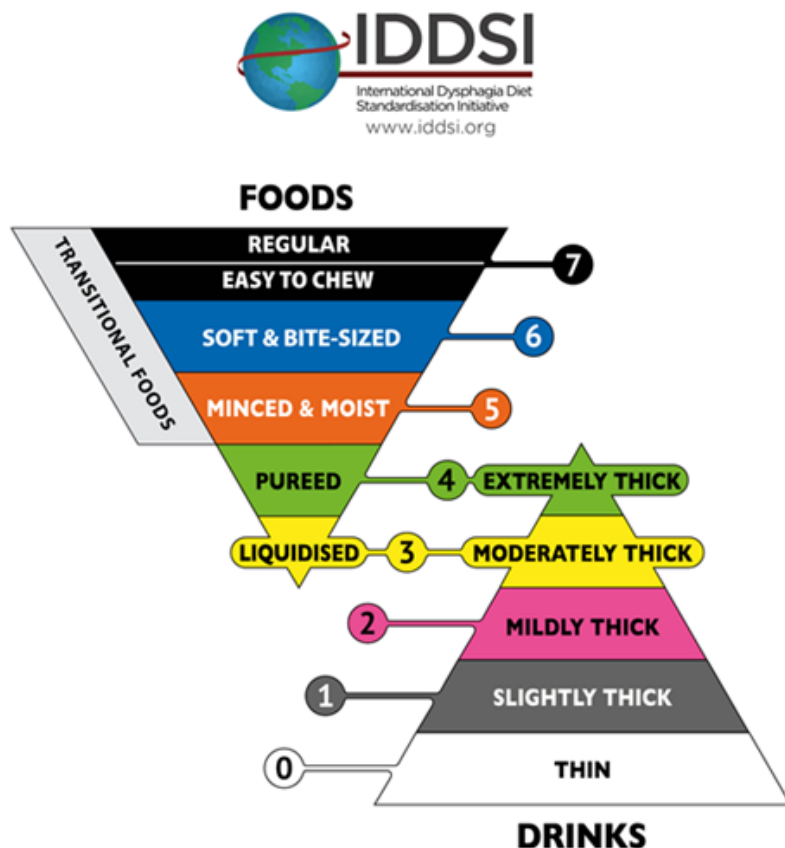
Use the blue "Related Client Ed" link to the right of the title of this page to access the Finger Foods Sample 1-Day Menu and Foods List.

IDDSI Framework and Levels

The International Dysphagia Diet Standardisation Initiative (IDDSI) established a globally consistent diet terminology and definitions for dysphagia diet therapy in all settings, for all ages, and for all cultures (IDDSI, 2024). The IDDSI Framework is used to prepare, evaluate, and serve food that is safe and easily tolerated.

Using IDDSI allows for consistent communication among health professionals, care providers, researchers, and industry partners to improve quality of care and safety for patients across the world. The transition to the IDDSI framework has been ongoing and is at different stages throughout the United States. In order to provide consistent international dysphagia care and further advance dysphagia research internationally, it is critical that clinicians use the same terms, definitions, and measurable characteristics for each food and liquid texture. The Nutrition Care Process Terminology includes the IDDSI terminology to facilitate adoption and documentation.

The IDDSI Framework



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Access the IDDSI Framework for more information about the following:

- Detailed level definitions
- IDDSI testing methods
- Evidence statement

The IDDSI Resources page provides audit tools, patient handouts, implementation guides, and other helpful information. Periodic visits to this page to check for updates and new content are recommended.

IDDSI Levels

The IDDSI Framework consists of eight levels (0-7), which include a number, diet name that describes food consistency, and a color. IDDSI levels may also be identified using an icon/symbol with its associated color, number, and abbreviation.

IDDSI testing confirms that a food or liquid meets the IDDSI level definitions according to the following characteristics:

- Food piece size and shape
- Food texture: Softness, tenderness, stickiness, and mashability
- Food moisture: Hardness and dryness
- Food items with dual textures: Liquid must not separate from solid
- Liquid thickness

IDDSI is not considered a diet order and is not intended to be prescriptive.

Evaluating and Confirming IDDSI Diet Orders

There are three parts to every diet order: Therapeutic or House, food texture, and liquid thickness.

Examples of Diet Orders:

- House, Level 7 Regular, Level 0 Thin
 - This diet order is the most liberal
- House, Level 6 Soft and Bite-Sized, Level 2 Mildly Thick
- NAS, Level 5 Minced and Moist, Level 3 Moderately Thick

Level 7 Regular, and Level 0 Thin are not texture-modified diets, as they do not restrict texture nor liquid thickness.

Patients whose condition necessitates a change in diet order require an individualized and interdisciplinary approach and an assessment by the speech-language pathologist (SLP), registered dietitian nutritionist (RDN), and medical team. In most health care settings, the SLP determines which diet texture will best meet the patient's physical ability to consume food safely and will present the lowest risk for aspiration. The RDN is responsible for implementing a balanced nutrition plan that meets the consistency specifications identified by the SLP. Often the entire health care team (including physicians and nurses) is involved in monitoring patients for difficulties related

to chew/swallow; in reporting concerns per facility policy; monitoring diet texture effectiveness, adherence, and understanding; and evaluating needs related to patient education.

Consider the following in determining the most appropriate textures and diet order for the patient with chewing or swallowing difficulties.

- Determine why the diet is ordered. Chewing difficulties or avoidance of some foods may indicate a safety and swallow concern. Consult the medical record and the health care team, including the SLP.
- Evaluate diet orders to ensure they are appropriate for the patient; for example, “bland”, “GI soft” or “soft” diet orders may be ordered for patients with chewing or swallowing issues instead of a texture-modified diet. (Note: A bland diet order is considered obsolete.)
- Determine intention of “cut-up” diets. The term “cut-up” may be used interchangeably for diets in patients with limited self-eating skills or with general food safety concerns. A cut-up diet is inappropriate for patients with chewing difficulties related to insufficient or impaired dentition or fatigue, as they are at high risk for choking. Consider an SLP assessment if there is an underlying neurological or cognitive issue.

As for all diets in health care settings, texture-modified diets should be at the most liberal level tolerated and should be frequently reassessed for potential adjustments/advancements.

Mixed-Consistency Foods

Mixed-consistency foods can separate into two components: liquid and solid food. Examples of these foods are soups, stews, and cold cereal with milk. Watermelon is also a mixed-consistency food since the liquid and solid separate once it enters the mouth.

Mixed consistency foods are much more challenging for a patient to eat because they require more skill to process and swallow safely. The individual may swallow the liquid component with poor control. The solid components may be swallowed before they are properly chewed into a safe bolus. Mixed consistency foods are allowed on IDDSI Level 7 Regular. They may be safe for a person on Level 7 Easy-to-Chew if a clinician has deemed them safe for the individual. They are not allowed in the IDDSI Framework for any patients on IDDSI Levels 1-6.

Use IDDSI testing methods to test both the liquid and the solid part. Determine if the item must be avoided or if the recipe can be modified to ultimately pass IDDSI testing.

The statements listed below are used in the IDDSI Framework to describe some mixed consistency foods that are safe to serve. If a food item does NOT behave this way, (liquid DOES separates from solid), modification is required before serving safely.

- No separate thin liquid
- Liquid must not separate from solid
- No thin liquid should drain from food
- Sauce must not separate from food/solid part

Listed below are modifications to mixed consistency foods described in the IDDSI Framework as guidance to assure the item is safe:

- Serve in mildly, moderately or extremely thick sauce AND drain excess liquid
- Drain excess milk, liquid or juice
- Thicken to thickness level recommended by clinician
- Liquid portion must be thick per clinician recommendation
- Assess individual ability to manage fruit with high water content (ex watermelon) where juice separates from solid in the mouth during chewing

As always, IDDSI emphasizes testing to confirm the level of the food item. Additionally, consider the patient's abilities along with clinician evaluation to determine the safest foods and liquids to serve.

Complex Foods

A complex food texture requires a patient to be able to chew and move

different food textures in one mouthful. This task can be challenging and increases the risk for choking. Menu items that may be considered as a complex food texture include pizza, spaghetti and meatballs, sandwiches, hamburgers, and hot dogs. More information is available in the Food Textures That Pose A Choking Risk section in the IDDSI Framework document.

Transitional Foods

Transitional foods start as one texture and change to another texture in the mouth. This effect can happen when moisture is added to the food or if there is a significant temperature change that modifies the food consistency, such as melting or heating.

Transitional foods require minimal chewing and can be broken down by tongue pressure after changes in moisture or temperature occur. Transitional foods are often used for developmental teaching or for swallowing rehabilitation. These transitional foods are allowed on IDDSI Levels 5, 6, 7 Easy to Chew and 7 Regular but only if assessed as safe by a clinician. Use IDDSI testing methods for transitional foods to confirm the food is safe for the individual.

Test by:

1. Take a 15 mm x 15 mm sample size, which is about the size of a thumbnail
2. Add 1 mL of water and wait 1 minute
3. Use fork to apply pressure until the thumbnail turns white
4. Evaluate results: It is a transitional food if the following are true:
 - The sample has been squashed
 - The sample does not return to its original shape
 - The sample has melted (as in the case of ice chips)

Classification for Ice Cream and Gelatin

Ice cream Ice cream is considered a transitional food, but it also has mixed consistency characteristics. As the ice cream stays in the mouth, depending on time and the product composition, liquid and solid components may become separated.

Determine the appropriateness of serving ice cream products by allowing them to melt down to room temperature and then conduct syringe testing, fork drip testing, and spoon tilt testing to evaluate if safe for an individual. As always, clinical assessment is key to determine which foods are safe.

Gelatin Gelatin has long been considered a difficult food product for individuals with swallowing issues. Gelatin-based jelly is also a concern because it can melt quickly in the mouth. Gelatin breaks apart in the mouth, becoming harder to control in the mouth before swallowing. Gelatin also presents the additional safety risk of breaking apart in clumps that can be swallowed whole.

Thickeners may improve the safety of ice cream, jelly, and gelatin. There are also pre-thickened ice cream and gelatin products available for purchase. Use IDDSI testing methods to evaluate any of these products.

Diet Manual > NCM Diet Manual > IDDSI Framework and Levels > Transitioning Texture > Modified Diet Terminology and Definitions to the IDDSI Framework

Transitioning Texture-Modified Diet Terminology and Definitions to the IDDSI Framework

Transitioning from prescribing the National Dysphagia Diet (NDD) and other texture-modified diets to prescribing the International Dysphagia Diet Standardisation Initiative (IDDSI) levels requires careful planning. Using the IDDSI implementation resources on the IDDSI website will help you design your transition plan.

Part of the IDDSI implementation process is to customize your diet manual to your institution which can be done by following the process outlined in the Diet Manual Implementation section. The NCM Diet Manual Crosswalk clarifies any differences between the diet name terminology used by your institution and the NCM Diet Manual Reference names. The NCM Diet Order Terminology and Definitions document provides descriptions of the NCM Diet Manual diets that are offered. Note: It is important to use official IDDSI level terminology and abbreviations.

Transitioning texture-modified diets to the IDDSI framework can be completed in stages or all at once depending on staff competency, policies and procedures, and individual readiness. However, it is recommended that institutions use a continuous improvement process approach during the implementation process. This approach may include adapting recipes, menus, and diet spreadsheets to the new framework.

Learn about the IDDSI framework, terms, and definitions.

At IDDSI.org, you can do the following:

- Review the Complete IDDSI Framework + Detailed Definitions
- Learn about IDDSI Testing Methods, Subscribe to IDDSI's YouTube Channel to view IDDSI testing method videos.
- Examine the numerous free resources available under the Resources section:
 - Printable posters
 - Presentations
 - Publications
 - Videos and webinars
- Read IDDSI Frequently Asked Questions
- Sign up for the free monthly newsletter, "E-Bites"
- Link to an app store to download the free IDDSI app

The United States IDDSI Reference Group provides additional information and guidance about IDDSI.

Review current texture-modified diet terminology and definitions at your facility and compare to IDDSI

- With other members of the health care team, particularly the speech-language pathologist, review the diet names and definitions of the texture-modified diets and thickened liquids that are used at your health care facility.
- Compare current diet terminology and definitions to the IDDSI framework using the following references: Mapping NDD Diets to IDDSI Diets and Summary of Texture-Modified Diets.
- Document and track your decisions by using the NCM Crosswalk, adjusting menus, recipes and diet spreadsheets.

Mapping NDD Diets to IDDSI Levels

The IDDSI has established eight levels of food and liquid consistencies. Detailed descriptions are available in the Complete IDDSI Framework + Detailed Definitions. Each NDD diet name and definition is somewhat comparable to an IDDSI level, though there are differences. During the mapping process—that is, identifying an IDDSI level comparable to an NDD diet — practitioners evaluate diet names, allowable textures, and foods currently served and then connect it to an IDDSI level that reflects similar expectations.

Neither IDDSI Level 7 Regular (Black) nor IDDSI Level 0 Thin (White) are texture-modified diets. There are no restrictions on foods for Level 7 Regular nor liquids for Level 0 Thin. Details regarding IDDSI Level 7, Easy to Chew through Level 1 are shown below adjacent to its similar (but not exact) NDD level counterpart.

NDD Diet Name	Similar to IDDSI Level
NDD Level 3: Advanced	Level 7 Easy to Chew
NDD Level 3: Advanced	Level 6 Soft & Bite-Sized: Adults, 15 mm x 15 mm* food particle size Pediatrics, 8 mm x 8 mm* food particle size
NDD Level 2: Mechanically altered	Level 5 Minced & Moist: Adults, 4 mm x 15 mm* food particle size Pediatrics, 2 mm x 8 mm* food particle size
NDD Level 1: Pureed	Level 4 Pureed Foods
Pudding or Spoon Thick	Level 4 Extremely Thick Drinks
Honey Thick Liquid	Level 3 Liquidized Foods

	Most other moist breads and bread products allowed. Salad, raw vegetables, and most fresh fruit not allowed. Adequate dentition and chewing ability expected.	Soft & Bite-Sized (Blue)	diet order) and drain excess liquids. Should be able to bit off food and chew without tiring easily. Soft, tender, moist foods; no mixed consistency; Foods should have a particle size no greater than 15 mm length by 15 mm width for adults or 8 mm length by 8 mm width for pediatrics. Bread is not allowed unless pureed or gelled throughout. Rice, couscous, and quinoa if suspended in sauce/liquid and does not separate in individual grains. Casseroles/stews: Serve foods in mildly, moderately or extremely thick sauce (per liquid diet order) and drain excess liquids. Some chewing ability is required and tongue force to move food for oral processing is expected.
Dysphagia level 2: Mechanically Altered or Ground	Foods should be moist, of soft-texture and can be formed into a bolus easily.	Level 5 Minced & Moist (Orange)	Minced, soft and moist foods; can be scooped and shaped (e.g., into a ball shape and

Meats are ground then softened and moistened by using gravies and sauces (served no larger than ¼ inch pieces).

Vegetables and fruits are soft cooked, and bite sized (served no larger than ¼ inch pieces).

Soft breads, grains and cereals, well moistened, blended, chopped or ground. Slurred or pureed bread/grain products are often used.

Desserts softened with milk or liquid. Pureed rice.

Avoid large chunks of foods or foods too hard to chew thoroughly, including raw vegetables, most fresh fruits, uncooked dried fruits, sticky, gummy, stringy, crunchy, dry, crisp or coarse foods.

Some chewing required and tolerance of mixed textures should be assessed.

easily mashed by fork. No mixed consistency. Food pieces with particle size no greater than **4 mm wide by 15 mm** long for adults or **2 mm wide by 8 mm** long for pediatrics.

Most foods are served in mildly, moderately or extremely thick sauce (per liquid diet order) and drain excess liquids

Bread is not allowed unless pureed or gelled throughout. Rice, couscous, and quinoa if suspended in sauce/liquid and does not separate in individual grains.

Foods require minimal chewing and tongue force to move bolus and to separate the particles. Biting is not required.

Dysphagia Level 1: Pureed	All foods are pureed, blended, or strained for a smooth consistency with no lumps. Some softened desserts and bread items allowed if gelled throughout. Soups, casseroles should be pureed and strained. Foods that require chewing or a bolus formation not allowed.	Level 4 Pureed Food (Green)	Smooth, lump free; not firm or sticky; does not require chewing or bolus formation. Falls off spoon as intact spoonful and holds shape. No mixed consistency. For liquids at this level: cannot be poured or drunk from a cup; nor sucked through a straw. No chewing or biting required nor ability to form bolus.
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Comparison of Thickened Liquids

NDD		IDDSI	
Pudding Thick Liquids	All thickened liquid diets provide pureed, strained soups and fortified ice cream product (thick at room temperature) instead of regular soup or ice cream offerings. Liquids should meet liquid consistency specifications, which may require use of thickening products. Avoid gelatin, some fruits with high water content.	Level 4 Extremely Thick liquids/drinks (Green)	Liquids at this level: cannot be poured or drunk from a cup; nor sucked through a straw. Smooth, lump free and there is no mixed consistency.
Honey Thick Liquids	All thickened liquid diets provide pureed, strained soups and fortified ice cream product (thick at room temperature) instead of regular soup or ice cream offerings. Liquids should meet liquid consistency specifications, which may require use of thickening products. Avoid gelatin, some fruits with high water content.	Level 3 Liquidized food (Yellow) and Level 3 Moderately Thick liquids/drinks (Yellow)	Smooth foods and fluids without lumps; cannot be molded on plate; requires no chewing or oral processing and can be directly swallowed. Drips slowly through slots of fork. Requires some effort to suck through straw

Nectar Thick Liquids	Some mixed-consistency items require thickening.	Level 2 Mildly Thick (Pink)	Liquids that require effort to be sipped through straw; flows off a spoon at a slower rate than thin liquid.
No NDD level		Level 1 Slightly Thick (Grey)	Liquids thicker than water, flows through straw or syringe
No NDD level		Level 0 Thin (White)	Flows like water. Fast flow. No restrictions. This is not a texture-modified diet.

Identify required modifications during IDDSI transition

Potential modifications required when transitioning to the IDDSI diet include the following:

- Terminology
- Diet definitions
- Forms and technologies (such as electronic medical records, diet office software, order sets, etc.) using NDD terminology
- Abbreviations for diet/menu tickets
- Policies and procedures including updating the diet manual
- Recipes, menu items, and diet spreadsheets
- Thickening products purchased and facility-specific thickened product recipes

Review and test current texture-modified recipes and menu items at your facility

- Use the IDDSI Testing Methods and audit tools to evaluate recipes and menu items
- Compare existing recipes and menu items with IDDSI definitions to identify differences
 - Determine which recipes and menu items are safe to serve at each IDDSI Level
- Re-create recipes that do not meet IDDSI specifications and test again
- Use the NCM Diet Manual section “Using IDDSI Testing Methods and Audit Tools” for food and fluid texture descriptions and testing methods
 - Subscribe to IDDSI’s YouTube Channel to view IDDSI testing method videos.
 - Practice each technique with a syringe or utensil for a visual, hands-on learning experience.

Consider your facility’s needs when purchasing products

It is important to thicken foods, drinks, and liquids to meet IDDSI specifications. Pre-thickened products and powder/liquid/gel thickeners are available. Consider your facility's needs when purchasing products.

- IDDSI language used on product labels
- Preferred packaging of the thickener: bulk or single serve
- Product ingredients used could vary:
 - Modified food starch (eg corn)
 - Gums (eg xanthan, gellan, guar, etc.)
- Products that accommodate food allergies
- Possibility of additional carbohydrates, which can affect blood glucose
- Time and temperature, which may affect product viscosity
- Ease of mixing

After initially confirming appropriateness of IDDSI classification of foods and liquids at your facility, frequent routine checks and monitoring including at point of service should be conducted per established policy. If a patient is not tolerating a thickened liquid, removal from service to perform immediate testing is necessary to determine if the product is consistent with the diet order or if additional clinician evaluation is indicated.

Many products are available for direct purchase by consumers. Distribute IDDSI Consumer Handouts and NCM Client Education handouts to provide texture modification education to patients purchasing their own products.

Using IDDSI Test Methods and Audit Tools

IDDSI testing procedures to verify food textures and liquid consistencies require use of items commonly available in kitchens and health care facilities, such as forks, spoons, syringes, and IDDSI funnel. These tools provide practical, consistent evaluation of specific food and fluid texture consistency, which ensures standardization of food and liquid consistency and improves patient safety. It is essential to complete all applicable testing methods when testing a menu item.

The following tables describe each IDDSI level and its testing methods. Subscribe to IDDSI's YouTube Channel to view IDDSI testing method videos.

IDDSI Food Texture Descriptions and Testing Methods		
Food/Drink Consistency	Description	Testing Method
Level 7 Regular (Black)	All textures allowed if developmentally appropriate and age appropriate. Any size and crunchy foods allowed; mixed consistency allowed.	
Level 7 Easy to Chew (Black)	Soft, tender foods; no hard, tough, stringy, crunchy or chewy foods. Mixed consistency allowed if safe for Level 0 and with clinician approval. Size of food pieces are not restricted. Some soft bread allowed per clinician approval.	Fork Pressure Test: Press the base of a fork into a food sample the size of a thumb nail until thumb nail blanches to white. Sample should change shape when pressure applied and not return to original shape when pressure and fork removed. Fork/Spoon Separation Test: Fork held on side can easily cut or break item into smaller pieces.
Level 6 Soft & Bite-Sized (Blue)	Soft, tender, moist foods that are easily mashed; no mixed consistency. Foods should be bite sized with particle size no greater than 15 mm	Fork Pressure Test: Press the base of a fork into a food sample the size of a thumb nail until thumb nail blanches to white. Sample should change

	= 1.5 cm x 1.5 cm for adults and no greater than 8 mm for pediatric.	shape when pressure applied and not return to original shape when pressure and fork removed. Fork/Spoon Separation Test: Fork held on side can easily cut or break item into smaller pieces. Fork Test: For particle size. Measure food pieces to confirm they are no greater than 15 mm for adults (approximate width of standard fork and is about the size of an adult human thumb nail). Note: Food pieces for pediatric is smaller.
Level 5 Minced & Moist (Orange)	Soft and moist foods; can be scooped and shaped; easily mashed by fork. No mixed consistency. Lumps should be easily squashed with tongue and have a particle size no greater than 4 mm wide by 15 mm long for adults and no greater than 2 mm wide by 8 mm long for pediatrics. Biting not required; Minimal chewing required and able to move bolus.	Fork Pressure Test: Food particles easily fit between the standard slots of fork prongs and come through when pressed. Easier to mash food with fork than Level 6 and thumb nail does not blanch white. Fork Test: For particle size. Measure food pieces to confirm size requirement. The space between tines on a standard metal fork is 4 mm and that the width of the base of fork is approximately 15 mm. Note: Food pieces for pediatric is smaller. Fork Drip Test: Scooped sample sits in pile and

		does not fall through prongs. Spoon Tilt Test: Sample holds shape on spoon. When spoon tilted, sample slides off spoon with little residual left behind (not sticky). Sample should not separate into individual grains/pieces.
Level 4 Pureed Food (Green)	Smooth, lump free; not firm or sticky; does not require chewing or bolus formation. Falls off spoon as intact spoonful and holds shape. No mixed consistency. Liquids cannot be poured, drank from a cup; or sucked through a straw.	Fork Drip Test: No lumps; stays as mound on fork with very little/no flow through prongs. Spoon Tilt Test: Holds shape; not firm or sticky; sample slides off easily. If it is too thick, sample will not fall off spoon when tilted or will stick to spoon. Fork Test: When fork is pressed into the food sample, tines make pattern and/or indentation stays. IDDSI Flow test not applicable to this level.
Level 3 Liquidized Food (Yellow)	Smooth foods and fluids without lumps; does not hold shape on plate; requires no chewing or oral processing and can be directly swallowed. Drips slowly through slots of fork. Requires moderate effort to suck through straw. Examples for this category: runny puree fruit or infants “first foods”; fruit syrup; and	IDDSI Flow Test: See instructions in IDDSI Testing Methods Fork Drip Test: Drips slowly in dollops; spreads out if spilled onto flat surface (plate). Fork Test: When fork is pressed into the food sample, the tines do not leave a clear pattern. Spoon Tilt Test: Easily pours; no sticking to

	some sauces/gravies.	spoon.
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IDDSI Liquid Consistencies and Testing Methods		
Food/Drink Consistency	Description	Testing Method
Level 4 Extremely thick liquid (Green)	Smooth, lump free; not firm or sticky; does not require chewing or bolus formation. Falls off spoon as intact spoonful and holds shape. No mixed consistency. Liquids cannot be poured, drank from a cup; or sucked through a straw.	Fork Drip Test: no lumps; stays as mound on fork with very little/no flow through prongs. Spoon Tilt Test: Holds shape; not firm or sticky; sample slides off easily. If it is too thick, sample will not fall off spoon when tilted or will stick to spoon. Fork Test: When fork is pressed into the food sample, tines make pattern and/or indentation stays. IDDSI Flow test not applicable to this level.
Level 3 Moderately thick liquid (Yellow)	Smooth foods and fluids without lumps; does not hold shape on plate; requires no chewing or oral processing and can be directly swallowed. Drips slowly through slots of fork. Requires moderate effort to suck through straw.	IDDSI Flow Test: See instructions in IDDSI Testing Methods Fork Drip Test: Drips slowly in dollops; spreads out if spilled onto flat surface (plate). Fork Test: When fork is pressed into the food sample, the tines do not leave a clear pattern. Spoon Tilt Test: Easily pours; no sticking to spoon.

Level 2 Mildly thick liquid (Pink)	Flows off spoon; pours slower than thin drinks; Sippable; some effort needed for straw use; May be suitable if tongue control slightly reduced	IDDSI Flow Test: See instructions in IDDSI Testing Methods
Level 1 Slightly thick liquid (Grey)	Thicker than water and flows through straw, syringe or nipple; similar to infant formula. Little more effort required to drink than thin liquids.	IDDSI Flow Test: See instructions in IDDSI Testing Methods
Level 0 Thin liquids (White)	Flows like water; can drink through any teat/nipple, cut or straw per age/skill	IDDSI Flow Test: See instructions in IDDSI Testing Methods

Additional IDDSI Testing Resources

This chart will assist you in converting IDDSI size requirements from millimeters to inches.

IDDSI Measurement Conversions		
Measurement in Millimeters (mm)	Measurement in Inches (in)	Approximate Measurement
15 mm	.59 in	Just less than 5/8 inch
14 mm	.55 in	Just over 1/2 inch
8 mm	.31 in	Just less than 1/3 inch or 5/16 inch
4 mm	.15 in	Slightly greater than 1/8 inch
3 mm	.11 in	1/8 inch
2 mm	.07 in	Slightly greater than 1/16 inch

This chart summarizes which testing methods are used while testing each level. All applicable food testing methods completed on a food item confirms its appropriateness.

Appropriate IDDSI Testing Methods for Food Levels

IDDSI Testing Methods	Level 7 Easy to Chew (Black)	Level 6 Soft & Bite Sized (Blue)	Level 5 Minced & Moist (Orange)	Level 4 Puree (Green)	Level 3 Liquidized (Yellow)
Fork/Spoon Separation Test: Checks if food breaks apart easily.	Yes	Yes	N/A	N/A	N/A
Fork Pressure Test: Checks for softness and this mimics tongue pressure.	Yes, and thumb nail may turn white.	Yes, and thumb nail may turn white.	Yes, and thumb nail does NOT turn white.	Use fork to press on surface of food. Tines make pattern and/or indentation stays.	Use fork to press on surface of food. Tines do not leave a pattern on food surface.
Fork Test: Checks for particle size.	N/A	15 mm size = width of fork or size of thumbnail Size for adults-15 mm Size for pediatrics-8 mm	4 mm size = space between tines; mimics chewed solids before swallowing. Size for adults-4mm wide x 15 mm long Size for pediatrics- 2 mm wide x 8 mm long	N/A	N/A
Fork Drip Test: Determines if too thin or too thick; and if there is separate liquid from solid.	N/A	N/A	Food sits on top of fork; does not fall through.	Food stays as a mound. Only small amount drips through; no dollop.	Drips slowly or in dollops through spaces between tines.

Spoon Tilt Test: Measures stickiness and how well food holds together (cohesiveness).	N/A	N/A	Foods hold its shape; not sticky; does not separate into individual grains/pieces.	Food slips off spoon, may need slight flick; may leave small film; may spread slightly on plate.	Easily pours from spoon and does not stick to the spoon.

Audit Sheets

IDDSI has developed the following audit sheets which are available on the IDDSI Resources page under Implementation Resources:

- Liquids (Levels 0-3)
- Level 4 - Pureed
- Level 4 - Extremely Thick
- Level 5 - Minced and Moist
- Level 6 - Soft and Bite Sized
- Level 7 - Easy to Chew

The audit sheets will help you determine if certain foods at your facility meet IDDSI criteria.

Level 7 Regular (Black)

Definition: This diet allows all foods and is intended for individuals without chewing or swallowing difficulties. All food textures, consistencies, and transitional foods are permitted. This is not a texture-modified diet.

The regular diet can be modified to specify the allowed/not allowed foods for an individual patient, which are determined by the speech-language pathologist (SLP). Foods should further meet the complete descriptive and testing specifications of the International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b). See House/Regular Diet.

Examples of diet terminology to use and what foods to avoid include:

- House (or any therapeutic diet), regular Level 7 with Level 6 soft & bite-sized fruits/vegetables
- Consistent Carbohydrate Diet, regular Level 7 no hard breads; or use no breads/sandwiches
- 2 g Low Sodium Diet, regular Level 7 no mixed consistencies
- Lactose-controlled Diet, regular Level 7 with level 6 soft & bite-sized meats

Level 7 Regular Easy-to-Chew (Black)

Definition: The Level 7 Regular Easy-to-Chew diet is prescribed to people who may have difficulty chewing hard, tough, stringy, or crunchy foods. It requires the ability to bite, chew, and orally process foods without easily tiring and the ability to eat independently. This texture-modified diet has no food size restriction and is NOT intended for individuals with swallowing problems or at a clinically identified risk for choking. The Level 7 Regular Easy-to-Chew diet may include mixed-consistency foods and some bread products if the clinician assessment deems it as safe. Foods should further meet the complete descriptive and testing specifications of the International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Food lists are based on the International Dysphagia Diet Standardization Initiative (IDDSI) Framework. Use IDDSI testing methods to confirm foods and drinks are safe.

The following table is not a complete list of foods. Other foods may be OK to eat as long as they meet the IDDSI food testing requirements.

Food Group	Foods Recommended	Foods Not Recommended
Grains	All well-moistened soft, cooked, dry, hot, or cold cereals with excess liquid drained All starches including couscous, pasta, quinoa, rice, wild rice, and moist bread dressing (as tolerated) Bread, toast, and sandwiches that can be cut or broken apart into smaller pieces with the side of a spoon/fork*	Dry bread, toast, crackers Tough, thick crusty breads such as French bread or baguettes Dry bread dressing Bread with nuts or seeds Coarse or dry cereals such as shredded wheat or bran flakes Dry cakes, cookies that are chewy, sticky, or very dry
	Chop or cut into pieces as needed: Prepared, moistened, soft/tender red meat, including beef, pork, or lamb served with gravy or sauce; chop or cut up as	Tough, dry, red meats (beef, pork, lamb) or meats with bone and/or gristle Tough, dry poultry (chicken and turkey) Poultry with bone/gristle

Protein
Foods

needed

Prepared, moistened, soft/tender poultry, including chicken or turkey served with gravy or sauce that is chopped or cut up as needed

Prepared, moistened, tender seafood, including fish (salmon, herring, and sardines), shrimp, lobster, clams, and scallops served with gravy or sauce

Tender, thin sliced deli meats such as bologna, ham, salami (made of meat or poultry) that is cut up or chopped as needed; salad-type sandwich filling

Bacon and sausage (as tolerated)

Eggs and egg substitutes, prepared in any way you prefer

Casseroles with tender, soft ingredients including meat or ground meats served in mildly, moderately, or extremely thick sauce with excess liquid drained

Smooth, nut and seed butters, such as peanut butter, almond butter, and sunflower seed butter (as tolerated)

Prepared, moistened soy foods, such as tofu or tempeh

Prepared, moistened meat alternatives, such as

Tough, dry fish

Fish with bones

Tough, dry, stringy deli meats, such as pastrami, corned beef

Bacon and sausage (if not tolerated)

Chunky nut seed butters

Whole nuts, such as peanuts and almonds, pistachios

Seeds such as sunflower, sesame, and pumpkin seeds

	veggie burgers, and sausages based on plant protein Prepared, moistened legumes, such as dried beans, lentils, or peas	
Dairy and Dairy Alternatives	Milk, yogurt (without nuts or coconut), cottage cheese, and soft, not sticky or chewy cheeses Fortified soymilk Cream cheese, sour cream, and whipped topping Pudding, custard Frozen desserts such as ice cream, sherbet, malts, and frozen yogurt*	Yogurt with nuts or coconut Hard or dry cheese Chunks of cheese that are hard to eat because they are very sticky or very chewy Cream cheese spread with coarse, difficult to chew, or chunky additives like nuts or pineapple
Vegetables	All cooked, tender vegetables soft enough to be broken apart with the side of a spoon/fork All tender potatoes soft enough to be broken apart with the side of a spoon/fork Vegetable juices	All raw vegetables Undercooked vegetables including some stir-fried vegetables Cooked vegetables that are fibrous, tough, or stringy, such as green beans, peapods, and rhubarb Tough, crisp fried vegetables such as potato skins or crisp-fried potatoes
Fruit	All canned and cooked fruits, soft enough to be broken apart with the side of a spoon/fork Fruit drained of excess liquid if recommended by clinician Soft, peeled fresh fruits such as peaches, nectarines, kiwi, cantaloupe, honeydew, and seedless	Fresh fruits that are difficult to chew such as apples or pears Stringy, high-pulp fruits such as papaya, pineapple, or mango Fresh fruits with difficult-to-chew peels or skins such as grapes Dried fruits such as prunes, raisins, or apricots

	watermelon**, *** Soft berries with small seeds (such as strawberries) 100% fruit juice Moist, cooked raisins (in recipes)	Sticky and chewy fruit items such as fruit leather, fruit roll-ups, fruit snacks, dried fruits
Oils	Olive, peanut, and canola oils Margarines and spreads Salad dressing Butter, gravy, cream sauces	All fats with coarse, difficult to chew, or chunky additives
Beverages	Coffee, tea, water, 100% fruit juice	
Other	Prepared foods, including all soups with tender meats, casseroles, baked goods, and snacks made from recommended ingredients All seasonings and sweeteners, including honey, jams, jellies, and preserves Non-chewy candies without nuts, seeds, or coconut; non-sticky, non-gummy candies	Nuts, seeds, coconut Candies with nuts, seeds, or coconut Chewing gum Chewy caramel or taffy-style candies and marshmallows Corn chips and potato chips Popcorn and sticky rice cakes

* Bread, bread products, sandwiches, and frozen desserts are OK to eat if approved by the clinician.

** Watermelon and other fruits with high water content may be difficult to chew because the juice separates from the solids in the mouth.

*** Hard seeds from fruit may be accidentally ingested.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Level 7 Regular Easy-to-Chew Sample 1-Day Menu with nutrient analysis.

Level 6 Soft and Bite-Sized (Blue)

Definition: The Level 6 Soft and Bite-Sized diet is prescribed for people who have difficulty chewing hard, tough, stringy or crunchy food and are unable to safely bite off pieces of food. Some chewing ability is required to break food further into pieces and to move food around for a safe swallow. This diet requires a texture modification so that foods are soft, tender, moist and have no separate thin liquids. Foods should have a particle size no larger than 1.5 centimeters x 1.5 centimeters. Foods should further meet the complete descriptive and testing specifications of the International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Food lists are based on the International Dysphagia Diet Standardization Initiative (IDDSI) Framework. Use IDDSI testing methods to confirm foods and drinks are safe.

The following table is not a complete list of foods. Other foods may be OK to eat as long as they meet the IDDSI food testing requirements.

Food Group	Foods Recommended	Foods Not Recommended
Grains	All well-moistened, soft cooked hot cereals with lumps no larger than 1.5 cm x 1.5 cm pieces, drained and served without excess liquid. Cold cereal (if approved by your clinician) with lumps no larger than 1.5 cm x 1.5 cm, softened, drained and served without excess liquid. Pregelleted, soaked bread that is consistently moist throughout, with no sections that have hardened. Starches including	Dry bread, toast, crackers, biscuits, muffins, pancakes, waffles, bread dressing Grainy, sticky, or glutinous rice

	pasta, potato, and rice dishes with pieces no larger than 1.5 cm x 1.5 cm. Not sticky. Couscous, quinoa, or rice held together with smooth, thick sauce that keeps rice from separating into individual grains. Soft, cooked pasta dishes with pieces no larger than 1.5 cm x 1.5 cm; sauce does not separate from pasta.	
	A note about red meat, pork, lamb, chicken, and casseroles: If food cannot be served soft at 1.5 cm by 1.5 cm, instead mince the food and serve moist by following Level 5, Minced & Moist guidelines. Cooked, moist, soft, and tender red meat, including beef, pork, or lamb, chopped no larger than 1.5 cm x 1.5 cm pieces. Cooked, moist, soft, and tender poultry, including skinless chicken or turkey, chopped no larger than 1.5 cm pieces x 1.5 cm. Cooked, moist, soft, and tender seafood, including fish	Protein foods in sizes larger than 1.5 cm x 1.5 cm pieces Chicken, turkey and fish with skin on or with bones Sausage skin Protein foods served with undrained thin liquids, such as beef stew, chowders, and casseroles Fish with skin on or with bones; fish that you cannot break into small pieces (1.5 cm x 1.5 cm) Fried eggs Whole nuts and seeds (including peanuts and almonds), pistachios, and sunflower seeds Nut and seed butters, unless they are smooth and used in a recipe that meets testing guidelines Dried beans, lentils or peas that have hard shells or skins; or do not meet size expectations or

Protein Foods

(salmon, herring, and sardines), shrimp, lobster, clams, and scallops; these foods should be soft enough to break and serve in pieces no larger than 1.5 cm x 1.5 cm pieces.

Eggs and egg substitutes, no larger than 1.5 cm x 1.5 cm pieces.

Casseroles with soft chunks of meat, ground meats, or tender meats no larger than 1.5 cm x 1.5 cm pieces served in mildly, moderately, or extremely thick sauce with excess liquid drained

Moistened soy foods, such as tofu or tempeh; soft enough to break and serve in pieces no larger than 1.5 cm x 1.5 cm pieces

Moistened meat alternatives, such as veggie burgers, and sausages based on plant protein; chopped no larger than 1.5 cm x 1.5 cm pieces

Sausages without skin cut into pieces no larger than 1.5 cm x 1.5 cm or ground

do not pass IDDSI testing methods

	Moistened legumes with soft skins/shells, such as dried beans, lentils, or peas no larger than 1.5 cm x 1.5 cm pieces and without tough skins. If incorporated into a recipe, must pass IDDSI testing methods.	
Dairy and Dairy Alternatives	Yogurt (without nuts or coconut), cottage cheese with lumps no larger than 1.5 cm x 1.5 cm, and soft cheeses that aren't sticky or chewy Cream cheese, sour cream, pudding, custard and whipped topping Frozen desserts such as, ice cream, sherbet, malts, and frozen yogurt if recommended by a clinician Milk, fortified soymilk, fortified nut milk in the liquid consistency recommended by a clinician	Yogurt with nuts or coconut Hard or dry cheese, cheeses in pieces bigger than 1.5 cm x 1.5 cm pieces, and cheese that is sticky or chewy Frozen desserts such as, ice cream, sherbet, malts, and frozen yogurt, unless approved by a clinician
	Soft cooked vegetables and starchy vegetables that are served steamed or boiled in pieces no larger than 1.5 cm x 1.5 cm Peas, creamed peas and creamed corn with soft skins and free of husks; may be	All raw vegetables Stir-fried or fried vegetables Undercooked vegetables Peas with tough skins or shells Corn with husks Cooked vegetables that are fibrous, tough, firm, or stringy such as green beans, peapods,

Vegetables	blended or incorporated into IDDSI safe stews and soups. If incorporated into a recipe, must pass IDDSI testing methods. Vegetable juices must meet the IDDSI liquid consistency recommended by a clinician	and mashes Sticky or crunchy potatoes such as mashed potatoes and potato casseroles with hard tops
Fruit	Canned and cooked fruits, drained of excess juices, served cut up into pieces no larger than 1.5 cm x 1.5 cm or served minced or mashed Soft, peeled fresh fruits such as peaches, nectarines, kiwi, cantaloupe, and, soft berries with small seeds (for example, strawberries), cut up into pieces no larger than 1.5 cm x 1.5 cm or serve minced or mashed 100% fruit juice in the liquid consistency recommended by a clinician	Fresh fruits that are difficult to chew, have peels/skins/pits/seeds that are difficult to chew, or are hard and crunchy such as apples, grapes, or pears Stringy, high-pulp fruits, fibrous fruits such as papaya, pineapple, rhubarb or mango Fruits with high water content or juice that separates from solid in the mouth such as melons (unless assessed as safe by clinician) Round or long fruits such as grapes Uncooked dried fruits such as prunes, raisins or apricots Fruit leather, fruit roll-ups, fruit snacks, dried fruits
Oils	Any with a consistency recommended by a clinician	Liquid consistencies other than the liquid consistency recommended by a clinician

Beverages	Coffee, tea, water, 100% fruit juice, and nutritional supplements in the liquid consistency recommended by a clinician	Any with a consistency not recommended by a clinician
Other	Prepared foods, including casseroles, salads, baked goods, and snacks made from recommended ingredients and served in 1.5 cm x 1.5 cm pieces Pureed strained soups or soups with small soft pieces (1.5 cm x 1.5 cm) that meet testing guidelines All seasonings and sweeteners, including honey	Soups, stews, casseroles, and prepared foods that have pieces bigger than 1.5 cm x 1.5 cm and that contain liquids that separate from the solid foods Jelly

Use the blue "Related Client Ed" link to the right of the title of this page to access the Level 6 Soft & Bite-Sized Sample 1-Day Menu with nutrient analysis.

Level 5 Minced and Moist (Orange)

Definition: The Level 5 Minced and Moist diet is prescribed to people who are unable to bite foods, have pain or difficulty chewing foods, or easily tire when chewing foods. This diet requires a texture modification so that foods are minced, soft, and moist and can be scooped and easily shaped into a ball. Small lumps of food are allowed for adults if the lumps are easy to squash with the tongue and if they are no larger than 4 mm wide by 15 mm long. Foods on this diet should meet the following requirements:

- Minimal chewing necessary
- Easily mashed with gentle fork pressure
- Meets size requirements
- No separation of thin liquids

Foods should further meet the complete descriptive and testing specifications of the International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Food lists are based on the International Dysphagia Diet Standardization Initiative (IDDSI) Framework. Use IDDSI testing methods to confirm foods and drinks are safe.

The following table is not a complete list of foods. Other foods may be OK to eat as long as they meet the IDDSI food testing requirements

Food Group	Foods Recommended (Foods should be finely minced or mashed, no larger than 4 mm wide x 15 mm long, and pass the IDDSI test for Level 5)	Foods Not Recommended (Foods larger than 4 mm wide x 15 mm long)
Grains	All well-moistened, soft cooked hot cereals that meet the size requirements. Serve without excess liquid. Cold cereal (if approved by a clinician) that meets the size requirements, drained and served without	Any cereal that doesn't meet size requirements and is served with excess liquid. Regular-consistency, dry bread or toast. Sandwiches with regular-consistency bread.

	excess liquid. Pregelged, soaked breads that are consistently moist throughout with no hardened sections. Muffins, pancakes, waffles, biscuits and bread dressing served moist and that meet the size requirements. These foods may be best served puree. Soft cooked pasta and potato dishes that meet the size requirements. Serve in a thick and smooth sauce that does not separate from pasta. Rice, couscous, quinoa that is not sticky or glutinous and does not separate into individual grains when cooked. Served with a thick and smooth sauce that does not separate away from food.	Pasta and potato dishes with pieces that don't meet size requirements and are served in thin sauces. Grainy, sticky, or glutinous rice. Rice that separates into individual grains when cooked or served and is served dry without sauce.
Protein Foods	Note: If protein foods cannot be processed soft and at the appropriate size, puree the foods instead. Cooked, moist, soft and tender meats minced or finely mashed to meet the size requirements: Red meat, including beef, pork, sausage or lamb Poultry, including skinless chicken or	Protein foods that don't meet size requirements. Protein foods served with un- drained thin liquids. Chicken, turkey and fish with skin on or with bones. Fish with skin on or with bones; or that is not soft enough to be broken into size requirements. Chunky and smooth nut

SKINNED CHICKEN or turkey

Seafood, including fish (salmon, herring, and sardines), shrimp, lobster, clams, and scallops

Eggs and egg substitutes finely mashed to meet the size requirements.

Casseroles with small chunks of meat, ground meats, or tender meats finely mashed to meet the size requirements. Serve in a thick and smooth sauce or gravy, draining excess.

Prepared, moistened soy foods, such as tofu or tempeh finely mashed to meet the size requirements.

Prepared, moistened meat alternatives, such as plant protein veggie burgers and sausages finely mashed to meet the size requirements. Serve in a thick and smooth sauce or gravy, draining excess, if necessary.

Cooked, moistened legumes, such as dried beans, lentils, or peas without shells mashed to meet the size requirements. Serve in a thick and smooth sauce or gravy, draining excess.

seed butters, unless recipe meets requirements.

Whole nuts and seeds, such as peanuts and almonds; pistachios and sunflower seeds.

Dairy and Dairy Alternatives	Yogurt (without nuts or coconut), cottage cheese finely mashed to meet the size requirements. Cream cheese, sour cream, pudding, custard and whipped topping, with thickness adjusted as needed to meet the size requirement. Frozen desserts such as ice cream, sherbet, malts, and frozen yogurt if recommended by a clinician. Milk, fortified soymilk, fortified nut milk in the liquid consistency recommended by a clinician.	Yogurt with nuts or coconut. Hard, dry, chewy or sticky cheese. Cheese chunks that do not meet size requirements. Frozen desserts such as ice cream, sherbet, malts, and frozen yogurt, unless approved by a clinician.
Vegetables	Cooked, tender vegetables and potatoes finely mashed to meet the size requirements and with liquid drained. If indicated, serve in a thick and smooth sauce or gravy, draining excess. There should not be thin liquid separating from food. Peas, creamed peas and creamed corn with soft skins and free of husks; may be blended or incorporated into IDDSI safe stews and soups. If incorporated into a recipe, must pass IDDSI testing methods. Vegetable juices in the liquid consistency recommended by a	All raw vegetables including lettuce. Cooked vegetables larger than diet requirements. Cooked vegetables that are fibrous, tough, firm, or stringy such as green beans and peapods. Peas with tough skins or shells Corn with husks Sticky or crunchy potatoes such as potato/mashed potato casseroles with hard tops. Stir-fried, undercooked or fried vegetables.

	Summary	
Fruits	Canned and cooked fruits, drained of excess juices, served finely minced or mashed to meet the size requirement. If indicated, serve in a thick and smooth sauce or liquid, draining excess. There should not be thin liquid separating from food. Soft, peeled fresh fruits such as peaches, nectarines, kiwi, cantaloupe, and honeydew, soft berries with small seeds (ie, strawberries). If indicated, serve in a thick and smooth liquid, draining excess. There should not be thin liquid separating from food. 100% fruit juice in the liquid consistency recommended by a clinician.	Soft fruit pieces larger than required of diet, or fruit served with excess juice. Most fresh fruits with difficult-to-chew peels or pits/seeds such as grapes and apples; and difficult-to-chew or hard/crunchy fresh fruits such as apples or pears. Stringy, high-pulp fibrous fruits such as papaya, pineapple, rhubarb or mango unless prepared to meet diet requirements. Fruits with high water content, that juice separates from solid in the mouth such as watermelon unless approved by a clinician. Dried fruits such as raisins, prunes or apricots. Fruit leather, fruit roll-ups, fruit snacks, dried fruits.
Beverages	Coffee, tea, water, 100% fruit juice, and nutritional supplements in the liquid consistency recommended by a clinician.	Liquids not approved by a clinician.
Other	Prepared foods, including casseroles, cooked salads, baked goods, and snacks that meet the size requirement. If indicated, serve in a	Jelly

thick and smooth sauce, gravy or liquid, draining excess. There should not be thin liquid separating from food.

Pureed soups with small chunks that meet size requirement.

All seasonings and sweeteners, including honey.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Level 5 Minced and Moist Sample 1-Day Menu with nutrient analysis.

Level 4 Pureed (Green)

Definition: The Level 4 Pureed Food diet is prescribed to people who have pain when chewing or swallowing or are unable to bite or chew foods. This diet requires a texture modification so that foods are smooth and lump-free and foods should not be firm or sticky. Foods should fall off spoon as an intact spoonful, hold its shape on a plate, and liquid must not separate from solid. Foods do not require chewing or bolus formation. Foods and liquids should further meet the complete descriptive and testing specifications of International Dysphagia Diet Standardization Initiative (IDDSI, 2019a; IDDSI, 2019b).

Food lists are based on the International Dysphagia Diet Standardization Initiative (IDDSI) Framework. Use IDDSI testing methods to confirm foods and drinks are safe.

The following table is not a complete list of foods. Other foods may be OK to eat as long as they meet the IDDSI food testing requirements.

Food Group	Foods Recommended	Foods Not Recommended
Grains	Pureed soft-cooked hot cereals with no lumps. Served without excess liquid. Soft breads, rolls, pastries, pancakes, French toast, muffins, donuts and bread dressing that have been pureed. Pre-gelled, soaked bread, cakes, cookies, and other grains that are consistently moist throughout without hard parts formed during sitting. Pureed, moist pasta, potatoes, and rice without lumps. Liquids/sauces do not separate from food.	Grains Any item that is not pureed or has lumps. Dry cereal, cooked cereal with lumps, cereal with seeds. Grainy, sticky, or glutinous rice. Rice that separates into individual grains when cooked or served. Pre-gelled, shaped, and molded puree foods that are too firm or sticky at serving temperature.

Protein Foods	Pureed, moistened, tender protein foods that meet IDDSI Level 4 puree testing expectations: Red meat, including beef, pork, or lamb. Poultry, including skinless chicken or turkey. Seafood, including fish (salmon, herring, and sardines), shrimp, lobster, clams, and scallops. Pureed eggs and egg substitutes. Pureed, smooth casseroles with no liquid separating from the solid; moist with incorporated sauces/gravies. Pureed, moistened soy foods, such as tofu or tempeh. Pureed, moistened meat alternatives, such as veggie burgers, and sausages based on plant protein. Pureed, smooth, moistened legumes, such as dried beans, lentils, or peas.	Protein foods not pureed into smooth, lump free items. Protein foods served with undrained thin liquids. Chicken, turkey and fish with skin on or with bones. Chunky and smooth nut seed butters, unless used in a recipe that is pureed and meets testing expectations. Whole nuts and seeds, such as peanuts and almonds; pistachios and sunflower seeds.
Dairy and Dairy Alternatives	Smooth yogurt (without nuts or coconut) or pureed; and pureed cottage cheese. Whipped cream cheese, sour cream,	Yogurt with lumps, seeds, fruit pieces, nuts or coconut; yogurt that is too thin, separates into liquid or is too thick. Cheeses unless pureed into allowed recipe.

	and whipped topping used in allowed recipes and as condiments. Frozen desserts such as ice cream, sherbet, malts, and frozen yogurt if recommended by your clinician. Milk, fortified soy milk, fortified nut milk in the liquid consistency recommended by your clinician.	Frozen desserts such as ice cream, sherbet, malts, and frozen yogurt unless approved by your clinician.
Vegetables	Pureed cooked tender vegetables and potatoes. If indicated, serve in a thick and smooth sauce or gravy, draining excess. There should not be thin liquid separating from food. Smooth tomato sauce without seeds. Mashed potatoes and whipped sweet potatoes without skin Vegetable juices in the liquid consistency recommended by your clinician.	All raw vegetables. Stir-fried or fried vegetables that do not puree into smooth, lump free product.
Fruit	Pureed canned and cooked fruits, drained of excess juices; pureed fresh fruit if smooth and lump free with no separate of liquids. 100% fruit juice in the liquid consistency recommended by your clinician	All non-pureed fruits; seeds and skins. Stringy, high-pulp fruits such as papaya, pineapple, or mango that do not puree into smooth, lump free product. Uncooked dried fruits such as raisins, prunes, or apricots.

	Smooth, lump free pureed prunes and apricots that pass IDDSI puree testing expectations.	Fruit leather, fruit roll-ups, fruit snacks, dried fruits.
Beverages	Coffee, tea, water, and nutritional supplements in the liquid consistency recommended by your clinician.	Liquids not approved by your clinician.
Other	Pureed foods, including all soups with tender meats, casseroles, baked goods, and snacks made from recommended ingredients that are smooth and lump free. All seasonings and sweeteners; honey if mixed into food that passes IDDSI testing expectations.	Jelly

Use the blue ["Related Client Ed"](#) link to the right of the title of this page to access the **Level 4 Pureed Sample 1-Day Menu with nutrient analysis**.

Thickened Liquids

Thickeners help to slow the movement of liquids/drinks, allowing patients to have better control over their swallow. All liquids/drinks should be thickened to meet the prescribed order.

- Thickened liquids/drinks may be purchased pre-mixed or may be made by adding commercial thickeners.
- Puree all soups and strain the liquid to remove chunks or lumps. Once these liquids are smooth, thicken them with flour, cornstarch, potato flakes, or commercial thickeners to the prescribed order.
- Beverages such as milk, juices without pulp, coffee, tea, soda, carbonated beverages, alcoholic beverages, eggnog and nutritional supplements should be thickened to the prescribed order.
- Beverages and supplements made with ice cream, frozen yogurt, or sherbet such as malts and milk shakes are not recommended. These beverages become thin liquids at body temperature (98°F).
- Frozen foods such as sherbet, frozen yogurt, and ice cream are not recommended. These foods become thin liquids at body temperature (98°F).
 - Some ice cream and frozen products are designed to meet IDDSI guidelines for liquid/drink levels.
 - These products are labeled including the level number and color.
- Gelatin is not recommended. It becomes thin liquid at body temperature (98°F).
- Yogurt may need to be thickened to the prescribed order.

Utilize IDDSI Thickened Liquid Nutrition Therapy client education handout and individualize with IDDSI Consumer Handouts or per facility policies and procedures.

Use the following table to familiarize yourself with the IDDSI guidelines for liquid/drink levels, descriptions, and testing methods.

IDDSI Liquid/Drink Levels and Testing Methods		
Food/Drink Level	Description	Testing Method
	Smooth, lump free; not firm or sticky; does not require chewing or bolus formation. Falls off spoon as intact spoonful and holds shape. No mixed consistency. Liquids cannot be poured, drank from a cup; or sucked through a	Fork Drip Test: no lumps; stays as mound on fork with very little/no flow through prongs. Spoon Tilt Test: Holds shape; not firm or sticky; sample slides off easily. If it is too thick, sample will not fall off

Level 4 Extremely thick liquid (Green)	straw.	spoon when tilted or will stick to spoon. Fork Test: When fork is pressed into the food sample, tines make pattern and/or indentation stays. IDDSI Flow test not applicable to this level.
Level 3 Moderately thick liquid (Yellow)	Smooth foods and fluids without lumps; does not hold shape on plate; requires no chewing or oral processing and can be directly swallowed. Drips slowly through slots of fork. Requires moderate effort to suck through straw.	IDDSI Flow Test: See instructions in IDDSI Testing Methods Fork Drip Test: Drips slowly in dollops; spreads out if spilled onto flat surface (plate). Fork Test: When fork is pressed into the food sample, the tines do not leave a clear pattern. Spoon Tilt Test: Easily pours; no sticking to spoon.
Level 2 Mildly thick liquid (Pink)	Flows off spoon; pours slower than thin drinks; Sippable; some effort needed for straw use; May be suitable if tongue control slightly reduced	IDDSI Flow Test: See instructions in IDDSI Testing Methods

Level 1 Slightly thick liquid (Grey)	Thicker than water and flows through straw, syringe or nipple; similar to infant formula. Little more effort required to drink than thin liquids.	IDDSI Flow Test: See instructions in IDDSI Testing Methods
Level 0 Thin liquids (White)	Flows like water; can drink through any teat/nipple, cut or straw per age/skill	IDDSI Flow Test: See instructions in IDDSI Testing Methods

Level 3 Liquidized Food (Yellow)

Definition: A diet used in the dietary management of dysphagia with food texture modification described as smooth foods without lumps that cannot be molded on a plate, require no oral processing or chewing, can be directly swallowed, and further meet the complete descriptive and testing specifications of International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Refer to Using IDDSI Testing Methods for additional information and IDDSI testing methods.

Level 4 Extremely Thick Liquid (Green)

Definition: A diet used in the dietary management of dysphagia with liquid thickness modification described as liquids that show slow movement under gravity, cannot be poured, cannot be drunk from a cup or sucked through a straw, liquid must not separate from solid, and further meet the complete descriptive and testing specifications of International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Refer to Thickened Liquids for additional information and IDDSI testing methods.

Level 3 Moderately Thick Liquid (Yellow)

Definition: A diet used in the dietary management of dysphagia with liquid thickness modification described as liquids that are smooth in texture, no lumps or fibers, and drips slowly in strands through slots of a fork. Requires some effort to suck through a standard or large bore straw, and further meets the complete descriptive and testing specifications of International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Refer to Thickened Liquids for additional information and IDDSI testing methods.

Level 2 Mildly Thick Liquid (Pink)

Definition: A diet used in the dietary management of dysphagia with liquid thickness modification described as liquids that are sippable, flow off a spoon at a slower rate than thin liquid, requires effort to suck through a standard straw, and further meet the complete descriptive and testing specifications of International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Refer to Thickened Liquids for additional information and IDDSI testing methods.

Level 1 Slightly Thick Liquid (Grey)

Definition: A diet used in the dietary management of dysphagia with liquid thickness modification described as liquids that are thicker than water, flows through a straw or syringe or nipple, similar in thickness to commercially available anti-regurgitation infant formula, and further meet the complete descriptive and testing specifications of International Dysphagia Diet Standardisation Initiative (IDDSI, 2019a; IDDSI, 2019b).

Refer to Thickened Liquids for additional information and IDDSI testing methods.

Clear Liquid Diet

Definition: The clear liquid diet is composed of fluids and foods that are transparent and that are liquid at body temperature. Foods allowed in this diet primarily consist of sugar and water or water and salt. Portion sizes may be modified to meet the needs of postoperative bariatric surgery patients.

Indication: The clear liquid diet is used to maintain hydration while minimizing residue in the bowel in the following situations: acute gastrointestinal illness, preparation for surgery or procedure, recovery from surgery, or food reintroduction after procedure. This diet is intended for short-term use and should be used only when absolutely necessary. For individuals with chewing or swallowing difficulties and who have a thickened liquid diet order, refer to the IDDSI Levels section. For individuals with diabetes, blood glucose levels must be closely monitored. Medication and insulin may require adjustment while following a clear liquid diet.

Nutritional Adequacy: The clear liquid diet does not meet the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients. The sample meal plan provides 200 g carbohydrate per day, which aligns with the recommended carbohydrate amount for patients with diabetes on a clear liquid diet (American Diabetes Association, 2008). Individualize the patient's meal plan to meet their nutrition needs. Clear liquid, fortified beverages may improve protein intake while on a clear liquid diet. Patients receiving this diet are monitored closely so they can progress to solid foods (or nutrition support, where necessary) as rapidly as tolerated.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Clear Liquid Diet Sample 1-Day Menu with nutrient analysis and Foods List.

References

American Diabetes Association. Nutrition recommendations and interventions for diabetes: A position statement of the American Diabetes Association. Diabetes Care 2008;31(Supplement_1):S61–S78. Full text.

Full Liquid Diet

Definition: The full liquid diet includes the foods allowed on the clear liquid diet plus milk and foods with a minimal amount of fiber. Thin cereal, strained cream soups, milkshakes, custard, pudding, and juices/nectars that do not contain pulp are appropriate for this diet. Nondairy substitutes are indicated for patients with lactose-intolerance. Portion sizes and specific nutrients (eg, carbohydrates, sodium, fat) may be modified to meet the needs of postoperative bariatric surgery patients and individuals with specific medical conditions.

Indication: A full liquid diet has historically been used as a short-term transition step between a clear liquid diet and a soft diet following gastrointestinal surgery. However, this practice has become less common and a more individualized approach for postoperative diet progression is often utilized. A full liquid diet may be indicated for individuals with impaired gastrointestinal function. For individuals with chewing or swallowing difficulties and who have a diet order for thickened liquids or Level 3 Liquidized Foods, refer to the IDDSI Levels section.

Nutritional Adequacy: The full liquid diet meets the Dietary Reference Intakes (DRIs) for macronutrients but may be inadequate for vitamins, minerals, and fiber. The sample meal plan provides 200 g carbohydrate per day, which aligns with the recommended carbohydrate amount for patients with diabetes on a full liquid diet (American Diabetes Association, 2008). Individualize the patient's meal plan to meet their nutrition needs. Oral nutrition supplements to meet nutrient needs may be ordered for patients who require a full liquid diet for more than 3-4 days.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Full Liquid Diet Sample 1-Day Menu with nutrient analysis and the Foods List.

References

American Diabetes Association. Nutrition recommendations and interventions for diabetes: A position statement of the American Diabetes Association. *Diabetes Care* 2008;31(Supplement_1):S61–S78. Full text.

Consistent Carbohydrate Diet

Definition: A consistent carbohydrate diet provides the same amount of total carbohydrate daily, distributed evenly across all meals and snacks. A consistent amount of carbohydrate at each daily meal is the goal. The exact amount and type of carbohydrate served is individualized by the registered dietitian nutritionist (RDN).

The name of the diet used in your facility should no longer emphasize the restriction of sugar or sweets but rather emphasize consistent carbohydrates or carbohydrate controlled (refer to the list of Obsolete Diets as you determine the appropriate nomenclature for your facility). Establish a general, consistent carbohydrate meal plan for your facility and individualize for patients when possible. To monitor the carbohydrate content of meals, count carbohydrates with the traditional exchange system or with assistance from computer-generated menu programs.

A consistent carbohydrate diet requires initial and ongoing collaborative planning with the patient to incorporate their individualized goals, preferences, and any comorbidities. Together, designing a healthy eating pattern that a person with diabetes can sustain is the most effective intervention.

Components of a Consistent Carbohydrate Diet

- There is no ideal amount of carbohydrate recommended for everyone (Evert, 2019; Briggs, 2018; ADA 2022).
- The quantity of carbohydrate is determined by the RDN's assessment (Evert, 2019; Briggs, 2018; ADA, 2022).
- The consistent carbohydrate diet provides a range of three to five carbohydrate servings (45 g-75 g) at each meal along with zero to two carbohydrate servings (0 g-30 g) with snacks. The following ranges are often used as a starting point in meal planning for people with diabetes:

Carbohydrate Servings for Meals/Snacks

Approximate Energy (kcal)	Carbohydrate Servings Per Meal Per Day	Carbohydrate Servings Per Snack Per Day (if recommended)
1,200-1,500 (for weight loss)	3 (45 g)	1 (15 g)
1,600-2,000 (for weight control)	4 (60 g)	1-2 (15-30 g)
2,100-2,400 (for active individuals)	5 (75 g)	1-2 (15-30 g)

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- The carbohydrate amount can be adjusted based on long-term and blood glucose goals. The RDN is responsible for using this information about the patient's clinical condition, meal planning, and lifestyle habits for establishing treatment goals to determine a realistic plan for nutrition therapy (ADA, 2022; Evert, 2019; Briggs, 2018).
 - Carbohydrate "counts" are based on the exchange system for diabetes meal planning where 1 carbohydrate serving = 15 g carbohydrate. Foods with 6 g-10 g carbohydrate per serving are considered a 0.5 carbohydrate serving and should be included in the total for the meal.
 - The amount of carbohydrate is divided between meals and snacks. Per-meal and per-snack quantities may vary but are consistent day to day for individual meals. For example, carbohydrate content at breakfast is the same day to day but may be more or less than the amount of carbohydrate at lunch and dinner. Patients should consume each daily meal at the same time.
 - No foods are omitted from the consistent carbohydrate meal plan unless required by a comorbidity. Options for reducing sodium or fat are frequently added to meal plans and the diet order when there is a comorbidity.
 - Regular desserts should be allowed on a consistent carbohydrate meal plan however it is important to control added sugars (ADA, 2022; Evert, 2019). If the patient/client prefers to eat sugar-sweetened items, portion sizes can be modified to meet carbohydrate goals.
 - Using sugar substitutes to replace calorie sweeteners, without caloric compensation, may be useful in reducing energy and carbohydrate intake; however, research is limited. There is insufficient evidence that consuming artificially sweetened beverages leads to long-term weight loss (Evert, 2019). Water should be encouraged in place of sugar sweetened beverages including fruit juice and non-nutritive sweetened beverages (ADA, 2022).
 - All sources of carbohydrates should be counted when meal planning, including starches, grains, fruits, dairy, legumes/pulses, vegetables, sauces, commercial thickeners (for dysphagia), breadings, desserts, and snacks. It is recommended to increase sources of fiber, emphasizing intake of non-starchy vegetables, and decrease consumption of refined grains, added sugars and processed foods. Carbohydrate intake from vegetables, fruits, whole grains, legumes, and dairy products should be emphasized over carbohydrate sources that contain added fats, sugars, and sodium (ADA, 2022; Evert, 2019; Briggs, 2018).
 - The Exchanges for Diabetes Meal Planning Chart and NCM Choices for Diabetes Caloric Intake Calculator are helpful resources in menu planning. See Choose Your Foods: Foods Lists for Diabetes for comprehensive exchange lists to use for individualization of patient diets.

Exchanges for Diabetes Meal Planning Chart				
Food Group	Carbohydrate (g)	Protein (g)	Fat (g)	Energy (kcal)
Starches (approx. ½ cup or 1-ounce slice bread)	15	3	0-1	80
Fruit (one small fruit or ½ cup)	15	0	0	80
Milk (1 cup)				
Skim	12-15	8	0-1	90
Low Fat	12-15	8	5	120
Whole	12-15	8	8	150
Other carbohydrates	15	Varies	Varies	–
Vegetables (½ cup cooked, 1 cup raw)	5	2	–	25
Meat (approx. 1 oz)				
Very lean	–	7	0-1	35
Lean	–	7	3	55
Medium Fat	–	7	5	75
High Fat	–	7	8	100
Fat	–	–	5	45

Indication: A consistent carbohydrate diet is indicated for individuals diagnosed with diabetes, prediabetes, or obesity. This diet may also be ordered to accommodate those who choose to follow a controlled or lower-carbohydrate meal pattern or for people who experience temporary elevated blood glucose levels related to acute medical conditions.

Although no specific dietary pattern is recommended for the management of diabetes, controlling the amount and type of carbohydrate consumed has been found to be helpful in managing blood glucose and reducing hemoglobin A1c levels (ADA, 2022; Evert, 2019). Many House diets meet the requirements for consistent carbohydrate meal plans, and RDN modification of the House diet may be sufficient to meet an individual patient's needs. See Planning the House

Diet for more information. The liberalization of the diet order to House may be appropriate for some older adult patients since intensive glycemic, blood pressure, and lipid control may not be warranted (ADA, 2022; Briggs, 2018; Dorner, 2018).

Consistent carbohydrate meal plans are preferred by many health care facilities as they match the prandial insulin dose to the amount of carbohydrate consumed, improving blood glucose control (ADA, 2022). Patients may also prefer to follow a consistent carbohydrate diet regardless of the presence of diabetes or hyperglycemia. Individuals with uncontrolled blood glucose levels, elevated hemoglobin A1c levels, or those receiving insulin may benefit from this diet while at a health care facility (Evert, 2019; ADA, 2022). The RDN should collaborate with the patient to individualize a meal plan that controls the amount and type of carbohydrates and to determine the best nutrition care plan (Briggs, 2018).

Nutritional Adequacy: The consistent carbohydrate diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Carbohydrate Counting for People with Diabetes Sample 1-Day Menu with nutrient analysis.

References

Briggs Early K, Stanley K. Position of the Academy of Nutrition and Dietetics: The role of medical nutrition therapy and registered dietitian nutritionists in the prevention and treatment of prediabetes and type 2 diabetes. *J Acad Nutr Diet.* 2018;118(2):343-353. Full text.

Evert AB, Dennison M, Gardner CD, et al. Nutrition therapy for adults with diabetes or prediabetes: a consensus report. *Diabetes Care.* 2019;42(5):731-754. Full text.

American Diabetes Association. Standards of medical care in diabetes – 2022. *Diabetes Care.* 2022;45(Suppl 1). Full text.

Dorner B, Friedrich EK. Position of the Academy of Nutrition and Dietetics: Individualized nutrition approaches for older adults: Long-term care, post-acute care, and other settings. *J Acad Nutr Diet.* 2018;118(4):724-735. Full text.

Heart-Healthy (Cardiac) Diet

Definition: The heart-healthy diet limits the amount of saturated fat (5% to 6% of total energy) and sodium (2000-3000 mg per day) and encourages foods containing unsaturated fats and unprocessed foods low in added sugars (AHA, 2021; EAL, 2017, Jacobson, 2015). The registered dietitian nutritionist (RDN) should individualize the sodium intake (EAL, 2017). Foods with trans fats are eliminated or limited as much as possible (USDA, 2020). A heart-healthy diet encourages vegetables, fruits, legumes, nuts, whole grains, low-fat or fat-free dairy, and fish (Lichtenstein, 2021; Arnett, 2019; Jacobson, 2015). A heart-healthy diet includes 22- 38 g total fiber per day and 5-10 g viscous fiber per day (USDA, 2020; Jacobson, 2015).

Indication: The heart-healthy diet is indicated for individuals with cardiovascular disease such as coronary artery disease, coronary artery bypass graft, myocardial infarction, stroke, dyslipidemia, and heart failure. This diet should be individualized by the RDN.

Nutritional Adequacy: The heart healthy diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Heart Healthy Sample 1-Day Menu with nutrient analysis and the Foods List.

References

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Lichtenstein AH, Appel LJ, Vadiveloo M, et al. 2021 Dietary Guidance to Improve Cardiovascular Health: a scientific statement from the American Heart Association. *Circulation*. 2021;144(23):e472-e487. Full text.

US Department of Agriculture, US Department of Health and Human Services. Dietary Guidelines Advisory Committee. Dietary Guidelines for Americans, 2020-2025. Updated 2020. Accessed March 24, 2022. Available at: Website

Sodium-Restricted Diet

Definition: A sodium-restricted diet is composed of foods and fluids with limited sodium content. These foods may be naturally low in sodium, or they may have been produced specifically to contain less sodium. Foods allowed in this diet consist of whole, unprocessed foods, as sodium is often high in processed foods. Many facilities will modify the House diet to meet the nutritional requirements for the sodium-restricted diet.

The 2020-2025 US Dietary Guidelines recommend limiting sodium to less than 2,300 mg per day (USDA, 2020). Reduction in sodium intake to 2000 mg to 3000 mg sodium per day has been effective in patients with heart failure (EAL, 2017). The registered dietitian nutritionist (RDN) should individualize the sodium intake (EAL, 2017). The degree of sodium restriction should be based on many factors including age, medical conditions, previous eating habits, willingness to change, and appropriateness of diet liberalization. A more liberalized approach to sodium intake may benefit older adults.

Many institutions offer diet orders with varying degrees of sodium restriction. The most liberal sodium diet is the no-salt-packet diet, which removes the salt packet or salt shaker from trays and tables. The no-added-salt (NAS) (3 g to 4 g sodium) diet limits high-sodium foods and eliminates the salt packet. The low-sodium diet (1.5 g to 2.3 g sodium) restricts some high-sodium foods and replaces other higher-sodium foods with lower-sodium items. Salt substitutes should not be offered without a physician’s order because of their potassium content.

Indication: The sodium-restricted diet is indicated for individuals with cardiovascular disease, heart failure, kidney disease, or edema. The sodium restriction will vary depending on medical conditions and clinical status.

Nutritional Adequacy: The sodium-restricted diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Types of Sodium-Modified Diets

In the setting of a health care facility, there are multiple ways to meet individual needs for reduced sodium intake.

Name of Diet	Diet Description
No-salt-shaker or no-salt-packet	This diet is the most liberal sodium-restricted diet to use when a salt restriction is indicated. This diet uses the House diet/menu and eliminates the presence of salt packets on meal trays and salt shakers at the table. The exact amount of sodium on this diet will vary depending on which foods are on the menu and whether the foods are processed or fresh. This diet may help improve meal

	acceptability and intake.
No added-salt (NAS)	The NAS diet restricts sodium to 3 g-4 g or 3000 to 4000 mg sodium per day. This diet limits high-sodium foods and eliminates the salt packet from trays. This diet is more palatable than the low-sodium diet may help improve meal acceptability and intake.
Low-sodium	This diet restricts sodium to 1.5 -2.3 g or 1500 -2300 mg per day. This diet requires substitution of standard menu items with low-sodium options and eliminates the salt packet from trays to achieve sodium limitation goals.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Low Sodium Sample 1-Day Menu with nutrient analysis and the Foods List. Access the Sodium (Salt) Content of Foods handout to view the amount of sodium in foods.

References

Academy of Nutrition and Dietetics, Evidence Analysis Library. Heart failure. Updated 2017. Accessed August 5, 2020. Available at: Website

US Department of Agriculture, US Department of Health and Human Services. Dietary Guidelines Advisory Committee. Dietary Guidelines for Americans, 2020-2025. Updated 2020. Accessed March 24, 2022. Available at: Website

Fluid-Restriction Diet

Definition: The fluid-restriction diet order limits an individual's daily fluid intake. Most foods are allowed in this diet, but extra planning is required to coordinate appropriate serving sizes of foods and liquids throughout the day. Note: items that are traditionally considered "foods" may be considered liquids in this diet.

A no-free-water, fluid restriction diet order limits or eliminates liquids that are mostly composed of water, including plain water, coffee, tea, and tonic.

Indication: A fluid restriction diet is indicated in individuals with congestive heart failure, hypertension, fluid retention, pulmonary edema, hyponatremia, and renal failure (Ikizler, 2020; Patel, 2020). Fluid restriction may be ordered in addition to a sodium restriction to help maintain fluid balance. A no-free water restriction diet order is often ordered for hyponatremia to provide a more liberalized approach.

Nutritional Adequacy: A fluid-restriction diet order may be nutritionally inadequate. Refer to the diet order to determine if the diet is adequate in energy, macronutrients, and micronutrients.

Counting Liquids and Foods for a Fluid Restriction Diet

A fluid is anything that is liquid or any foods that melt at room temperature. These foods and liquids must be counted as part of daily fluid intake. Examples of fluids include the following:

Fluids and Beverages	Foods with High Water Content/Foods That Melt
• Coffee, tea, and other hot beverages • Ice cubes, ice chips (1 cup = ½ cup fluid = 4 oz) • Milk, liquid creamer • Oral nutritional supplements • Soft drinks, lemonade, limeade • Vegetable and fruit juices • Juice/water from canned fruit • Syrups	• Gelatin (½ cup = ½ cup fluid = 4 oz) • Watermelon • Gravy • Ice cream, sherbet, sorbet (½ cup = ½ cup fluid = 4 oz) • Popsicles (1 twin stick = ¼ cup fluid = 2 oz) • Yogurt • Pudding • Soups

Fluid restrictions are ordered in milliliters or liters but often converted to household measurements as these units tend to be more familiar.

Fluid Volume Equivalents

2 tablespoons	1 ounce	30 mL
$\frac{1}{4}$ cup	2 ounces	60 mL
$\frac{1}{2}$ cup	4 ounces	120 mL
$\frac{3}{4}$ cup	6 ounces	180 mL
1 cup	8 ounces	240 mL

Tips for Processing a Diet Order with a Fluid Restriction

Determine how to distribute total fluids throughout the day:

- Determine the amount of fluid required for distribution of medications, nourishments/snacks, and liquid oral nutritional supplements for 24 hours/3 shifts. Consider the individual's preferences or usual routine at home (if applicable).
- Subtract this amount of fluid from total daily fluid allowed to establish amount reserved for dietary trays (mealtime).
 - Example: 1500 mL fluid restriction – 500 mL for medications = 1000 mL for dietary trays
- Confirm diet order with health care team and patient per facility policy to determine amount of fluid available for each nursing shift and for the dietary meal trays.
 - Example: 1500 mL fluid restriction: 1000 mL dietary trays and 500 mL nursing/snacks as 200 mL 7am-3pm; 200 mL 3pm-11pm; 100 mL 11pm-7am.

Options for Fluid Restriction Diet Orders

The following tables provide options for how to distribute fluids throughout the day based on fluid restriction order and dietary and nursing fluid amounts.

1000 mL Fluid Restriction

Meal	Option 1	Option 2	Option 3
Breakfast	8 ounces/240 mL 4 ounces/120 mL	8 ounces/240 mL	8 ounces/240 mL 4 ounces/120 mL

Lunch	8 ounces/240 mL	8 ounces/240 mL	4 ounces/120 mL 6 ounces/180 mL (soup)
Evening meal	8 ounces/240 mL	8 ounces/240 mL	4 ounces/120 mL
Dietary Fluid Amount =	840 mL	720 mL	780 mL
Nursing Fluid Amount =	160 mL	280 mL	220 mL

1200 mL Fluid Restriction

Meal	Option 1	Option 2	Option 3
Breakfast	8 ounces/240 mL 4 ounces/120 mL	8 ounces/240 mL	8 ounces/240 mL 4 ounces/120 mL
Lunch	8 ounces/240 mL	8 ounces/240 mL	4 ounces/120 mL
Evening meal	8 ounces/240 mL	8 ounces/240 mL 6 ounces/180 mL (soup)	8 ounces/240 mL
Dietary Fluid Amount =	840 mL	900 mL	720 mL
Nursing Fluid Amount =	360 mL	300 mL	480 mL

1500 mL Fluid Restriction

Meal	Option 1	Option 2	Option 3
Breakfast	8 ounces/240 mL 4 ounces/120 mL	8 ounces/240 mL 4 ounces/120 mL	8 ounces/240 mL 8 ounces/240 mL
Lunch	8 ounces/240 mL 4 ounces/120 mL	8 ounces/240 mL 4 ounces/120 mL	8 ounces/240 mL 8 ounces/240 mL
Evening meal	6oz/180mL (soup) 4 ounces/120 mL	8 ounces/240 mL 4 ounces/120 mL	8 ounces/240 mL

Dietary Fluid Amount =	1020 mL	1080 mL	1200 mL
Nursing Fluid Amount =	480 mL	420 mL	300 mL

Refer to the interactive NCM Metric Converter Calculator for a convenient unit conversion tool.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Fluid-Restricted Sample 1-Day Menu with nutrient analysis. Access the Water Content Of Common Foods And Beverages handout to view the amount of water in foods and beverages.

References

Ikizler TA, Burrowes JD, Byham-Gray LD et al. KDOQI Nutrition in CKD Guideline workgroup. KDOQI clinical practice guideline for nutrition in CKD: 2020 Update. Am J Kidney Dis. 2020;76(3)(suppl 1):S1-S107. Available at: Website

Patel Y, Joseph J. Sodium intake and heart failure. Int J Mol Sci. 2022;21(24):9474. Full text.

Diet for Chronic Kidney Disease

Definition: The diet for chronic kidney disease (CKD) modifies specific nutrients based on individual needs, stage of kidney disease, clinical data, and treatment modalities. The diet for individuals with CKD needs to be individualized (Gonyea, 2022, Ikizler, 2020, EAL, 2020). Nutrients that may be modified in a diet for CKD include protein, sodium, potassium, phosphorous, and fluids. The specific nutrients needing modification must be identified in the diet order. Other medical conditions may indicate additional diet considerations, such as consistent carbohydrate for individuals with CKD who also have diabetes.

Indication: The diet for CKD is indicated for individuals with CKD. The registered dietitian nutritionist (RDN) will determine the need for modification in protein, sodium, potassium, phosphorous, and fluid. Evidence of hyperkalemia, hyperphosphatemia, and edema may indicate a decreased intake of potassium, phosphorus, sodium, and fluid is necessary.

Nutritional Adequacy: The diet for CKD may meet the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients. Diet modifications may intentionally restrict macronutrients and micronutrients to thresholds below DRIs.

Individualizing the CKD Diet: A diet for CKD may be ordered upon admission. It is expected that the diet order will be individualized by the RDN.

A preliminary diet order before the RDN is able to perform an assessment may comprise the following:

- Protein: 80 g
- Phosphorus: 1000-1200 mg
- Potassium: 2000-2500 mg
- Sodium: 2300 mg

A house diet or a low-sodium diet can be modified to meet the nutritional requirements for a diet for CKD. An alternative option is a no-added-salt diet order that also modifies the amount of allowed potassium in foods (such as orange juice, bananas, tomatoes, and potatoes).

The following table is a guide for planning a diet for CKD which modifies (or controls) sodium, potassium, and phosphorous.

Food Group	Foods to Choose	Foods to Limit
Grains	Whole grain cereal Oats, oatmeal Whole wheat bread, pita	Grains with more than 200 milligrams of sodium per serving Boxed biscuit, cake,

	English muffin Corn tortillas Whole wheat pasta Brown rice Quinoa Couscous Grits Popcorn Rice cakes Whole wheat crackers	pancake/waffle mixes and other convenience foods Snacks and sweets should be eaten in moderation
Protein Foods	Eggs or egg whites Lean beef, wild game, and “all natural” chicken, fish, pork, seafood, or turkey Legumes/Pulses: Beans (such as black, kidney, or white beans), lentils, split peas, black-eyed peas Soy: Tofu, edamame Nuts and nut butters	Protein with more than 200 mg sodium per serving Processed or frozen protein foods Salty processed meats (such as bacon, bologna, salami and other lunch meats), ham, hot dogs, sausage, breakfast sausage, and pre-seasoned meats
Dairy	Low-fat or fat-free milk (may need to be limited) Unfortified dairy alternates such as almond or rice beverages, soy beverage Cottage cheese with no “phos” ingredients Yogurt (all-natural or plain preferred) Natural cheese such as feta, Swiss, cheddar, or mozzarella Regular or low-fat cream	High sodium cheeses or cheese sauces Non-dairy creamers and half and half creamers that have phosphorus containing additives Enriched dairy alternates such as almond, oat or rice beverages Processed cheese, such as American Processed cheese spreads and dips

	cheese, Neufchatel, or sour cream Sherbet, sorbet, fruit ice, or popsicles	Fat free cream cheese or sour cream Ice cream, pudding or frozen yogurt Milk-based or cheese-based soups or sauces
Vegetables	Fresh, frozen, or no-salt added canned vegetables Lower-potassium vegetables: Asparagus, beans (green, wax, yellow), bean sprouts, broccoli, cabbage (red or green), carrots, cauliflower, celery, corn, cucumbers, eggplant, greens (kale, mustard greens, endive, watercress), peppers (green, red, orange), kale, lettuce (all types), leeks, mushrooms, onions, peas (green, sugar snap, snow peas), radishes, summer squash, turnip	Processed vegetables or vegetable juice with more than 200 milligrams sodium per serving. Pickled foods, such as olives, sauerkraut, pickles, kimchi Vegetables with added sauces Higher-potassium vegetables: Artichokes, avocado, bamboo shoots (raw), beets (canned, fresh), brussels sprouts, chard (cooked), Chinese cabbage/bok choy (cooked), corn, kohlrabi, parsnips, potatoes (white, sweet, yams), pumpkin, rutabagas, spinach, squash (winter, acorn, butternut, rhubarb), tomatoes (raw, boiled, canned, purees, sauces), vegetable or tomato juice
Fruits	Fresh, frozen, or canned fruit Lower-potassium fruits: Apples, applesauce, berries (blackberries blueberries cranberries, raspberries, strawberries), cherries, clementine, dried fruits (apples, blueberries, cherries, cranberries), fruit cup, grapes, grapefruit, lemon, limes, plum, pear,	Fresh, frozen, or canned fruit Higher-potassium fruits: Bananas, dried fruits (dates, figs, prunes, raisins, currants), kiwi, honeydew melon, cantaloupe, nectarine, orange, papaya, peach, plantain, pomegranate Fruit juice: pomegranate, prune, orange, carrot

	watermelon 100% fruit juice: apple, cranberry, grape, pineapple Nectars: mango, papaya, peach, pear	
Oils	Healthy fats such as olive oil, vegetable oil or lower sodium salad dressings Butter, margarine, mayonnaise, and sour cream in moderation	Dressing, condiments, and other sauces with more than 200 mg sodium per serving
Beverages	Water Coffee Tea Lemonade Seltzer	Processed beverages (such as most colas, sports drinks, energy drinks, some flavored waters, drink mixes., some bottled teas and others). Check for added potassium and phosphorus.
Other	Herbs, spices, lemon juice, vinegars to flavor food instead of salt Bouillon, broths, canned soups, condiments, and sauces with less than 200 mg per serving Stocks or broths labeled as "no salt added" Check for added potassium as a replacement for sodium. Prepared frozen meals with < 600 mg sodium per serving	Salt, and salt substitutes Bouillon, broths, canned soups, condiments, and sauces with more than 200 mg per serving Broths and stocks labeled as "low sodium" Prepared frozen meals with > 600 mg sodium per serving

Use the blue "Related Client Ed" link to the right of the title of this page to access the Chronic Kidney Disease Stage 3-5 Nutrition Therapy Sample 1-Day Menus with nutrient analysis. The Chronic Kidney Disease Stage 5 (on Dialysis) Sample 1-Day Menu has about 80 grams of protein. The Chronic Kidney Disease Stage 5 (Not on Dialysis) Sample 1-Day Menu has about 60 grams of protein. The link

can also be used to access the Potassium Content of Foods and the Phosphorus Content of Foods handouts.

References

Academy of Nutrition and Dietetics Evidence Analysis Library. Chronic Kidney Disease. 2020. Available at: Website

Gonyea JE, Phillips SC, eds. Clinical Guide to Nutrition Care in Kidney Disease. 3rd ed. Academy of Nutrition and Dietetics; 2022.

Ikizler TA, Burrowes JD, Byham-Gray LD et al. KDOQI Nutrition in CKD Guideline workgroup. KDOQI clinical practice guideline for nutrition in CKD: 2020 Update. Am J Kidney Dis. 2020;76(3)(suppl 1):S1-S107. Available at: Website

General Food Allergy Guidelines

Milk, eggs, wheat, soy, fish, crustacean shellfish, peanuts, and tree nuts have represented the eight major food allergens since the Food Allergen Labeling and Consumer Protection Act (FALCPA) became law (FDA, 2022). Sesame is the ninth food allergen, as of 2023 (FDA, 2022). It only takes a trace amount of an allergen to trigger a reaction in some people with food allergy, so vigilance in monitoring ingredients is required (CDC, 2022).

A food avoidance diet is indicated once a food allergy has been evaluated and diagnosed by the medical team. A food intolerance may also require a food avoidance diet. Food allergies and food intolerances share similar symptomology, but a food allergen can result in a life-threatening reaction (AAAAI, 2022).

Because celiac disease involves the immune system and preventing cross-contact is essential, it is similar to a true food allergy. Thus, gluten is often treated as an allergen in a facility. A gluten-free diet is ordered for an individual with celiac disease, gluten intolerance, or non-celiac gluten sensitivity. See Gluten-Free Diet for additional guidance.

Keeping the Facility and Patients Safe

Strict omission of the allergy-causing food is the only way for patients to avoid a food allergy reaction. In a health care setting, health care staff members are responsible for identifying the allergen and informing the appropriate network of individuals about their patients' food allergies and providing safe meals.

The registered dietitian nutritionist may screen for food allergies or intolerances and coordinates care with the health care team. Determining a true food allergen from a food intolerance may affect the diet order.

Food allergies can be managed in all facilities with proper training and education. All institutions should have policies and procedures that outline each staff person's role in keeping the patient safe.

Food allergy awareness in-services with foodservice, nursing, and other staff are recommended. The requirements for food allergen training may vary based on state and local regulations and facility policy. Note that the ServSafe certification program often includes food allergen education, which may be sufficient food allergen training.

Food may be served in nontraditional settings such as community rooms and rec rooms that are not managed by foodservice, or food may be brought in from outside parties. It is important to inform and educate each patient's care team about their food allergies.

Avoiding Cross-Contact in the Kitchen

It is critical to prevent cross-contact in the foodservice kitchen. Cross-contact occurs when a non-allergenic food comes into contact with an allergenic food and becomes unsafe. The following precautions are necessary to prevent cross-contact in foodservice kitchen environments:

- Wash hands thoroughly and put on a fresh pair of gloves before preparing an allergen-free meal.
- Prepare allergen-free food first, cover it, label it, and remove it from the cooking area before preparing allergen-containing foods.
- Use hot, soapy water to clean kitchen surfaces, cutting boards, and utensils. Soap and water or commercial wipes will remove a food allergen. Sanitizing gels or water alone will not remove an allergen.
- When possible, maintain a separate allergy-free food preparation station. Wash kitchen countertops often, including immediately before and after food preparation.
- Clean toaster oven rack often and consider the option of using commercially available reusable toaster bags. Alternatively, use a separate toaster strictly for allergen-free food.
- Use squeeze bottles for condiments whenever possible. Any knife or other utensil that has previously contacted foods containing the allergen proteins should not be reused in a condiment jar.
- Do not reuse frying oil after it has been used to fry a food containing the allergen.
- Use color coding (eg, stickers) for easy identification of allergen-free foods
- Store allergen-free food on separate shelves in the refrigerator and pantry. Allergen-free food should be stored on higher shelves above the allergen-containing foods.
- Wrap or cover allergen-free meals to prevent accidental cross-contact via spills and splashes.
- During meal delivery, place allergen-free meals on a shelf higher than all other meals in the delivery cart or carry allergen-free meals on a tray directly from the kitchen to the patient.
- Do not attempt to make a food allergen-free by picking out the allergen (eg, removing croutons from a salad), as such efforts will not make the food safe for the patient to consume.
- Consider using foil (wrapping 2 times) to prepare allergen-free foods that are cooked on a grill. A grill should be fully cleaned before using it to cook for the patient with food allergy.

Label Reading Tips for Health Care Facilities

Processed foods can vary greatly in their ingredients. Always look at the ingredient list to identify sources of food allergens. When it's not clear if a food contains an allergen, call the manufacturer. For convenience, it is recommended that facilities keep a list of contact information for food manufacturers.

Ingredients must be listed on the label if they are present in any amount, even in colors, flavors, or spice blends. Manufacturers must list the specific nut (eg, almond, walnut, or cashew) or seafood (eg, tuna, salmon, shrimp, or lobster) that is present in the food.

To identify food allergens using a food label, look for the following identifiers:

- The food is listed in the precautionary “contains” statement. Example: “Contains: egg, wheat, milk, and soy.”
- The food ingredient is listed in parentheses. Example: “albumen (egg)”
- The food allergen appears in bold type in the list of ingredients. Example: “Ingredients: Enriched flour (wheat flour), whey (milk), eggs, lecithin (soy).”

The use of advisory statements such as “may contain,” “processed in a facility that also processes,” or “made on equipment with” is voluntary for manufacturers. There are no laws governing or requiring these statements, so manufacturers may or may not indicate if a product contains a specific allergen. The US Food and Drug Administration advises manufacturers in the food industry that “may contain” advisory labels should be accurate and not used as a substitute for following the current good manufacturing practices (FDA, 2022).

Indication: Food allergy diets are indicated for individuals with food allergies.

Nutritional Adequacy: Food allergy diets can meet the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients despite the intentional restrictions required of a food allergy or food intolerance.

Resources for Health Care Facilities:

Centers for Disease Control (CDC) Food Allergies Publications and Resources

Food Allergy Research and Education (FARE)

Nutrition.gov Food Allergies and Intolerances

The National Institute of Health (NIH) Allergy and Infectious Diseases

USDA’s National Agricultural Library Food and Nutrition Information Center

Specific Food Allergies

Egg Allergy

Read nutrition facts labels to identify allergens. Avoid any foods prepared with egg ingredients.

Food Group	Foods to Avoid
Grains	Egg bread, egg rolls, challah, egg matzo, egg noodles Any foods that contain egg such as breads, rolls, baked goods, waffles, pasta

Protein Foods	Eggs, egg whites, egg yolks, egg substitutes, surimi
Oils	Mayonnaise or salad dressings
Beverages	Eggnog
Other	Custard or meringue

Egg Allergy Sample 1- Day Menu

Meal	Menu
Breakfast	1 cup oatmeal 1 banana 1 tablespoon peanut butter 1 cup 1% milk
Morning Snack	$\frac{3}{4}$ cup yogurt $\frac{1}{2}$ cup blueberries
Lunch	2 slices whole wheat bread 3 ounces turkey 2 teaspoons mustard 2 cups lettuce for salad 1 tablespoon olive oil and vinegar dressing 1 apple 1 cup 1% milk
Afternoon Snack	$\frac{1}{4}$ cup hummus 3 celery sticks 3 carrot sticks
Dinner	3 ounces chicken, baked $\frac{1}{2}$ cup broccoli, cooked 1 baked sweet potato 1 whole wheat roll 2 teaspoons margarine, soft, tub

Menu Nutrient Analysis

Sesame Allergy

Read nutrition facts labels to identify allergens. Avoid any foods prepared with sesame ingredients.

Food Group	Foods to Avoid
Grains	Sesame flour
Protein Foods	Sesame seeds, tahini
Fats and Oils	Sesame oil
Other	Sesame salt, sesame paste

Sesame Seeds Allergy Sample 1-Day Menu

Breakfast	1 hard-boiled egg 1 cup oatmeal 1 banana 1 cup 1% milk
Morning Snack	$\frac{3}{4}$ cup yogurt $\frac{1}{2}$ cup blueberries
Lunch	2 slices whole wheat bread (without sesame seeds) 3 ounces turkey 2 teaspoons mayonnaise 2 cups lettuce for salad 1 tablespoon olive oil and vinegar dressing 1 apple 1 cup 1% milk
Afternoon Snack	1 tablespoon peanut butter 5 celery sticks

Evening Meal	3 ounces chicken, baked ½ cup broccoli, cooked 1 baked sweet potato 1 whole wheat roll (without sesame seeds) 2 teaspoons margarine, soft, tub
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Menu Nutrient Analysis

Soy Allergy

Read nutrition facts labels to identify allergens. Avoid any foods prepared with soy ingredients.

Food Group	Foods to Avoid
Grains	Soy fiber, flour, grits, sprouts
Protein Foods	Processed meats and sausage that may have added soy ingredients Tofu, tempeh, edamame, soynuts, textured vegetable protein, soy protein (concentrate, hydrolyzed, isolate) Meat alternative, vegetarian and vegan products that contain soy ingredients
Dairy	Soymilk, soy yogurt, soy cheeses, soy ice cream
Oils	Salad dressings that contain soy
Other	Salt and seasonings that soy ingredients Hoisin, miso, natto, shoyu sauce, soy sauce, tamari Broths, bullions, marinades with soy ingredients Nutritional bars and supplements with soy ingredients such as soy protein beverages, high-protein supplements, high-energy supplements, breakfast bars and drinks

Soy Allergy Sample 1- Day Menu

Breakfast	1 hard-boiled egg 1 cup oatmeal 1 banana 1 cup 1% milk
Morning Snack	$\frac{3}{4}$ cup yogurt $\frac{1}{2}$ cup blueberries
Lunch	2 slices whole wheat bread 3 ounces turkey 2 teaspoons mayonnaise 2 cups lettuce for salad 1 tablespoon olive oil and vinegar dressing 1 apple 1 cup 1% milk
Afternoon Snack	$\frac{1}{4}$ cup hummus 3 celery sticks 3 carrot sticks
Evening Meal	3 ounces chicken, baked $\frac{1}{2}$ cup broccoli, cooked 1 baked sweet potato 1 whole wheat roll 1 teaspoon olive oil

Menu Nutrient Analysis

Tree Nut Allergy

Read nutrition facts labels to identify allergens. Avoid any foods prepared with tree nut ingredients.

Food Group	Foods to Avoid
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Protein Foods	Tree nuts: Almond, beechnut, Brazil nut, butternut, cashew, chestnut, chinquapin, coconut, filbert/hazelnut, ginkgo, hickory, lychee nut, macadamia nut, pecan, pili nut, pine nut (pignolia nut), pistachio, shea nut, walnut Mortadella (a type of luncheon meat that contains pistachios) Tree nut butters
Beverages	Nut-flavored coffees (made with natural nut extracts)
Other	Natural extracts such as almond or hazelnut extract Barbecue sauces and other marinades with tree nut ingredients Marzipan Nougat Artificial nuts Pesto Nut meal Gianduja

Tree Nut Allergy Sample 1-Day Menu

Breakfast	1 hard-boiled egg 1 cup oatmeal 1 banana 1 cup 1% milk
Morning Snack	$\frac{3}{4}$ cup yogurt $\frac{1}{2}$ cup blueberries
Lunch	2 slices whole wheat bread 3 ounces turkey 2 teaspoons mayonnaise 2 cups lettuce for salad 1 tablespoon olive oil and vinegar dressing 1 apple 1 cup 1% milk

Afternoon Snack	$\frac{1}{4}$ cup hummus 3 celery sticks 3 carrot sticks
Evening Meal	3 ounces chicken, baked $\frac{1}{2}$ cup broccoli, cooked 1 baked sweet potato 1 whole wheat roll 2 teaspoons margarine, soft, tub

Menu Nutrient Analysis

Fish Allergy

Use the blue "Related Client Ed" link to the right of the title of this page to access the Fish Allergy Nutrition Therapy to view the Foods to Avoid table and the Sample 1-Day Menu with nutrient analysis.

Milk Allergy

Use the blue "Related Client Ed" link to the right of the title of this page to access the Milk Allergy Nutrition Therapy to view the Foods to Avoid table and the Sample 1-Day Menu with nutrient analysis.

Peanut Allergy

Use the blue "Related Client Ed" link to the right of the title of this page to access the Peanut Allergy Nutrition Therapy to view the Foods to Avoid table and the Sample 1-Day Menu with nutrient analysis.

Shellfish Allergy

Use the blue "Related Client Ed" link to the right of the title of this page to access the Shellfish Allergy Nutrition Therapy to view the Foods to Avoid table and the Sample 1-Day Menu with nutrient analysis.

Wheat Allergy

Use the blue "Related Client Ed" link to the right of the title of this page to access the Wheat Allergy Nutrition Therapy to view the Foods to Avoid table and the Sample 1-Day Menu with nutrient analysis.

References

American Academy of Allergy Asthma and Immunology. Food Allergy vs. Food Intolerance. Reviewed September 28, 2020. Available at: Website. Accessed

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Centers for Disease Control and Prevention. Food allergies in schools. Reviewed August 23, 2022. Available at: Website. Accessed September 30, 2022.

US Food and Drug Administration. Food allergies. Updated October 10, 2022. Access November 16, 2022. Available at: Website

Fiber-Restricted Diet

Definition: The fiber-restricted diet comprises less than 13 g of fiber daily. A fiber-restricted diet includes limited amounts of indigestible foods and limits whole grains, seeds, whole nuts, raw fruits, raw vegetables, and the connective tissues of meats.

Indication: A fiber-restricted diet may be indicated for individuals with acute diverticulitis, Crohn's disease, or ulcerative colitis, or for patients recovering from gastrointestinal tract surgery (such as new colostomy or ileostomy) (Makharia, 2022; Bridges, 2020; Levine, 2020). This diet should decrease the volume and bulk of the stool and cause slower movement of foods through the intestines. It may also decrease symptoms of diarrhea, abdominal pain, gas, and bloating. For individuals who have recently undergone gastrointestinal surgery, slowly and progressively increasing fiber intake may help to prevent bowel obstructions.

Nutritional Adequacy: A fiber-restricted diet may not meet the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients. Fiber intake on this diet may be inadequate and other micronutrient and macronutrient deficiencies are possible because this diet intentionally restricts intake of whole grains, fruits, and vegetables. This diet is not intended for long-term use.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Fiber-Restricted (13 grams) Sample 1-Day Menu with nutrient analysis and the Foods List.

References

Bridges M. Low-fiber diet. National Institutes of Health. U.S. National Library Medline Plus. Updated August 20, 2020. Accessed March 9, 2021. Available at: Website

Levine A, Rhodes JM, Lindsay JO, et al. Dietary guidance from the International Organization for the Study of Inflammatory Bowel Diseases. Clin Gastroenterol Hepatol. 2020;18(6):1381-1392. Abstract.

Makharia G, Gibson PR, Bai JC, et al. World Gastroenterology Organisation Global Guidelines: Diet and the gut. J Clin Gastroenterol. 2022;56(1):1-15. Full text.

High-Fiber Diet

Definition: A high-fiber diet includes added sources of fiber from fruits, vegetables, legumes, nuts and seeds, and whole grains. A fiber intake of 25 g to 38 g per day reflects a general, healthful eating pattern and an adequate fiber intake (USDA, 2020). If fiber needs to be increased in an individual's diet, it should be done gradually. Increased fluid intake may also be necessary when dietary fiber content is increased.

Indication: A high-fiber diet may be indicated for individuals with heart disease, constipation, diabetes, diverticular conditions, and Crohn's disease (Makharia, 2022; Levine, 2020; Evert 2019; Liu, 2017; Paquette, 2016; Crowe, 2011). A high-fiber diet promotes intestinal health and growth of gut microbiota (Makharia, 2022).

Nutritional Adequacy: The high-fiber diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Use the blue "Related Client Ed" link to the right of the title of this page to access the High-Fiber Diet Sample 1-Day Menu with nutrient analysis and the Foods List.

References

Crowe FL, Appleby PN, Allen NE, Key TJ. Diet and risk of diverticular disease in Oxford cohort of European Prospective Investigation into Cancer and Nutrition (EPIC): Prospective study of British vegetarians and non-vegetarians. *BMJ*. 2011;343:1-15. Full text.

Evert AB, Dennison M, Gardner CD, et al. Nutrition therapy for adults with diabetes or prediabetes: a consensus report. *Diabetes Care*. 2019;42(5):731-754. doi: 10.2337/dci19-0014

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Liu PH, Cao Y, Keeley BR, et al. Adherence to a healthy lifestyle is associated with a lower risk of diverticulitis among men. *Am J Gastroenterol*. 2017;112(12):1868-1876. Abstract.

Makharia G, Gibson PR, Bai JC, et al. World Gastroenterology Organisation Global Guidelines: Diet and the gut. *J Clin Gastroenterol*. 2022;56(1):1-15. Full text.

Paquette IM, Varma M, Ternent C, et al. The American Society of Colon and Rectal Surgeons' clinical practice guideline for the evaluation and management of constipation. *Dis Colon Rectum*. 2016;59(6):479-492. Full text.

US Department of Agriculture, US Department of Health and Human Services. Dietary Guidelines Advisory Committee. Dietary Guidelines for Americans, 2020-2025. Updated 2020. Accessed March 24, 2022. Available at: Website

Gluten-Free Diet

Definition: A gluten-free diet restricts all sources of gluten and gluten-containing products. Gluten is found in wheat (wheatberries, durum, emmer, semolina, spelt, farina, farro, graham, and einkorn), rye, and barley. Gluten-free grains, (such as amaranth, quinoa, buckwheat, brown rice, teff, millet, and corn), and starchy vegetables (such as potatoes, sweet potatoes, and winter squash) can be used to replace gluten-containing menu options to ensure adequate nutrient intake. Oats can be included on a gluten-free diet if they are gluten-free, uncontaminated, and tolerated by the individual.

Indication: A gluten-free diet is indicated for individuals with celiac disease or gluten sensitivity. Removal of gluten from the meal plan is essential for gastrointestinal and nutritional health in individuals with these conditions (EAL, 2022; Raiteri, 2022).

Nutritional Adequacy: A gluten-free diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients. Many gluten-free grain foods are made from refined flours and/or starches, and these products may not be enriched with B-vitamins and iron. As a result, an individual's intake may be lacking in B-vitamins, iron, and fiber, especially if there is heavy reliance on low nutrient density foods. Micronutrient and macronutrient intake assessment by a registered dietitian nutritionist is key in preventing potential micronutrient deficiencies. (Raiteri A, 2022)

Use the blue "Related Client Ed" link to the right of the title of this page to access the Celiac Disease Sample 1-Day Menu with nutrient analysis and Foods List.

References

Academy of Nutrition and Dietetics, Evidence Analysis Library. Celiac disease (CD) guideline. Updated 2021. Accessed February 15, 2022. Available at: Website

Raiteri A, Granito A, Giamperoli A, et al. Current guidelines for the management of celiac disease: A systematic review with comparative analysis. World J Gastroenterol. 2022;28(1):154-175. Full text.

Lactose-Controlled Diet

Definition: The lactose-controlled diet limits the amount of lactose in the diet. Lactose is main type of sugar in milk and dairy foods. A specific enzyme, lactase, in the small intestine is needed to digest lactose. People who do not produce enough lactase may have trouble digesting lactose. Limiting or avoiding milk and dairy products or consuming lactose-free products may help ease symptoms (Markharia, 2022).

Indications: A lactose-controlled diet is indicated for individuals with lactose intolerance or lactase enzyme deficiency. A lactose-controlled diet may be individualized based on symptoms (eg, diarrhea, bloating, stomach pain, and gas) and food tolerances.

Nutritional Adequacy: The lactose-controlled diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients. Milk and dairy foods are primary sources of calcium, vitamin D, and potassium. When milk and dairy foods are limited, the individual may need to get calcium, vitamin D, and potassium from other foods.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Lactose-Controlled Nutrition Therapy Sample 1-Day Menu with nutrient analysis and the Foods List.

References

Makharia G, Gibson PR, Bai JC, et al. World Gastroenterology Organisation Global Guidelines: Diet and the gut. J Clin Gastroenterol. 2022;56(1):1-15. Full text.

Fat-Restricted Diet

Definition: Fat-restricted diets limit the daily total fat intake to 20% to 30% of total energy. A 2000 kcal/day menu should limit total fat to approximately 45 g to 65 g. A fat-restricted diet may be individualized based on symptoms or diagnosis. A common low-fat diet order comprises 50 g fat/day.

Indication: A fat-restricted diet may be indicated if an individual is having trouble digesting or absorbing fat and may help prevent uncomfortable side effects—such as diarrhea, bloating, and cramping—that may occur with consumption of high-fat foods. This diet may be prescribed for disorders of the pancreas, gall bladder, liver, and gastrointestinal tract (Arvanitakis, 2020; Madden 2021).

Nutritional Adequacy: A fat-restricted diet meets the Dietary Reference Intakes (DRIs) for macronutrients and micronutrients.

Use the blue "Related Client Ed" link to the right of the title of this page to access the Fat-Restricted Sample 1-Day Menu with nutrient analysis and Foods List.

References

Arvanitakis M, Ockenga J, Bezmarevic M, et al. ESPEN guideline on clinical nutrition in acute and chronic pancreatitis. Clin Nutr. 2020;39(3):612-631. Full text.

Madden AM, Trivedi D, Smeeton NC, Culkin A. Modified dietary fat intake for treatment of gallstone disease. Cochrane Database Syst Rev. 2021;2021(6):CD012608. Full text.

Diet Manual > Obsolete Diets and Diet Terminology

Obsolete Diets and Diet Terminology

The table below lists conditions associated with obsolete diets and describes the outdated nutrition specifications. Use of the noted obsolete terms and associated recommendations should be avoided.

Condition	Obsolete Diet Name	Rationale for “Obsolete” Label
Acute Renal Failure	• Renal diet (a single diet for all renal diseases) • Low-protein diet	Previously, a standard 60 g protein, 2 g sodium, 2 g potassium diet was recommended for all individuals with renal disease. Malnutrition can develop during the course of chronic kidney disease and may be related to increased morbidity and mortality for those with renal disease. Inadequate protein and inadequate energy intake are critical causes of malnutrition in this population. Diet orders should be liberalized as much as possible to maximize oral intake and quality of life. See Renal client education handouts for current nutrition therapies.
		Research has shown that patients postoperatively had no difference in tolerance to a clear liquid diet or regular diet. Significantly more energy and protein were consumed on a regular diet compared with a clear liquid diet (74% vs 42% of energy and 58% vs 8% of protein) (Jeffery, 1996;

Bowel Surgery	• Clear liquid diet	Franklin 2011). Careful evaluation of diet progression is needed in patients with significant bowel resections, strictures, fistula, or motility disorders. Advancing from a clear liquid to a full liquid diet may not be necessary as patients may tolerate an oral diet more quickly than previously anticipated. Full liquid diets are often very high in fat and may not be well tolerated post bowel surgery. Clear liquid diets should be used with extreme caution and should not be used for more than 1-2 meals at a time; hospital protocols should be developed to monitor these parameters. Research has demonstrated that clear liquid diets are used longer than necessary (greater than 3 days) and with poor justification (Franklin, 2011). See Clear Liquid Diet Nutrition Therapy (this diet should not be used more than 1 day).
Chronic Kidney Disease	• Renal diet (nonspecific for stage of kidney disease)	Previously, a standard 60 g protein, 2 g sodium, 2 g potassium diet was recommended for all individuals with renal disease. Malnutrition can develop during the course of chronic kidney disease and may be related to increased morbidity and mortality for those with

(CKD)	• 60 g protein • 60 g protein, 2 g sodium, 2 g potassium • Giovanetti diet	renal disease. Inadequate protein and inadequate energy intake are critical causes of protein-energy malnutrition. Diet orders should be liberalized as much as possible to maximize oral intake and quality of life. See Renal Nutrition Therapy for current nutrition therapies.
Cirrhosis	• Low-protein diet	Traditionally, protein restrictions were imposed on patients with cirrhosis because of concern about their ability to metabolize protein. Protein has also been restricted because of the theory regarding its potential to provoke or cause hepatic encephalopathy. These protein restrictions can provoke an inadequate intake and subsequent malnutrition. Few patients are truly protein sensitive or intolerant. Most patients with cirrhosis do not need a dietary protein restriction. See Cirrhosis Nutrition Therapy for current nutrition recommendations.
	• Low-cholesterol diet • Low-fat diet • American Heart Association I or	Previously, patients may have been prescribed one diet emphasizing reduction of saturated fat, cholesterol, and sodium. Research suggests several dietary patterns are appropriate for cardiovascular disease, which emphasizes healthy food choices (Jacobson, 2015; Eckel, 2013).

Coronary Artery Bypass Graft (CABG) Cardiovascular Disease	American Heart Association II diet • National Cholesterol Education Program Step I or Step II diet • Low-sodium diet • Low-salt diet • Mild salt-restricted diet	Research suggests that for individual with heart failure, the RDN should individualize energy intake, assess need for purposeful weight loss (once fluid status is stable), individualize protein intake 1.1-1.4g/kg, and tailor sodium (2000-3000 mg) and fluid intake (1-2 L fluid per day) (Academy, 2017). See Cardiovascular section client education handouts for current nutrition therapies.
Diarrhea	• Clear liquid diet	Because of the high sugar content present in clear liquid diets, the resulting hyperosmolality may exacerbate diarrhea. See Diarrhea Nutrition Therapy for current nutrition recommendations.
Diverticulosis	• Low-fiber diet • Regular diet avoiding nuts, seeds, and hulls	Historically, dietary advice has recommended avoidance of nuts, seeds, and hulls. Recent literature has reevaluated this approach based on the lack of evidence in published research that supports these common nutrition recommendations (Strate, 2008; Eglash, 2006). See Diverticular Conditions for current nutrition recommendations.

Gastroesophageal Reflux Disease (GERD)	• Bland diet	The term "bland diet" and the use of milk to treat heartburn are considered to be obsolete. See Gastroesophageal Reflux Disease (GERD) Nutrition Therapy for current recommendations.
Hepatitis	• Low-protein diet • Low-fat diet • Low-caffeine diet	Previously, many patients were restricted in their protein intake with the diagnosis of chronic hepatitis/liver disease as a means of preventing or controlling hepatic encephalopathy. Other nutrients that may have been historically restricted include fat and caffeine; however, these do not require restriction unless symptoms of the liver disease dictate. See Hepatitis Nutrition Therapy for current recommendations.
Hypoglycemia (non-diabetic)	• No concentrated sweets	Traditional advice that is no longer appropriate is to avoid foods containing sugars and to eat protein-containing and fat-containing foods. Research on the glycemic index and sugar raises questions about the appropriateness of restricting only sugars, as many of these foods have a lower glycemic response than many of the starches that were recommended in the past. Furthermore, protein is also a stimulant of insulin, and a high-fat intake, especially saturated fats, may interfere with the body's

		use of insulin. See Hypoglycemia (Not Caused by Diabetes) Nutrition Therapy for current recommendations.
Nephrotic Syndrome	• High-protein diet • Renal diet (a single diet for all renal diseases)	Previously, a standard 60 g protein, 2 g sodium, 2 g potassium diet was recommended for all individuals with renal disease. Malnutrition can develop during the course of chronic kidney disease and may be related to increased morbidity and mortality for those with renal disease. Inadequate protein and inadequate energy intake are critical causes of malnutrition. Diet orders should be liberalized as much as possible to maximize oral intake and quality of life. See Nephrotic Syndrome Nutrition Therapy for current recommendations.
Organ Transplant	There are no obsolete diets or names of diets associated with transplantation alone.	Obsolete diet terms for organ transplant patients are only applicable with comorbidities—for example, "ADA (American Diabetes Association) diet" for an organ transplant patient with diabetes mellitus. There are, however, obsolete practices such as "no fresh fruits and vegetables" or "low-bacteria diet." These restrictions are not necessary for organ transplant recipients.

Pancreatitis	There are no obsolete diets or names for pancreatitis	Many institutions consider a progression in diet from clear liquids to full liquids to be standard. However, a full liquid diet may be problematic as it typically consists of high-fat, lactose-containing (ie, milk-based) foods that may exacerbate symptoms. Furthermore, patients who have had prolonged nothing by mouth (nil per os, or NPO) status or who have experienced a gastrointestinal insult may have temporarily reduced levels of the lactase enzyme. They may develop a transient but resolvable lactose intolerance. Progression to a solid, low-lactose, low-fat, high-protein nutrition therapy may prudent in achieving tolerance during recovery. See Pancreatitis Nutrition Therapy for current recommendations.
Preeclampsia/Eclampsia	• No added salt • Mild salt restricted	
Pressure Ulcers	A specific diet for wound healing has not previously existed.	Those with pressure ulcers are often ordered a high-protein, high-kilocalorie diet without consideration of individual protein and energy needs, increased fluid requirements, obesity, and changes in gastrointestinal tolerance that are associated with malnutrition.

Urinary Stones/Urolithiasis	• Low-calcium diet	
Type 1 Diabetes	• ADA (American Diabetes Association) diet • No concentrated sweets diet • No sugar added • Low sugar • Liberal diabetic diet	The term "ADA diet" has never been clearly defined but has usually meant a physician-determined energy level with a specified percentage of carbohydrate, protein, and fat based on exchange lists. This term should no longer be used because neither the Academy of Nutrition and Dietetics nor American Diabetes Association endorse any single meal plan or specified percentages of macronutrients. With regard to the no concentrated sweets, no sugar added, low sugar, and liberal diabetic diets: None of these approaches to food and meal planning is appropriate because each unnecessarily restricts sucrose. See Type 1 Diabetes Nutrition Therapy for current recommendations.
	• ADA diet • No	The term "ADA diet" has never been clearly defined but has usually meant a physician-determined energy level with a specified percentage of carbohydrate, protein, and fat based on exchange lists. This term should no longer be used because neither the Academy of Nutrition and Dietetics nor American

Type 2 Diabetes	concentrated sweets diet • No sugar added • Low sugar • Liberal diabetic diet	Diabetes Association endorse any single meal plan or specified percentages of macronutrients. With regard to the no concentrated sweets, no sugar added, low sugar, and liberal diabetic diets: None of these approaches to food and meal planning is appropriate because each unnecessarily restricts sucrose. See Type 2 Diabetes Nutrition Therapy for current recommendations.
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Franklin GA, McClave SA, Hurt RT, et al. Physician-delivered malnutrition: Why do patients receive nothing by mouth or a clear liquid diet in a university hospital setting?. JPEN J Parenter Enteral Nutr. 2011;35(3):337-342. Abstract.

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Overview/General Information

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Components of correctional foodservice —such as, nutrition care, menu planning, food production, and meal service—are generally similar to foodservice in other institutional health care settings. However, there are also many unique concepts, policies, and procedures that affect correctional foodservice and nutrition services; many of these practices are recognized or adopted nationally.

Correctional facilities (eg, prisons, jails, and other detention facilities) provide for all of an individual's needs, including nutrition and health care. In foodservice, menus are designed to provide foods and beverages to meet the needs of diverse populations and demographics, including individuals with medical/therapeutic diet needs. Often, a “heart-healthy” diet for the general population is planned to satisfy overall nutrition needs, meet regulatory requirements, and accommodate budget constraints. Special considerations for dining, such as resident safety and availability of therapeutic foods, may be required as meals are served in multiple settings within a facility, including dining halls, housing units, health services units, or individual cells.

Although terminology to describe populations in correctional settings may vary across facilities— for example, the term resident, patient, prisoner, or inmate—the term “resident” will be used throughout this section of the NCM Diet Manual.

The registered dietitian nutritionist (RDN) working in corrections can use the NCM Diet Manual as a resource to offer guidance to foodservice and medical staff providing care for residents, or it can be adapted as a correctional facility's diet manual. The educational materials in this manual can be used as resources to educate the residents as well. The following forms can be incorporated into correctional foodservice practice:

- Corrections Diet Terminologies
- Medical Diet Requisition
- Medical Diet Roster Sign-Off Sheet
- Menu Substitution Logs

Additional Online Resources

Association of Correctional Food Service Affiliates_

American Correctional Association_

American Jail Association_

National Commission on Correctional Health Care_

National Institute of Corrections

School Nutrition Association_

US Department of Agriculture National School Lunch Program/School Breakfast Program_

Diet manuals and/or other approved procedural publications are required in most correctional institutions and should be reviewed/approved per governing agency/facility protocol. They should be available to both the foodservice and the medical departments. Many state correctional systems, particularly prisons, provide a statewide diet manual for use in all institutions within their purview; the Nutrition Care Manual (NCM) Diet Manual can be adapted for this setting. Jails and other detention facilities that have created facility-specific manuals with approved diets for their institution, or use a simplified/liberalized diet manual, can use the NCM as an adjunct resource. The registered dietitian nutritionist (RDN) is usually instrumental in assisting facilities in diet regulations and provides detailed diet guidance upon request by a foodservice authority or administration.

Meeting Nutritional Standards

Many corrections regulations refer to the Dietary Reference Intakes (DRIs) and Recommended Dietary Allowances (RDAs) for nutrition standards in correctional facilities. Other resources and guidelines used include the following:

- Dietary Guidelines for Americans
- American Heart Association Diet and Lifestyle Recommendations
- MyPlate.gov
- US Department of Agriculture Child Nutrition Program (National School Lunch Program/National School Breakfast Program)
- DASH (Dietary Approaches to Stop Hypertension) Eating Plan

Governing Agencies

Standards, policies, and directives vary by state, facility, age, and sex of the populations served. Regulations range from very precise, in which meal patterns and macronutrients/micronutrients to be provided are explicitly defined, to very general noting that the goal is “ensuring adequate nutrition.”

The Bureau of Prisons maintains its own standards for federal facilities, whereas state governments set the regulations for state facilities. Only 38 states have nutrition standards or guidelines specifically for jail and detention facilities.

Standards for contracted and private facilities are based on the governing agency, contractual requirements, accreditations, and the type of facility (federal, state, US Marshalls, detention, etc.). If a given facility houses more than one type of resident, more than one set of standards may be required.

Accrediting Organizations

Accreditation for corrections foodservice is voluntary and is fee-based, but facilities often incorporate these standards in their policies and procedures, regardless of accreditation status. The American Correctional Association (ACA) and National Commission on Correctional Health Care (NCCHC) are the two most prominent organizations for standards and accreditations in the corrections industry. Both organizations have defined standards that address

nutrition, medical/therapeutic diets, and menu review and approval by the RDN. Following these practices helps to reduce potential litigation concerns in correctional facilities.

The Joint Commission is the accrediting organization for health services within correction facilities and publishes standards of care.

Menu Planning in Correctional Settings

The following considerations affect menu planning:

- Population
- Governing agency standards
- State standards
- Accreditations
- Facility policy
- Court mandates
- Contracts Safety and availability protocols related to food purchasing, delivery, storage, and preparation

When planning menus for an agency or facility, the registered dietitian nutritionist (RDN) has to be knowledgeable of these criteria, regulatory standards, and policies to ensure compliance and meet the needs of the populations served.

Menus in correctional settings are cyclical and most often provide higher energy ranges (average 2200 kcal to 2800 kcal for adults and higher for youth). These menus factor in age, sex, and activity levels of the populations; mandated regulations; tray capacity; foodservice equipment; budgets; and security constraints.

Menu cycles commonly range from 3 to 6 weeks, and up to 12-week cycles with seasonal changes in some agencies. Seasonal changes may correspond with facility garden produce as well.

Correctional menus may offer one, two, or three hot meals per day, although it is rare to offer only one hot meal. These variations are based on governing agency regulations, contracts, and accreditation standards. Following are different types of meals and meal schedules available at different correctional facilities:

Hot meals: A meal must include at least one hot item to be considered a hot meal. Menus may include one to three hot meals per day, although menus with only one hot meal are rare. Two hot meals are common and are mandated in ACA Standards.

Cold meals: A cold meal is often a sack meal to correspond with reduced staffing during those shifts, or to serve residents who leave the facility for work crews, court, or medical purposes.

Brunch: Some facilities serve brunch on weekends.

Two meals/day: There are a few facilities that serve only two meals per day, but they are still required to meet all the nutrient needs of the population in the

two meals.

Snacks: Snacks are included as part of the daily menu offerings in youth correctional facilities, but they are not routinely included in menus for adult residents unless part of a therapeutic regimen to address nutritional concerns.

Fortified food items such as beverages and baking mixes are often served as part of the standard menu to meet nutrient needs in the highly cost-controlled environment. These items are fortified with varying micronutrients such as calcium, potassium, zinc, and vitamins A, C, D, and E.

When possible, fresh and frozen produce is preferred over canned, processed items. Given the higher energy content of menus, some facilities/agencies may purchase reduced-sodium meats and soup bases as part of the heart-healthy menu for the general population to reduce overall sodium content and to be consistent with the US Dietary Guidelines and American Heart Association Guidelines.

Many facilities offer pork-free menus and/or meatless alternative entrees. A 4-week, pork-free menu cycle is common in jails and detention centers, whereas prisons may have a lengthier menu cycle and are more likely to offer meatless alternatives as an option to the entire resident population. The meatless alternatives may be acknowledged for a religious option, lifestyle choice or preference, depending on the facility or agency policy.

Menu Considerations to Meet Individual Needs

In the correctional setting, resident housing arrangements can be grouped according to age, sex, and classification of crime. Menus are planned to reflect the nutrition needs of these populations as dictated by governing agencies, contracts, and accreditation standards.

When an institutional (prisons, work release, boot camps) population comprises men only or women only, menus are developed to meet the population's nutrition needs. Youth and adolescents are usually defined as up to age 20, but this definition varies by state. The institution may be correctional/detention or community residential and typically adolescents are housed in the same building, but segregated by sex and classification of crime (not by age group). Menus are developed to meet the population's nutrition needs. If a facility that houses youth and adolescents is classified as a residential child care institutions, it could participate in Child Nutrition Programs such as National School Lunch Program (NSLP) and the School Breakfast Program (SBP).

In some institutions (jails, detention, community residential centers, work crew housing, and county prisons), adult male and female residents, and sometimes adolescents, are housed in one facility. Most often the menu is served that meets the highest nutrition needs for the reference population, which usually results in women provided more kilocalories than their normal requirement. Some facilities may offer a menu that is more appropriate for women. Accommodations should be made when adolescents are housed alongside adult residents to ensure their specific nutrition needs are met (eg, milk with all meals).

Pregnant residents may be housed in any the women's facilities mentioned

previously, or they may be housed separately by medical need or by classification of crime within a correctional facility. The menu may be modified to increase calorie and protein options (for example, serving milk or a fortified beverage with all meals) and snacks may be served to meet the nutrient needs of pregnant residents.

Adults age 55 and older are considered elderly residents. Some facilities have a separate unit to house older residents, while some jurisdictions have specialized geriatric facilities where all older residents are housed. Many facilities, such as jails or detention, may not differentiate and house by classification. Aging populations may need individualized diets and may require texture-modified diets.

Diets ordered for religious reasons are not considered medical/therapeutic diets. Religious diets are ideally addressed through a religious authority and administration. Residents with a religious diet order may be served foods that are not offered on the standard menu. Detailed lists of cultural and religious foods, including food practices and meal planning information, can be found in the NCM Cultural Foods section.

Security Risk Considerations

Foods that may be limited or prohibited for security risk include the following:

- Fresh fruit, canned fruit, fruit juice: Can be used to make alcohol (known in correctional facilities as “pruno” or “hooch”)
- Yeast, honey, and sugar: Can be used to make alcohol
- Powdered creamer: Flammable, can be used to set fire
- Pepper: Can be used as a weapon (to attack eyes)
- Nutmeg: Can provide the effect of a hallucinogenic drug
- Spinach: Can be dried and smoked, presenting risk for fire
- Bones: can be sharpened and used as weapons

Foodservice supplies that are limited or monitored for security purposes include the following:

- Metal eating utensils: Can be used as weapons
 - Eating utensils are most often plastic or made of other materials that cannot be broken or sharpened into a weapon.
 - Adaptive eating utensils may be considered for residents with specific, medically diagnosed conditions, but security protocols require documentation.
- Knives may be tethered to the counter for use in food preparation. Where knives are prohibited, knifeless kitchens may purchase pre-sliced or pre-shredded foods.
 - Containers with foil lids may be prohibited along with aluminum foil. Cellophane is the preferred substitute.
 - Glass containers.
 - Metal tabs from packaging and can lids may be controlled or tracked.

Menu Substitutions

Facilities must have clear, written policies regarding menu substitutions. Menu items should only be changed when there is a valid reason to alter the menu, such as if a vendor is out of product, an equipment failure, or a lockdown. If substitutions become necessary because of product quality or poor production, this change should happen between service for different cell blocks/housing units. Most often communication with administration is required if the menu is changed. Menu substitutions may be communicated to the resident population based on institutional protocol.

Menu substitutions should be documented on a Menu Substitution Log or a production record that is reviewed by the RDN. Some facilities complete the menu substitution log daily for each meal regardless of whether anything has changed. Just as in other health care facilities, menu changes can impact diets, so these must be documented as well.

Menu Reviews and Nutrition Analysis

Nationally recognized standards from the accrediting organizations the American Correctional Association and National Commission on Correctional Health Care require general menus and/or medical/therapeutic menus to be reviewed and approved by the RDN at least annually or when there is a significant menu change. Governing agency policies may also reflect these standards for menu review requirements whether accredited or not. A computerized nutrient analysis is the most common means to assess menus for nutritional adequacy.

Medical and Therapeutic Diets

Medical and therapeutic diets are ordered in correctional settings, as required. A standardized medical/therapeutic diet program, including definitions of medical and therapeutic diets and the criteria for ordering each type of order, should be established. The Corrections Diet Terminologies can be a useful resource.

Use the International Dysphagia Diet Standardisation Initiative (IDDSI) Level guidelines to determine texture-modified diet orders. The reason for ordering a texture-modified diet may be unique to the correctional setting, for example, psychological/security reasons (eg, risk for self-harm, suicide, or aggressive behavior). Snacks may be included or ordered as part of a medical/therapeutic diet for residents with the following conditions:

- Diabetes
- Hypoglycemia
- Pregnancy
- Unintended weight loss
- Eating disorder
- Long-term chronic illness

A general, healthful menu that provides consistent carbohydrates and accommodates food preferences can meet the nutrition needs of most residents and reduce the need for some specialized medical/therapeutic diets. There is not one standard set of diets that applies to all correctional settings as the restrictions may be contingent on the menu served to the general population and governing agency standards. Each facility will have standard dietary restrictions approved by the medical department

The registered dietitian nutritionist (RDN) can work with the medical department to review the list of approved diets at least annually or as per facility policy and accreditation requirements and to enforce consistent diet terminology and modifications for residents. The Process for Annual Diet Manual Review and Implementation can be a useful resource in this effort.

Allergy Diets and Intolerances

Food allergies and intolerances are often assessed on an individual basis to confirm validity of resident claims. Frequently, a stated food dislike or food preference becomes part of a medical diet order—for example, a stated aversion to soy or mayonnaise may be mistakenly documented as an allergy. Allergy testing for residents with suspected food allergies varies widely across institutions. Perform a thorough evaluation of resident-reported allergens via collection of allergy history data and laboratory testing to ensure residents with confirmed allergens receive necessary nutrients and are educated about self-management of allergen avoidance.

Meal Preparation in Correctional Settings

The meal preparation process in correctional settings differs from the process in other settings because residents often provide foodservice labor under the supervision of civilian or correctional foodservice staff. The extent to which residents participate in foodservice activities varies by facility based on policy; this work might primarily comprise sanitation or serving duties, baking, or full meal preparation.

Foodservice training is important for all kitchen staff. Many facilities/agencies offer ServSafe certification for foodservice staff and resident workers. Some facilities/agencies provide culinary certifications for resident foodservice workers. Following are some key areas that should be accommodated during the training and supervision of foodservice workers, both civilian and resident, and overall planning for all diet preparation (including medical and therapeutic diets):

- Foodservice skills
- Language barriers
- Literacy levels
- Math skills

Food Preparation Considerations

- Food prepared for medical/therapeutic diets should be as similar to the general population's menu as possible.
- Food prepared for medical diets/therapeutic is usually portioned from the general population's food before seasoning or adding any fat.
- Standardized recipe production guidelines may be developed to accommodate both the general population and those receiving medical/therapeutic diets to minimize production requirements.
- Occasionally, it may be necessary to offer alternative menu items depending on the nutrient content of the general population's menu and any agency or contract requirements.

Facilities that offer heart-healthy options may permit residents to self-select foods for their own diets. See NCM Diet Manual Heart-Healthy Diet.

The Process of Diet Order to Meal Delivery

Consistent with health care facilities, medical/therapeutic diets in the correctional setting are ordered by the medical department. Diet orders in the correctional settings should contain initiation and termination dates and are signed by authorized medical personnel. See Medical Diet Requisition Form.

Factors to address when establishing foodservice protocols in the correctional setting include staffing, preparation, and delivery. Diet trays may be premade, labeled with the resident's name and/or diet type and delivered to satellite feeding areas, or served upon request in the dining hall. If meals are served in the dining hall, it is the resident's responsibility to identify that they are on a specific diet and request the diet tray when coming through the line. Some institutions issue a diet identification card that presents dietary needs, whereas other facilities will establish a separate diet line, and some will employ the honor system and serve the diet when requested along with the rest of the population. Some facilities will serve medical/therapeutic diets first or last, depending on pill call/medication pass or insulin delivery schedules. Residents may consider some medical/therapeutic diets to have desirable foods, and some residents may request such a diet as a preference. Some facilities may serve medical/therapeutic diets before or after main line service to discourage requests based on preference rather than medical necessity.

Diet Compliance

Monitoring resident dietary compliance at meals (and via commissary purchases, as needed) is important for foodservice and for residents who are on a medical or therapeutic diet. A system to monitor diet compliance is a common part of dietary procedures in most correctional settings, particularly where residents coming to the dining hall to eat. The degree of a resident's dietary compliance can be used by the medical department to determine if dietary counseling and adjustment of the diet prescription are necessary.

In correctional settings, foodservice maintains a diet roster of all residents who are prescribed a medical/therapeutic diet along with other pertinent details related to the diet. See Medical Diet Roster Sign-Off Sheet. Protocols and procedures for measuring dietary compliance will vary by institution.

Diet Education and Counseling

Diet education enables residents to make independent menu selections, when possible, and/or understand the restrictions of their diets, empowering them to be responsible for their individual health. If requested by medical staff, the registered dietitian nutritionist (RDN) can provide diet education to residents one to one or in group settings. The RDN also provides the diet education materials to the medical department, which staff then reviews with residents. The NCM client education handouts provide adequate information to support nutrition education of residents.

When a medical/therapeutic diet is initially ordered for a resident, the diet should be reviewed with the resident. Depending on institutional protocols, the resident may be required to sign a form acknowledging the diet education was provided. Information to discuss with the resident includes the following:

- Content of the client education handout, including purpose of the diet as related to their diagnosis
- Possible complications from failure to follow the prescribed diet
- Facility's noncompliance policy

The Role of the Registered Dietitian Nutritionist

A prison may have a medical unit where medical nutrition therapy clinical documentation is standard protocol or requested by medical staff. Within some agencies, the registered dietitian nutritionist (RDN) may complete nutritional assessments and provide inmate diet instructions. Typically, in a medical correctional facility, where the RDN may be on site full time, the latter is more common. RDN functions in the ambulatory (1:1 visits, group classes) care, inpatient care, and long-term care settings in corrections are similar to RDN functions in other clinical care settings. The RDN's role may expand if additional credentials (eg certified diabetes care and education specialist) are held (ACFSA, 2001).

Nutrition counseling, especially when performed by an RDN, may help residents with self-management of diabetes. Group or individual counseling on carbohydrates in food that affect diabetes control is an important component of enabling residents in selecting better food options in the dining hall or commissary (ADA, 2014).

References: Correctional Facilities

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